Unary primitive recursive functions in Z/EVES

Formalization of Section 6 of D. E. Severin, *Unary primitive recursive functions*, J. Symbolic Logic 73(4), 2008, at the level of closures (as in the Mizar version), and of the two open questions answered in the README.

Conventions: one-argument functions are the elements of $\mathbb{N} \to \mathbb{N}$. A function or operator X of the paper is introduced by an axiomatic box (`funX` or `opX`) whose defining equations are labelled rewrite rules. Every axiomatic box is followed by a consistency theorem `XEx`, which exhibits a witness (`XWit`, defined by an explicit term) satisfying the box; so no definition adds an axiom that could be inconsistent.

1 Building functions

Z/EVES has no rules showing that a λ-term is a total function; these four lemmas provide them. A relation that is single-valued and total on X is a total function.

theorem disabled rule RelPow $[X, Y]$
$X \leftrightarrow Y = \mathbb{P}(X \times Y)$

proof
use relDefinition$[X,\ Y]$;
prove;
■

theorem disabled rule PfunIntro $[X, Y]$
$f \in X \leftrightarrow Y \land (\forall x : X;\ y, z : Y \mid (x, y) \in f \land (x, z) \in f \bullet y = z) \Rightarrow (f \in X \pfun Y \Leftrightarrow \mathit{true})$

proof
prove;
use pfunDef$[X,\ Y][R\ :=\ f]$;
prove;
with enabled (*inPower*, *compDef*) *prove*;
with enabled (*invDef*, *inId*) *prove by reduce*;
instantiate $x__0\ ==\ y,\ y__0\ ==\ x,\ z__0\ ==\ z$;
prove;
■

theorem disabled rule DomTotal $[X, Y]$
$f \in X \leftrightarrow Y \land (\forall x : X \bullet \exists y : Y \bullet (x, y) \in f) \Rightarrow \mathrm{dom}\, f = X$

proof
apply extensionality;
prove;
with enabled (*inDom*) *prove*;
instantiate $x\ ==\ y$;
prove;
instantiate $y__2\ ==\ y__0$;
prove;
■

theorem disabled rule FunIntro $[X, Y]$
$f \in X \leftrightarrow Y \land (\forall x : X \bullet \exists y : Y \bullet (x, y) \in f) \land (\forall x : X;\ y, z : Y \mid (x, y) \in f \land (x, z) \in f \bullet y = z) \Rightarrow (f \in X \to Y \Leftrightarrow \mathit{true})$

proof
use pfunIsFun$[X,\ Y][A\ :=\ X,\ B\ :=\ Y,\ f\ :=\ f]$;
with enabled (*PfunIntro*, *DomTotal*) *prove*;
■

theorem rule FunIsPfun $[X, Y]$
$f \in X \to Y \Rightarrow (f \in X \nrightarrow Y \Leftrightarrow true)$

proof
use fun_type$[X,\ Y]$;
with enabled (*inPower*) *prove*;
■

Relational composition and iteration of one-argument functions (the witness for iteration $F^{\square(a)}$ uses the Toolkit's iteration F^n).

theorem rule FunIsRel $[X, Y]$
$f \in X \to Y \Rightarrow (f \in X \leftrightarrow Y \Leftrightarrow true)$

proof
use inPowerTransitive$[X := X \to Y,\ Y := X \nrightarrow Y,\ Z := X \leftrightarrow Y]$;
use fun_type$[X,\ Y]$;
use pfun_type$[X,\ Y]$;
prove;
■

theorem DomFunGen $[X, Y]$
$\forall f : X \to Y \bullet \operatorname{dom} f = X$

proof
prove;
■

theorem BigcapSub $[X]$
$\forall T : \mathbb{P}(\mathbb{P} X);\ Y : \mathbb{P} X \mid Y \in T \bullet \bigcap T \subseteq Y$

proof
prove;
■

theorem SubAntisym $[X]$
$\forall A, B : \mathbb{P} X \mid A \subseteq B \land B \subseteq A \bullet A = B$

proof
prove;
apply extensionality to predicate $A = B$;
with enabled (*subsetDef*, *inPower*) *prove*;
■

theorem InPair $[X]$
$\forall u, p, q : X \mid u \in \{p, q\} \bullet u = p \lor u = q$

proof
prove;
■

theorem rule CompRelFun
$\forall f, g : \mathbb{N} \to \mathbb{N} \bullet f \fatsemi [\mathbb{N}, \mathbb{N}, \mathbb{N}] g \in \mathbb{N} \to \mathbb{N}$

proof
prove;
with enabled (*easyDomComp*) *prove*;
■

theorem rule CompRelApp
$\forall f, g : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet (f \fatsemi [\mathbb{N}, \mathbb{N}, \mathbb{N}] g)\, x = g\,(f\, x)$

proof
prove;
■

$IterSet == \{n : \mathbb{N} \mid \forall F : \mathbb{N} \to \mathbb{N} \bullet iter[\mathbb{N}]\, n\, F \in \mathbb{N} \to \mathbb{N}\}$

proof [IterSet$domainCheck]
prove;
use applyInRanFun$[\mathbb{Z},\ \mathbb{P}\,(\mathbb{N} \times \mathbb{N}) \to \mathbb{P}\,(\mathbb{N} \times \mathbb{N})][f := iter[\mathbb{N}],\ a := n]$;
use DomFunGen$[\mathbb{P}\,(\mathbb{N} \times \mathbb{N}),\ \mathbb{P}\,(\mathbb{N} \times \mathbb{N})][f := iter[\mathbb{N}]\, n]$;
prove;
equality substitute;
prove;
■

theorem rule IterBase
$0 \in IterSet$

proof
invoke IterSet;
prove;
■

theorem rule IterStep
$\forall n : IterSet \bullet 1 + n \in IterSet$

proof
invoke IterSet;
prove;
use iterPositive$[\mathbb{N}][R__1 := F,\ n__0 := n]$;
instantiate $F__0 == F$;
with enabled (*easyDomComp*) *prove*;
■

theorem IterSub
$\mathbb{N} \subseteq IterSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq IterSet$;
prove;
invoke IterSet;
prove;
■

theorem rule IterFun
$\forall F : \mathbb{N} \to \mathbb{N};\ n : \mathbb{N} \bullet iter[\mathbb{N}]\, n\, F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use IterSub;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
invoke IterSet;
prove;
instantiate $F__1 == F$;
prove;
■

theorem IterSucR
$\forall F : \mathbb{N} \to \mathbb{N};\ k, a : \mathbb{N} \bullet iter[\mathbb{N}]\,(1 + k)\,F\,a = iter[\mathbb{N}]\,k\,F\,(F\,a)$

proof
prove;
use composePositiveIterates$[\mathbb{N}][n := 1,\ k := k,\ R := F,\ Y := \mathbb{N}]$;
use CompRelApp$[f := F,\ g := iter[\mathbb{N}]\,k\,F,\ x := a]$;
use IterFun$[F := F,\ n := k]$;
prove;
■

theorem rule IterInDom
$\forall F : \mathbb{N} \to \mathbb{N};\ n : \mathbb{N} \bullet F \in \mathrm{dom}(iter[\mathbb{N}]\,n)$

proof
prove;
use applyInRanFun$[\mathbb{Z},\ \mathbb{P}\,(\mathbb{N} \times \mathbb{N}) \to \mathbb{P}\,(\mathbb{N} \times \mathbb{N})][f := iter[\mathbb{N}],\ a := n]$;
use DomFunGen$[\mathbb{P}\,(\mathbb{N} \times \mathbb{N}),\ \mathbb{P}\,(\mathbb{N} \times \mathbb{N})][f := iter[\mathbb{N}]\,n]$;
prove;
■

theorem rule IterZero
$\forall F : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \bullet iter[\mathbb{N}]\,0\,F\,a = a$

proof
prove;
■

theorem rule IterSuc
$\forall F : \mathbb{N} \to \mathbb{N};\ n, a : \mathbb{N} \bullet iter[\mathbb{N}]\,(1 + n)\,F\,a = F\,(iter[\mathbb{N}]\,n\,F\,a)$

proof
prove;
use composePositiveIterates$[\mathbb{N}][n := n,\ k := 1,\ R := F,\ Y := \mathbb{N}]$;
use CompRelApp$[f := iter[\mathbb{N}]\,n\,F,\ g := F,\ x := a]$;
use IterFun$[F := F,\ n := n]$;
prove;
■

The domain of a one-argument function, and extensionality.

theorem rule DomUF
$\forall f : \mathbb{N} \to \mathbb{N} \bullet \mathrm{dom}\, f = \mathbb{N}$

proof
prove;
■

theorem rule UFAppNat
$\forall f : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet f\ x \in \mathbb{N}$

proof
prove;
use applyInRanFun[$\mathbb{N}$, $\mathbb{N}$][$f := f,\ a := x$];
prove;
■

theorem LeOrGt
$\forall a, b : \mathbb{Z} \bullet a \leq b \lor b < a$

proof
prove;
■

theorem UFNonneg
$\forall f : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet f\ x \geq 0$

proof
prove;
use applyInRanFun[$\mathbb{N}$, $\mathbb{N}$][$f := f,\ a := x$];
with disabled (*UFAppNat*) *prove*;
■

theorem rule NatAdd
$\forall a, b : \mathbb{N} \bullet a + b \in \mathbb{N}$

proof
prove;
■

theorem rule NatMul
$\forall a, b : \mathbb{N} \bullet a * b \in \mathbb{N}$

proof
prove;
■

theorem rule DomUOP
$\forall u : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \mathrm{dom}\ u = \mathbb{N} \to \mathbb{N}$

proof
prove;
■

theorem rule DomBOP
$\forall b : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \mathrm{dom}\ b = (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N})$

proof
prove;
■

theorem rule AppliesUOP
$\forall\, u : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F : \mathbb{N} \to \mathbb{N} \bullet \mathit{uapplies\$toF}$

proof
prove;
■

theorem rule AppliesBOP
$\forall\, b : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{bapplies\$to}(F, G)$

proof
prove;
■

theorem rule AppliesUF
$\forall\, f : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{fapplies\$tox}$

proof
prove;
■

theorem rule UOPApp
$\forall\, u : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F : \mathbb{N} \to \mathbb{N} \bullet u\, F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
■

theorem rule BOPApp
$\forall\, b : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F, G : \mathbb{N} \to \mathbb{N} \bullet b(F, G) \in \mathbb{N} \to \mathbb{N}$

proof
prove;
■

theorem rule NatUOPApp
$\forall\, b : \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ a : \mathbb{N} \bullet b\, a \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
prove;
■

theorem disabled rule UFExt
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet F = G \Leftrightarrow (\forall\, x : \mathbb{N} \bullet F\, x = G\, x)$

proof
use funExtensionality$[\mathbb{N},\ \mathbb{N}][f\ :=\ F,\ g\ :=\ G]$;
prove;
■

2 Section 2: functions and operators

2.1 The identity and the constants

$$funI : \mathbb{N} \to \mathbb{N}$$

$\langle\!\langle$ rule IVal $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funI\ x = x$

proof [funI$domainCheck]
prove;
■

$IWit == (\lambda x : \mathbb{N} \bullet x)$

theorem IWitFun
$IWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke IWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule IWitVal
$\forall x : \mathbb{N} \bullet IWit\ x = x$

proof
prove;
use IWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := IWit, y := x]$;
invoke IWit;
prove by reduce;
■

theorem IEx
$\exists funI : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funI\ x = x$

proof
instantiate funI $==$ *IWit*;
use IWitFun;
prove;
■

$$cst : \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$$

$\langle\!\langle$ rule CstVal $\rangle\!\rangle$
$\forall n, x : \mathbb{N} \bullet cst\ n\ x = n$

proof [cst$domainCheck]
prove;
■

$CstWit == (\lambda n : \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet n))$

theorem CstWitFun
$CstWit \in \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$

proof
invoke CstWit;
apply FunIntro to predicate $(\lambda\ n : \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet n)) \in \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem CstWitApp
$\forall\, n : \mathbb{N} \bullet CstWit\ n = (\lambda\, x : \mathbb{N} \bullet n)$

proof
prove;
use CstWitFun;
use pairInFunction$[\mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := CstWit,\ x := n,\ y := (\lambda\ x : \mathbb{N} \bullet n)]$;
invoke CstWit;
prove by reduce;
■

theorem rule CstWitVal
$\forall\, n, x : \mathbb{N} \bullet CstWit\ n\ x = n$

proof
prove;
use CstWitApp;
prove;
equality substitute CstWit n;
prove;
■

theorem CstEx
$\exists\, cst : \mathbb{N} \to (\mathbb{N} \to \mathbb{N}) \bullet \forall\, n, x : \mathbb{N} \bullet cst\ n\ x = n$

proof
instantiate cst == *CstWit*;
use CstWitFun;
prove;
■

2.2 Arithmetic functions

$funS : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule SVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funS\ x = x + 1$

proof [funS$domainCheck]
prove;
■

$SWit == (\lambda\, x : \mathbb{N} \bullet x + 1)$

theorem SWitFun
$SWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke SWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x + 1) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule SWitVal
$\forall x : \mathbb{N} \bullet SWit\ x = x + 1$

proof
prove;
use SWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := SWit, x := x, y := x + 1]$;
invoke SWit;
prove by reduce;
■

theorem SEx
$\exists funS : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funS\ x = x + 1$

proof
instantiate funS $==$ *SWit*;
use SWitFun;
prove;
■

$funD : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule DVal $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funD\ x = 2 * x$

proof [funD$domainCheck]
prove;
■

$DWit == (\lambda x : \mathbb{N} \bullet 2 * x)$

theorem DWitFun
$DWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke DWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet 2 * x) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule DWitVal
$\forall x : \mathbb{N} \bullet DWit\ x = 2 * x$

proof
prove;
use DWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := DWit, x := x, y := 2 * x]$;
invoke DWit;
prove by reduce;
■

theorem DEx
$\exists\, funD : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funD\, x = 2 * x$

proof
instantiate funD == DWit;
use DWitFun;
prove;
■

$funSq : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule SqVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funSq\, x = x * x$

proof [funSq$domainCheck]
prove;
■

$SqWit == (\lambda\, x : \mathbb{N} \bullet x * x)$

theorem SqWitFun
$SqWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke SqWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x * x) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule SqWitVal
$\forall\, x : \mathbb{N} \bullet SqWit\, x = x * x$

proof
prove;
use SqWitFun;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := SqWit,\ x := x,\ y := x * x]$;
invoke SqWit;
prove by reduce;
■

theorem SqEx
$\exists\, funSq : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funSq\, x = x * x$

proof
instantiate funSq == SqWit;
use SqWitFun;
prove;
■

$funHf : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule HfVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funHf\, x = x \operatorname{div} 2$

proof [funHf$domainCheck]
prove;
■

$HfWit == (\lambda\, x : \mathbb{N} \bullet x \;\mathsf{div}\; 2)$

proof [HfWit$domainCheck]
prove;
■

theorem HfWitFun
$HfWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke HfWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x\ \mathsf{div}\ 2) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule HfWitVal
$\forall\, x : \mathbb{N} \bullet HfWit\, x = x \;\mathsf{div}\; 2$

proof
prove;
use HfWitFun;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := HfWit,\ x := x,\ y := x\ \mathsf{div}\ 2]$;
invoke HfWit;
prove by reduce;
■

theorem HfEx
$\exists\, funHf : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funHf\, x = x \;\mathsf{div}\; 2$

proof
instantiate funHf $==$ *HfWit*;
use HfWitFun;
prove;
■

$funO : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule OVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funO\, x = \mathbf{if}\ x = 0\ \mathbf{then}\ 1\ \mathbf{else}\ 0$

proof [funO$domainCheck]
prove;
■

$OWit == (\lambda\, x : \mathbb{N} \bullet 0) \oplus \{0 \mapsto 1\}$

theorem OWitFun
$OWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke OWit;
prove by reduce;
■

theorem rule OWitVal
$\forall x : \mathbb{N} \bullet OWit\, x = \textbf{if } x = 0 \textbf{ then } 1 \textbf{ else } 0$

proof
invoke OWit;
prove by reduce;
■

theorem OEx
$\exists funO : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funO\, x = \textbf{if } x = 0 \textbf{ then } 1 \textbf{ else } 0$

proof
instantiate funO == OWit;
use OWitFun;
prove;
■

$funSgn : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule SgnVal $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funSgn\, x = \textbf{if } x = 0 \textbf{ then } 0 \textbf{ else } 1$

proof [funSgn$domainCheck]
prove;
■

$SgnWit == (\lambda x : \mathbb{N} \bullet 1) \oplus \{0 \mapsto 0\}$

theorem SgnWitFun
$SgnWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke SgnWit;
prove by reduce;
■

theorem rule SgnWitVal
$\forall x : \mathbb{N} \bullet SgnWit\, x = \textbf{if } x = 0 \textbf{ then } 0 \textbf{ else } 1$

proof
invoke SgnWit;
prove by reduce;
■

theorem SgnEx
$\exists funSgn : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funSgn\, x = \textbf{if } x = 0 \textbf{ then } 0 \textbf{ else } 1$

proof
instantiate funSgn == SgnWit;
use SgnWitFun;
prove;
■

2.3 Integer square root and the excess over a square

$$\begin{array}{|l} \mathit{funRt} : \mathbb{N} \to \mathbb{N} \\ \hline \langle\!\langle\, \mathrm{RtSpec} \,\rangle\!\rangle \\ \forall\, x : \mathbb{N} \bullet \mathit{funRt}\, x * \mathit{funRt}\, x \leq x \land x < (\mathit{funRt}\, x + 1) * (\mathit{funRt}\, x + 1) \end{array}$$

proof [funRt$domainCheck]
prove;
■

theorem RtUnique
$\forall\, r, s, x : \mathbb{N} \mid r * r \leq x \land x < (r+1) * (r+1) \land s * s \leq x \land x < (s+1) * (s+1) \bullet r = s$

proof
prove;
split $r < s$;
cases;
use *timesMonotonic2*$[a := r + 1,\ b := r + 1,\ c := s,\ d := s]$;
prove;
next;
split $s < r$;
cases;
use *timesMonotonic2*$[a := s + 1,\ b := s + 1,\ c := r,\ d := r]$;
prove;
next;
prove;
next;
■

$\mathit{RtExSet} == \{x : \mathbb{N} \mid \exists\, r : \mathbb{N} \bullet r * r \leq x \land x < (r+1) * (r+1)\}$

theorem rule RtExBase
$0 \in \mathit{RtExSet}$

proof
invoke *RtExSet*;
prove;
instantiate $r == 0$;
prove;
■

theorem rule RtExStep
$\forall\, x : \mathit{RtExSet} \bullet 1 + x \in \mathit{RtExSet}$

proof
invoke *RtExSet*;
prove;
split $x + 1 < (r + 1) * (r + 1)$;
cases;
instantiate $r__0 == r$;
prove;
next;
instantiate $r__1 == r + 1$;
prove;
next;
■

theorem RtExSub
$\mathbb{N} \subseteq RtExSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq RtExSet$;
prove;
invoke RtExSet;
prove;
■

theorem RtExists
$\forall x : \mathbb{N} \bullet \exists r : \mathbb{N} \bullet r * r \leq x \land x < (r+1) * (r+1)$

proof
prove;
use RtExSub;
with enabled (*inPower*) *prove*;
instantiate $e == x$;
invoke RtExSet;
prove;
■

$RtWit == \{x, r : \mathbb{N} \mid r * r \leq x \land x < (r+1) * (r+1) \bullet (x, r)\}$

theorem RtWitFun
$RtWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke RtWit;
apply FunIntro to predicate $\{ x, r : \mathbb{N} \mid r * r \leq x \land x < (r+1) * (r+1) \bullet (x, r) \} \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use RtExists$[x := x]$;
use RtUnique$[r := y, s := z, x := x__0]$;
prove;
instantiate $y__0 == r$;
prove;
■

theorem rule RtWitLe
$\forall x : \mathbb{N} \bullet RtWit\, x * RtWit\, x \leq x$

proof
prove;
use RtExists$[x := x]$;
use RtWitFun;
prove;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := RtWit, x := x, y := r]$;
invoke RtWit;
prove by reduce;
■

theorem rule RtWitLt
$\forall x : \mathbb{N} \bullet x < (1 + RtWit\, x) * (1 + RtWit\, x)$

proof
prove;
use RtExists[$x := x$];
use RtWitFun;
prove;
use pairInFunction[$\mathbb{N}$, $\mathbb{N}$][$f := RtWit$, $x := x$, $y := r$];
invoke RtWit;
prove by reduce;
■

theorem RtEx
$\exists\, funRt : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funRt\, x * funRt\, x \leq x \land x < (funRt\, x + 1) * (funRt\, x + 1)$

proof
instantiate funRt $==$ *RtWit*;
use RtWitFun;
prove;
■

$\mathrm{Rt}(x) = r$ whenever $r^2 \leq x < (r+1)^2$.

theorem RtI
$\forall\, r, x : \mathbb{N} \mid r * r \leq x \land x < (r+1) * (r+1) \bullet funRt\, x = r$

proof
prove;
use RtSpec[$x := x$];
use RtUnique[$r := funRt\, x$, $s := r$, $x := x$];
use UFNonneg[$f := funRt$, $x := x$];
prove;
■

$funE : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ disabled rule EVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funE\, x = x - funRt\, x * funRt\, x$

proof [funE$domainCheck]
prove;
■

$EWit == (\lambda\, x : \mathbb{N} \bullet x - funRt\, x * funRt\, x)$

proof [EWit$domainCheck]
prove;
■

theorem EWitRel
$(\lambda\, x : \mathbb{N} \bullet x - funRt\, x * funRt\, x) \in \mathbb{P}(\mathbb{N} \times \mathbb{N})$

proof
with enabled (*inPower*) *prove by reduce*;
use RtSpec[$x := x$];
prove;
■

theorem EWitFun
$EWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke EWit;
use EWitRel;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x - funRt\, x * funRt\, x) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use RtSpec$[x := x]$;
prove;
■

theorem rule EWitVal
$\forall x : \mathbb{N} \bullet EWit\, x = x - funRt\, x * funRt\, x$

proof
prove;
use EWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := EWit, x := x, y := x - funRt\, x * funRt\, x]$;
invoke EWit;
prove by reduce;
■

theorem EEx
$\exists funE : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funE\, x = x - funRt\, x * funRt\, x$

proof
instantiate funE == *EWit*;
use EWitFun;
prove;
■

2.4 Operators

$F^+(x) = F(x) + x.$

$opPlusI : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

⟪ rule PlusIVal ⟫
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlusI\, F\, x = F\, x + x$

proof [opPlusI$domainCheck]
prove;
■

theorem rule PlusIInner
$\forall F : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\, x + x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F\, x + x) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

$PlusIWit == (\lambda F : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\, x + x))$

proof [PlusIWit$domainCheck]
prove;
■

theorem PlusIWitFun
$PlusIWit \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke PlusIWit;
apply FunIntro to predicate $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\,x + x)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

theorem PlusIWitApp
$\forall F : \mathbb{N} \to \mathbb{N} \bullet PlusIWit\ F = (\lambda x : \mathbb{N} \bullet F\,x + x)$

proof
use PlusIWitFun;
use pairInFunction$[\mathbb{N} \to \mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := PlusIWit,\ x := F,\ y := (\lambda\ x : \mathbb{N} \bullet F\,x + x)]$;
invoke PlusIWit;
prove by reduce;
■

theorem rule PlusIWitVal
$\forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet PlusIWit\ F\,x = F\,x + x$

proof
prove;
use PlusIWitApp$[F := F]$;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet F\,x + x),\ x := x,\ y := F\,x + x]$;
prove;
■

theorem PlusIEx
$\exists\, opPlusI : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlusI\ F\,x = F\,x + x$

proof
instantiate opPlusI $==$ *PlusIWit*;
use PlusIWitFun;
prove;
■

Binary operators are functions on pairs of one-argument functions: $F + G$ and the composition FG.

$opPlus : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ rule PlusVal $\rangle\!\rangle$
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlus(F, G)\,x = F\,x + G\,x$

proof [opPlus$domainCheck]
prove;
■

theorem rule PlusInner
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\,x + G\,x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F\,x + G\,x) \in \mathbb{N} \to \mathbb{N}$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

$PlusWit == (\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\, x + G\, x))$

proof [PlusWit$domainCheck]
prove;
■

theorem PlusWitFun
$PlusWit \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke PlusWit;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x)) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem PlusWitApp
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet PlusWit(F, G) = (\lambda\, x : \mathbb{N} \bullet F\, x + G\, x)$

proof
use PlusWitFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := PlusWit,\ x := (F,\ G),\ y := (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x)]$;
invoke PlusWit;
prove by reduce;
■

theorem rule PlusWitVal
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet PlusWit(F, G)\, x = F\, x + G\, x$

proof
prove;
use PlusWitApp$[F := F,\ G := G]$;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x),\ x := x,\ y := F\, x + G\, x]$;
prove;
■

theorem PlusEx
$\exists\, opPlus : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlus(F, G)\, x = F\, x + G\, x$

proof
instantiate $opPlus == PlusWit$;
use PlusWitFun;
prove;
■

$comp : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ rule CompVal $\rangle\!\rangle$
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet comp(F, G)\, x = F\, (G\, x)$

proof [comp$domainCheck]
prove;
■

theorem rule CompInner
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F(G x)) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F(G x)) \in \mathbb{N} \to \mathbb{N}$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

$CompWit == (\lambda F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F(G x)))$

proof [CompWit$domainCheck]
prove;
■

theorem CompWitFun
$CompWit \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke CompWit;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F(G x))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem CompWitApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet CompWit(F, G) = (\lambda x : \mathbb{N} \bullet F(G x))$

proof
use CompWitFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := CompWit,\ x := (F,\ G),\ y := (\lambda\ x : \mathbb{N} \bullet F(G x))]$;
invoke CompWit;
prove by reduce;
■

theorem rule CompWitVal
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet CompWit(F, G)\, x = F(G x)$

proof
prove;
use CompWitApp$[F := F,\ G := G]$;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet F(G x)),\ x := x,\ y := F(G x)]$;
prove;
■

theorem CompEx
$\exists comp : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet comp(F, G)\, x = F(G x)$

proof
instantiate comp $==$ *CompWit*;
use CompWitFun;
prove;
■

Iteration from a: $F^{\Box(a)}(0) = a$ and $F^{\Box(a)}(x+1) = F(F^{\Box(a)}(x))$ (in the defining equations Z/EVES's normal form $1 + x$ is used for $x + 1$).

$opBoxA : \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$

$\langle\!\langle$ rule BoxA0 $\rangle\!\rangle$
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet opBoxA\ a\ F\ 0 = a$
$\langle\!\langle$ rule BoxASuc $\rangle\!\rangle$
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opBoxA\ a\ F\ (1 + x) = F\ (opBoxA\ a\ F\ x)$

proof [opBoxA$domainCheck]
prove;
■

theorem rule BoxAInner
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a) \in \mathbb{N} \to \mathbb{N}$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

theorem rule BoxAMid
$\forall\, a : \mathbb{N} \bullet (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
apply FunIntro to predicate $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

$BoxAWit == (\lambda\, a : \mathbb{N} \bullet (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a)))$

proof [BoxAWit$domainCheck]
prove;
■

theorem BoxAWitFun
$BoxAWit \in \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$

proof
invoke BoxAWit;
apply FunIntro to predicate $(\lambda\ a : \mathbb{N} \bullet (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a))) \in \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

theorem BoxAWitApp
$\forall\, a : \mathbb{N} \bullet BoxAWit\ a = (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a))$

proof
use BoxAWitFun;
use pairInFunction$[\mathbb{N},\ (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})][f := BoxAWit,\ x := a,\ y := (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ n : \mathbb{N} \bullet iter[\mathbb{N}]\ n\ F\ a))]$;
invoke BoxAWit;
prove by reduce;
■

theorem BoxAWitApp2
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet \mathit{BoxAWit}\ a\ F = (\lambda\, n : \mathbb{N} \bullet \mathit{iter}[\mathbb{N}]\ n\ F\ a)$

proof
prove;
use *BoxAWitApp*$[a := a]$;
use *pairInFunction*$[\mathbb{N} \to \mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ n : \mathbb{N} \bullet \mathit{iter}[\mathbb{N}]\ n\ F\ a)),\ x := F,\ y := (\lambda\ n : \mathbb{N} \bullet \mathit{iter}[\mathbb{N}]\ n\ F\ a)]$;
prove;
■

theorem rule BoxAWitVal
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ n : \mathbb{N} \bullet \mathit{BoxAWit}\ a\ F\ n = \mathit{iter}[\mathbb{N}]\ n\ F\ a$

proof
prove;
use *BoxAWitApp2*$[a := a,\ F := F]$;
use *pairInFunction*$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ n : \mathbb{N} \bullet \mathit{iter}[\mathbb{N}]\ n\ F\ a),\ x := n,\ y := \mathit{iter}[\mathbb{N}]\ n\ F\ a]$;
prove;
■

theorem BoxAEx
$\exists\, \mathit{opBoxA} : \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet \mathit{opBoxA}\ a\ F\ 0 = a) \land (\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opBoxA}\ a\ F\ (1 + x) = F\ (\mathit{opBoxA}\ a\ F\ x))$

proof
instantiate *opBoxA* == *BoxAWit*;
use *BoxAWitFun*;
prove;
■

Pure iteration $F^{\Box}$, i.e. iteration from 0 (its witness is $F^{\Box(0)}$).

$\mathit{opBox} : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ rule Box0 $\rangle\!\rangle$
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet \mathit{opBox}\ F\ 0 = 0$
$\langle\!\langle$ rule BoxSuc $\rangle\!\rangle$
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opBox}\ F\ (1 + x) = F\ (\mathit{opBox}\ F\ x)$

proof [opBox$domainCheck]
prove;
■

theorem BoxEx
$\exists\, \mathit{opBox} : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet (\forall\, F : \mathbb{N} \to \mathbb{N} \bullet \mathit{opBox}\ F\ 0 = 0) \land (\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opBox}\ F\ (1 + x) = F\ (\mathit{opBox}\ F\ x))$

proof
instantiate *opBox* == *opBoxA* 0;
prove;
■

2.5 Bases and closures

A basis is given by a set I of initial functions, a set U of unary operators and a set B of binary operators; its closure is the least set containing I and closed under U and B, i.e. the intersection of all such sets. First, the sets closed under U and those closed under B.

$UClosed : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))$

⟪ disabled rule UClosedDef ⟫
$\forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet UClosed\ U = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, u : U;\ F : X \bullet u\, F \in X\}$

proof [UClosed$domainCheck]
prove;
■

$UClosedWit == (\lambda\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, u : U;\ F : X \bullet u\, F \in X\})$

proof [UClosedWit$domainCheck]
prove;
■

theorem UClosedWitFun
$UClosedWit \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))$

proof
invoke UClosedWit;
apply FunIntro to predicate $(\lambda\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \{ X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\ u : U;\ F : X \bullet u\, F \in X \}) \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem disabled rule UClosedWitVal
$\forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet UClosedWit\ U = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, u : U;\ F : X \bullet u\, F \in X\}$

proof
use UClosedWitFun;
use pairInFunction$[\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})), \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))][f := UClosedWit,\ x := U,\ y := \{ X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\ u : U;\ F : X \bullet u\, F \in X \}]$;
invoke UClosedWit;
prove by reduce;
■

theorem UClosedEx
$\exists\, UClosed : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N})) \bullet \forall\, U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet UClosed\ U = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, u : U;\ F : X \bullet u\, F \in X\}$

proof
instantiate UClosed $==$ *UClosedWit*;
use UClosedWitFun;
with enabled (*UClosedWitVal*) *prove*;
■

$BClosed : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))$

⟪ disabled rule BClosedDef ⟫
$\forall\, B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet BClosed\ B = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, b : B;\ F, G : X \bullet b(F, G) \in X\}$

proof [BClosed$domainCheck]
prove;
■

$BClosedWit == (\lambda\, B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, b : B;\ F, G : X \bullet b(F, G) \in X\})$

proof [BClosedWit$domainCheck]
prove;
■

theorem BClosedWitFun
$BClosedWit \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N}))$

proof
invoke BClosedWit;
apply FunIntro to predicate $(\lambda\ B : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \{\ X : \mathbb{P}\ (\mathbb{N} \to \mathbb{N}) \mid \forall\ b : B;\ F, G : X \bullet b(F, G) \in X\ \}) \in \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}\ (\mathbb{P}\ (\mathbb{N} \to \mathbb{N}))$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem disabled rule BClosedWitVal
$\forall\, B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet BClosedWit\ B = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, b : B;\ F, G : X \bullet b(F, G) \in X\}$

proof
use BClosedWitFun;
use pairInFunction$[\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})), \mathbb{P}\ (\mathbb{P}\ (\mathbb{N} \to \mathbb{N}))][f := BClosedWit, x := B, y := \{\ X : \mathbb{P}\ (\mathbb{N} \to \mathbb{N}) \mid \forall\ b : B;\ F, G : X \bullet b(F, G) \in X\ \}]$;
invoke BClosedWit;
prove by reduce;
■

theorem BClosedEx
$\exists\, BClosed : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{P}(\mathbb{N} \to \mathbb{N})) \bullet \forall\, B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet BClosed\ B = \{X : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid \forall\, b : B;\ F, G : X \bullet b(F, G) \in X\}$

proof
instantiate BClosed $==$ *BClosedWit*;
use BClosedWitFun;
with enabled (*BClosedWitVal*) *prove*;
■

Domains and applications of the curried closure function.

theorem rule ClosApp1
$\forall\, c : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})));\ I : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \bullet c\ I \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N}))$

proof
prove;
■

theorem rule ClosApp2
$\forall\, g : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N}));\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet g\ U \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
prove;
■

theorem rule ClosDom2
$\forall\, g : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})) \bullet \operatorname{dom} g = \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$

proof
prove;
■

theorem rule ClosApplies3
$\forall\, h : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N});\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet$ *happlies*$*toB*

proof
prove;
■

The closure, as a curried function of I, U and B.

$clos : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})))$

$\langle\!\langle$ ClosDef $\rangle\!\rangle$
$\forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet clos\ I\ U\ B = \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}$

proof [clos$domainCheck]
prove;
■

Consistency: the intersection is a set of one-argument functions, because the set of all one-argument functions belongs to the family.

theorem rule ClosFamUF
$\forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \mathbb{N} \to \mathbb{N} \in \{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}$

proof
with enabled (*UClosedDef*, *BClosedDef*) *prove by reduce*;
■

theorem rule ClosInPower
$\forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\} \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
prove;
with enabled (*inPower*, *inBigcap*) *prove*;
■

theorem rule ClosL3
$\forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda\, B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}) \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
apply FunIntro to predicate $(\lambda\ B : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \}) \in \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}\ (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove*;
■

theorem rule ClosL2
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \bullet (\lambda U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\})) \in \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N}))$

proof
apply FunIntro to predicate $(\lambda\ U : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda\ B : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \})) \in \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}\ (\mathbb{N} \to \mathbb{N}))$;
with enabled $(RelPow,\ inPower)$ *prove*;
■

$ClosWit == (\lambda I : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \bullet (\lambda U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\})))$

proof [ClosWit$domainCheck]
prove;
■

theorem ClosWitFun
$ClosWit \in \mathbb{P}(\mathbb{N} \to \mathbb{N}) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})))$

proof
invoke ClosWit;
apply FunIntro to predicate $(\lambda\ I : \mathbb{P}\ (\mathbb{N} \to \mathbb{N}) \bullet (\lambda\ U : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda\ B : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \})))\ \in \mathbb{P}\ (\mathbb{N} \to \mathbb{N}) \to (\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}\ (\mathbb{N} \to \mathbb{N})))$;
with enabled $(RelPow,\ inPower)$ *prove*;
■

theorem ClosWitApp1
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \bullet ClosWit\ I = (\lambda U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}))$

proof
use ClosWitFun;
use pairInFunction$[\mathbb{P}\ (\mathbb{N} \to \mathbb{N}),\ \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}\ (\mathbb{N} \to \mathbb{N}))][f := ClosWit,\ x := I,\ y := (\lambda\ U : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda\ B : \mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \}))]$;
invoke ClosWit;
prove;
■

theorem ClosWitApp2
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet ClosWit\ I\ U = (\lambda B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\})$

proof
use *ClosWitApp*1[$I := I$];
use *pairInFunction*[$\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})), \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N})$][$f := (\lambda\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet (\lambda\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \}), x := U, y := (\lambda\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \})$];
prove;
■

theorem rule ClosWitVal
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet$
$ClosWit\ I\ U\ B = \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}$

proof
use *ClosWitApp*2[$I := I, U := U$];
use *pairInFunction*[$\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})), \mathbb{P}(\mathbb{N} \to \mathbb{N})$][$f := (\lambda\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \}), x := B, y := \bigcap \{\ X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \}$];
prove;
■

theorem ClosEx
$\exists\ clos : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to (\mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \to \mathbb{P}(\mathbb{N} \to \mathbb{N}))) \bullet \forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet$
$clos\ I\ U\ B = \bigcap\{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\}$

proof
instantiate *clos* == *ClosWit*;
use *ClosWitFun*;
prove;
■

2.6 The closure rules

theorem rule ClosInit
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F : \mathbb{N} \to \mathbb{N} \mid F \in I \bullet F \in clos\ I\ U\ B$

proof
prove;
use *ClosDef*[$I := I, U := U, B := B$];
prove;
with enabled (*inBigcap*, *subsetDef*, *inPower*) *prove*;
instantiate *e__2* == *F*;
prove;
■

theorem disabled rule ClosMem
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \bullet F \in clos\ I\ U\ B \Leftrightarrow (\forall\ Y : \{X : UClosed\ U \cap BClosed\ B \mid I \subseteq X\} \bullet F \in Y)$

proof
use *ClosDef*[$I := I, U := U, B := B$];
prove;
■

theorem rule ClosUnop
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ u : (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F : \mathbb{N} \to \mathbb{N} \mid u \in U \wedge F \in clos\ I\ U\ B \bullet u\ F \in clos\ I\ U\ B$

proof
with enabled (*ClosMem*) *prove*;
instantiate $Y__0 == Y$;
with enabled (*UClosedDef*) *prove*;
instantiate $u__0 == u$, $F__0 == F$;
prove;
■

theorem rule ClosBinop
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ b : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N});\ F, G : \mathbb{N} \to \mathbb{N} \mid b \in B \land F \in clos\ I\ U\ B \land G \in clos\ I\ U\ B \bullet b(F, G) \in clos\ I\ U\ B$

proof
with enabled (*ClosMem*) *prove*;
instantiate $Y__0 == Y$, $Y__1 == Y$;
with enabled (*BClosedDef*) *prove*;
instantiate $b__0 == b$, $F__0 == F$, $G__0 == G$;
prove;
■

The closure is contained in every closed set containing I.

theorem ClosInPow
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet clos\ I\ U\ B \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use *applyInRanFun*$[\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})),\ \mathbb{P}\ (\mathbb{N} \to \mathbb{N})][f := clos\ I\ U,\ a := B]$;
prove;
■

theorem ClosElemUF
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\ I\ U\ B \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
use *applyInRanFun*$[\mathbb{P}\ ((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})),\ \mathbb{P}\ (\mathbb{N} \to \mathbb{N})][f := clos\ I\ U,\ a := B]$;
prove;
with enabled (*inPower*) *prove*;
■

theorem ClosLeast
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ Z : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid Z \in UClosed\ U \land Z \in BClosed\ B \land I \subseteq Z \bullet clos\ I\ U\ B \subseteq Z$

proof
use *ClosDef*$[I := I,\ U := U,\ B := B]$;
use *BigcapSub*$[\mathbb{N} \to \mathbb{N}][T := \{\ X :\ UClosed\ U \cap BClosed\ B \mid I \subseteq X\ \},\ Y := Z]$;
prove;
■

2.7 Arithmetic facts about Rt and E used silently in the paper

theorem RtSqAdd
$\forall\, m, k : \mathbb{N} \mid k \leq 2 * m \bullet \mathit{funRt}\,(m * m + k) = m$

proof
prove;
use $\mathit{RtI}[r := m,\ x := m * m + k]$;
prove;
■

theorem ESqAdd
$\forall\, m, k : \mathbb{N} \mid k \leq 2 * m \bullet \mathit{funE}\,(m * m + k) = k$

proof
prove;
use $\mathit{RtSqAdd}[m := m,\ k := k]$;
with enabled (EVal) *prove*;
■

theorem rule RtSq
$\forall\, m : \mathbb{N} \bullet \mathit{funRt}\,(m * m) = m$

proof
prove;
use $\mathit{RtSqAdd}[m := m,\ k := 0]$;
prove;
■

theorem ELe
$\forall\, x : \mathbb{N} \bullet \mathit{funE}\, x \leq x$

proof
prove;
use $\mathit{RtSpec}[x := x]$;
with enabled (EVal) *prove*;
■

theorem rule E0
$\mathit{funE}\, 0 = 0$

proof
use $\mathit{ESqAdd}[m := 0,\ k := 0]$;
prove;
■

theorem rule E1
$\mathit{funE}\, 1 = 0$

proof
use $\mathit{ESqAdd}[m := 1,\ k := 0]$;
prove;
■

theorem rule E2
$\mathit{funE}\, 2 = 1$

proof
use $ESqAdd[m := 1,\ k := 1]$;
prove;
■

theorem rule E3
$funE\,3 = 2$

proof
use $ESqAdd[m := 1,\ k := 2]$;
prove;
■

theorem rule E4
$funE\,4 = 0$

proof
use $ESqAdd[m := 2,\ k := 0]$;
prove;
■

3 Section 6: iteration and the unary operator F^+

The closure of the basis $\mathcal{F} = \langle \bar{1}, E, F^+, FG, F^\square \rangle$.

$C6 == clos\,\{cst\,1, funE\}\,\{opPlusI, opBox\}\,\{comp\}$

proof [C6$domainCheck]
prove;
■

theorem disabled rule C6Def
$C6 = clos\,\{cst\,1, funE\}\,\{opPlusI, opBox\}\,\{comp\}$

proof
invoke $C6$;
prove;
■

theorem disabled rule C6UF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in C6 \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $ClosElemUF[I := \{\ cst\,1,\ funE\ \},\ U := \{\ opPlusI,\ opBox\ \},\ B := \{\ comp\ \},\ F := F]$;
with enabled $(C6Def)$ *prove*;
■

theorem rule C6One
$cst\,1 \in C6$

proof
with enabled $(C6Def)$ *prove*;
■

theorem rule C6E
$funE \in C6$

proof
with enabled $(C6Def)$ *prove*;
■

theorem rule C6PlusI
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet opPlusI\ F \in C6$

proof
with enabled $(C6Def)$ *prove*;
■

theorem rule C6Box
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet opBox\ F \in C6$

proof
with enabled $(C6Def)$ *prove*;
■

theorem rule C6Comp
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \bullet comp(F, G) \in C6$

proof
with enabled $(C6Def)$ *prove*;
■

3.1 Auxiliary functions of Section 6

$funM3 : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule M3Val $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funM3\ x = 3 * x$

proof [funM3$domainCheck]
prove;
■

$M3Wit == (\lambda x : \mathbb{N} \bullet 3 * x)$

theorem M3WitFun
$M3Wit \in \mathbb{N} \to \mathbb{N}$

proof
invoke $M3Wit$;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet 3 * x) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled $(RelPow)$ *prove by reduce*;
■

theorem rule M3WitVal
$\forall x : \mathbb{N} \bullet M3Wit\ x = 3 * x$

proof
prove;
use $M3\,WitFun$;
use $pairInFunction[\mathbb{N},\ \mathbb{N}][f\ :=\ M3\,Wit,\ x\ :=\ x,\ y\ :=\ 3\ *\ x]$;
invoke $M3\,Wit$;
prove by reduce;
■

theorem M3Ex
$\exists\, funM3 : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funM3\, x = 3 * x$

proof
instantiate $funM3\ ==\ M3\,Wit$;
use $M3\,WitFun$;
prove;
■

$R(x) = x + 2\,\mathrm{Rt}(x)$, as in Lemma 5.7.

$funR : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule RVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funR\, x = x + 2 * funRt\, x$

proof [funR$domainCheck]
prove;
■

$RWit == (\lambda\, x : \mathbb{N} \bullet x + 2 * funRt\, x)$

proof [RWit$domainCheck]
prove;
■

theorem RWitFun
$RWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke $RWit$;
apply FunIntro to predicate $(\lambda\ x :\ \mathbb{N}\ \bullet\ x\ +\ 2\ *\ funRt\, x)\ \in\ \mathbb{N}\ \to\ \mathbb{N}$;
prove by reduce;
with enabled $(RelPow)$ *prove by reduce*;
■

theorem rule RWitVal
$\forall\, x : \mathbb{N} \bullet RWit\, x = x + 2 * funRt\, x$

proof
prove;
use $RWitFun$;
use $pairInFunction[\mathbb{N},\ \mathbb{N}][f\ :=\ RWit,\ x\ :=\ x,\ y\ :=\ x\ +\ 2\ *\ funRt\, x]$;
invoke $RWit$;
prove by reduce;
■

theorem REx
$\exists\, funR : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funR\, x = x + 2 * funRt\, x$

proof
instantiate funR == RWit;
use RWitFun;
prove;
■

The excess $E(x)$ vanishes exactly on the squares.

theorem ESquareIf
$\forall\, k : \mathbb{N} \bullet \mathit{funE}\,(k * k) = 0$

proof
prove;
use ESqAdd[$m := k,\ k := 0$];
prove;
■

theorem ESquareOnly
$\forall\, x : \mathbb{N} \mid \mathit{funE}\, x = 0 \bullet x = \mathit{funRt}\, x * \mathit{funRt}\, x$

proof
with enabled (*EVal*) *prove*;
■

theorem ESquare
$\forall\, x : \mathbb{N} \bullet \mathit{funE}\, x = 0 \Leftrightarrow (\exists\, k : \mathbb{N} \bullet x = k * k)$

proof
prove;
split funE x = 0;
cases;
use ESquareOnly[$x := x$];
prove;
instantiate k == funRt x;
prove;
next;
prove;
use ESquareIf[$k := k$];
prove;
next;
■

theorem rule OEVal
$\forall\, x : \mathbb{N} \bullet \mathit{funO}\,(\mathit{funE}\, x) = \textbf{if } (\exists\, k : \mathbb{N} \bullet x = k * k) \textbf{ then } 1 \textbf{ else } 0$

proof
prove;
use ESquare[$x := x$];
prove;
instantiate k__1 == k;
prove;
■

Q is the characteristic function of the squares.

$\mathit{funQ} : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule QVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet \mathit{funQ}\, x = \textbf{if } (\exists\, k : \mathbb{N} \bullet x = k * k) \textbf{ then } 1 \textbf{ else } 0$

proof [funQ$domainCheck]
prove;
■

theorem QEx
$\exists\, funQ : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funQ\, x = \textbf{if } (\exists\, k : \mathbb{N} \bullet x = k * k) \textbf{ then } 1 \textbf{ else } 0$

proof
instantiate $funQ == comp(funO,\ funE)$;
prove;
■

3.2 First functions in $\mathrm{clos}\,\mathcal{F}$

theorem SExpr
$opPlusI\ (cst\ 1) = funS$

proof
apply UFExt to predicate $opPlusI\ (cst\ 1) = funS$;
prove;
■

theorem ZeroExpr
$comp(funE, cst\ 1) = cst\ 0$

proof
apply UFExt to predicate $comp(funE,\ cst\ 1) = cst\ 0$;
prove;
■

theorem DExpr
$opPlusI\ (opPlusI\ (cst\ 0)) = funD$

proof
apply UFExt to predicate $opPlusI\ (opPlusI\ (cst\ 0)) = funD$;
prove;
■

theorem M3Expr
$opPlusI\ funD = funM3$

proof
apply UFExt to predicate $opPlusI\ funD = funM3$;
prove;
■

theorem OExpr
$comp(comp(comp(funE, funD), funS), funSgn) = funO$

proof
apply UFExt to predicate $comp(comp(comp(funE,\ funD),\ funS),\ funSgn) = funO$;
prove;
■

theorem QExpr
$comp(funO, funE) = funQ$

proof
apply UFExt to predicate $comp(funO,\ funE) = funQ$;
prove;
■

Robinson's difference formula: if $b \leq a$ then $a - b = E((a+b)^2 + 3a + b + 1)$.

theorem RobDiffE
$\forall a, b : \mathbb{N} \mid b \leq a \bullet funE\,((a+b) * (a+b) + 3 * a + b + 1) = a - b$

proof
prove;
use ESqAdd$[m := a + b + 1,\ k := a - b]$;
prove;
■

3.3 Division with remainder

theorem DivLe
$\forall x, d, q : \mathbb{N} \mid 0 < d \land x < d * q + d \bullet x \operatorname{\mathsf{div}} d \leq q$

proof
prove;
■

theorem DivGe
$\forall x, d, q : \mathbb{N} \mid 0 < d \land d * q \leq x \bullet q \leq x \operatorname{\mathsf{div}} d$

proof
prove;
■

theorem DivMod
$\forall x, d, q, r : \mathbb{N} \mid 0 < d \land r < d \land x = d * q + r \bullet x \operatorname{\mathsf{div}} d = q \land x \operatorname{\mathsf{mod}} d = r$

proof
prove;
use DivLe$[x := x,\ d := d,\ q := q]$;
use DivGe$[x := x,\ d := d,\ q := q]$;
use divModRelation$[x := x,\ y := d]$;
with disabled (*divLowerBound*, *divUpperBound*) *prove*;
■

x mod 3.

$funMod3 : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ rule Mod3Val $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funMod3\, x = x \operatorname{\mathsf{mod}} 3$

proof [funMod3$domainCheck]
prove;
■

$Mod3Wit == (\lambda x : \mathbb{N} \bullet x \operatorname{\mathsf{mod}} 3)$

proof [Mod3Wit$domainCheck]
prove;
■

theorem Mod3WitFun
$Mod3\,Wit \in \mathbb{N} \to \mathbb{N}$

proof
invoke $Mod3\,Wit$;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet x \bmod 3) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule Mod3WitVal
$\forall x : \mathbb{N} \bullet Mod3\,Wit\ x = x \bmod 3$

proof
prove;
use $Mod3\,WitFun$;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := Mod3\,Wit, x := x, y := x \bmod 3]$;
invoke $Mod3\,Wit$;
prove by reduce;
■

theorem Mod3Ex
$\exists\, funMod3 : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funMod3\ x = x \bmod 3$

proof
instantiate $funMod3 == Mod3\,Wit$;
use $Mod3\,WitFun$;
prove;
■

theorem DivModOf
$\forall x, d : \mathbb{N} \mid 0 < d \bullet x = d * (x \operatorname{div} d) + x \bmod d \land 0 \leq x \bmod d \land x \bmod d < d$

proof
prove;
■

theorem Mod3Cases
$\forall n : \mathbb{N} \bullet (n \bmod 3 = 0 \land (1 + n) \bmod 3 = 1) \lor (n \bmod 3 = 1 \land (1 + n) \bmod 3 = 2) \lor (n \bmod 3 = 2 \land (1 + n) \bmod 3 = 0)$

proof
prove;
use $DivModOf[x := n, d := 3]$;
split $n \bmod 3 = 2$;
cases;
use $DivMod[x := 1 + n, d := 3, q := n \operatorname{div} 3 + 1, r := 0]$;
prove;
next;
use $DivMod[x := 1 + n, d := 3, q := n \operatorname{div} 3, r := n \bmod 3 + 1]$;
prove;
next;
■

3.4 First functions in $\mathrm{clos}\,\mathcal{F}$, continued

theorem SgnExpr
$opBox\,(cst\,1) = funSgn$

proof
apply UFExt to predicate $opBox\,(cst\,1) = funSgn$;
prove;
split $x = 0$;
cases;
prove;
next;
use $BoxSuc[F__0 := cst\,1,\ x := x - 1]$;
prove;
next;
■

$Mod3Set == \{n : \mathbb{N} \mid opBox\,(comp(comp(funE, funS), funS))\,n = n \bmod 3\}$

proof [Mod3Set$domainCheck]
prove;
■

theorem rule Mod3Base
$0 \in Mod3Set$

proof
invoke $Mod3Set$;
prove;
■

theorem rule Mod3Step
$\forall\, n : Mod3Set \bullet 1 + n \in Mod3Set$

proof
invoke $Mod3Set$;
prove;
use $Mod3Cases[n := n]$;
split $n \bmod 3 = 0$;
cases;
prove;
next;
split $n \bmod 3 = 1$;
cases;
prove;
next;
prove;
next;
■

theorem Mod3Sub
$\mathbb{N} \subseteq Mod3Set$

proof
apply natInduction to predicate $\mathbb{N} \subseteq Mod3Set$;
prove;
invoke $Mod3Set$;
prove;
■

theorem Mod3All
$\forall\, n : \mathbb{N} \bullet opBox\,(comp(comp(funE, funS), funS))\, n = n \bmod 3$

proof
prove;
use Mod3Sub;
invoke Mod3Set;
with enabled (*inPower*) *prove*;
instantiate e == n;
prove;
■

theorem Mod3Expr
$opBox\,(comp(comp(funE, funS), funS)) = funMod3$

proof
apply UFExt to predicate $opBox\,(comp(comp(funE,\ funS),\ funS)) = funMod3$;
prove;
use $Mod3All[n := x]$;
prove;
■

3.5 Membership in $\mathrm{clos}\,\mathcal{F}$

theorem SInExpr
$opPlusI\,(cst\,1) \in C6$

proof
prove;
■

theorem rule SIn
$funS \in C6$

proof
use SInExpr;
use SExpr;
prove;
■

theorem SgnInExpr
$opBox\,(cst\,1) \in C6$

proof
prove;
■

theorem rule SgnIn
$funSgn \in C6$

proof
use SgnInExpr;
use SgnExpr;
prove;
■

theorem ZeroInExpr
$comp(funE, cst\,1) \in C6$

proof
prove;
■

theorem rule ZeroIn
$cst\,0 \in C6$

proof
use ZeroInExpr;
use ZeroExpr;
prove;
■

theorem DInExpr
$opPlusI\,(opPlusI\,(cst\,0)) \in C6$

proof
prove;
■

theorem rule DIn
$funD \in C6$

proof
use DInExpr;
use DExpr;
prove;
■

theorem M3InExpr
$opPlusI\,funD \in C6$

proof
prove;
■

theorem rule M3In
$funM3 \in C6$

proof
use M3InExpr;
use M3Expr;
prove;
■

theorem OInExpr
$comp(comp(comp(funE, funD), funS), funSgn) \in C6$

proof
prove;
■

theorem rule OIn
$funO \in C6$

proof
use *OInExpr*;
use *OExpr*;
prove;
■

theorem QInExpr
$comp(funO, funE) \in C6$

proof
prove;
■

theorem rule QIn
$funQ \in C6$

proof
use *QInExpr*;
use *QExpr*;
prove;
■

theorem Mod3InExpr
$opBox\,(comp(comp(funE, funS), funS)) \in C6$

proof
prove;
■

theorem rule Mod3In
$funMod3 \in C6$

proof
use *Mod3InExpr*;
use *Mod3Expr*;
prove;
■

3.6 $R(x) = x + 2\,\mathrm{Rt}(x)$ and Sq (Lemma 5.7 technique)

theorem QSq
$\forall\, x, k : \mathbb{N} \mid x = k * k \bullet funQ\, x = 1$

proof
prove;
instantiate $k__0 == k$;
prove;
■

theorem NotSquare
$\forall\, m, x : \mathbb{N} \mid m * m < x \wedge x < (m+1) * (m+1) \bullet \neg\, (\exists\, k : \mathbb{N} \bullet x = k * k)$

proof
prove;
split $k \leq m$;
cases;
use *timesMonotonic2*$[a := k,\ b := k,\ c := m,\ d := m]$;
prove;
next;
use *timesMonotonic2*$[a := m + 1,\ b := m + 1,\ c := k,\ d := k]$;
prove;
next;
■

theorem QNonSq
$\forall\, m, x : \mathbb{N} \mid m * m < x \wedge x < (m+1) * (m+1) \bullet \mathit{funQ}\ x = 0$

proof
prove;
use *NotSquare*$[m := m,\ x := x]$;
prove;
instantiate $k__0 == k$;
prove;
■

theorem RStep
$\forall\, n : \mathbb{N} \bullet \mathit{opPlusI}\ (\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funD}, \mathit{funQ}), \mathit{funS}), \mathit{funS}),$
$\mathit{funS}))\ (\mathit{funS}\ (\mathit{funR}\ n)) = \mathit{funR}\ (1 + n)$

proof
prove;
use *RtSpec*$[x := n]$;
split $(\mathit{funRt}\ n + 1) * (\mathit{funRt}\ n + 1) = n + 1$;
cases;
use *RtI*$[r := \mathit{funRt}\ n + 1,\ x := n + 1]$;
prove;
instantiate $k == 2 + \mathit{funRt}\ n$;
prove;
next;
use *RtI*$[r := \mathit{funRt}\ n,\ x := n + 1]$;
use *QNonSq*$[m := \mathit{funRt}\ n + 1,\ x := 4 + n + 2 * \mathit{funRt}\ n]$;
prove;
next;
■

$\mathit{RBoxSet} == \{n : \mathbb{N} \mid \mathit{opBox}\ (\mathit{comp}(\mathit{opPlusI}\ (\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funD}, \mathit{funQ}), \mathit{funS}), \mathit{funS}), \mathit{funS})),$
$\mathit{funS}))\ n = \mathit{funR}\ n\}$

proof [RBoxSet$domainCheck]
prove;
■

theorem rule RBoxBase
$0 \in \mathit{RBoxSet}$

proof
invoke *RBoxSet*;
prove;
use *RtSq*$[m := 0]$;
prove;
■

theorem rule RBoxStep
$\forall\, n : RBoxSet \bullet 1 + n \in RBoxSet$

proof
invoke RBoxSet;
prove;
use RStep[$n := n$];
prove;
■

theorem RBoxSub
$\mathbb{N} \subseteq RBoxSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq RBoxSet$;
prove;
invoke RBoxSet;
prove;
■

theorem RBoxAll
$\forall\, n : \mathbb{N} \bullet opBox\ (comp(opPlusI\ (comp(comp(comp(comp(funD, funQ), funS), funS), funS)), funS))\ n = funR\ n$

proof
prove;
use RBoxSub;
invoke RBoxSet;
with enabled (*inPower*) *prove*;
instantiate e $==$ n;
prove;
■

theorem RExpr
$opBox\ (comp(opPlusI\ (comp(comp(comp(comp(funD, funQ), funS), funS), funS)), funS)) = funR$

proof
apply UFExt to predicate $opBox\ (comp(opPlusI\ (comp(comp(comp(comp(funD,\ funQ),\ funS),\ funS),\ funS)),\ funS)) = funR$;
prove;
use RBoxAll[$n := x$];
prove;
■

$SqBoxSet == \{n : \mathbb{N} \mid opBox\ (comp(funS, funR))\ n = n * n\}$

proof [SqBoxSet$domainCheck]
prove;
■

theorem rule SqBoxBase
$0 \in SqBoxSet$

proof
invoke SqBoxSet;
prove;
■

theorem rule SqBoxStep
$\forall\, n : SqBoxSet \bullet 1 + n \in SqBoxSet$

proof
invoke SqBoxSet;
prove;
equality substitute opBox $(comp(funS,\ funR))\ n$;
rewrite;
■

theorem SqBoxSub
$\mathbb{N} \subseteq SqBoxSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq SqBoxSet$;
prove;
invoke SqBoxSet;
prove;
■

theorem SqBoxAll
$\forall\, n : \mathbb{N} \bullet opBox\ (comp(funS, funR))\ n = n * n$

proof
prove;
use SqBoxSub;
invoke SqBoxSet;
with enabled (*inPower*) *prove*;
instantiate e $==$ n;
prove;
■

theorem SqExpr
$opBox\ (comp(funS, funR)) = funSq$

proof
apply UFExt to predicate opBox $(comp(funS,\ funR)) = funSq$;
prove;
use SqBoxAll$[n := x]$;
prove;
■

theorem RInExpr
$opBox\ (comp(opPlusI\ (comp(comp(comp(comp(funD, funQ), funS), funS), funS)), funS)) \in C6$

proof
prove;
■

theorem rule RIn
$funR \in C6$

proof
use RInExpr;
use RExpr;
prove;
■

theorem SqInExpr
$opBox\,(comp(funS, funR)) \in C6$

proof
prove;
■

theorem rule SqIn
$funSq \in C6$

proof
use SqInExpr;
use SqExpr;
prove;
■

Inflationary functions ($F(x) \geq x$ for all x), and pairs with $G(x)^2 \leq F(x)$.

$Infl == \{F : \mathbb{N} \to \mathbb{N} \mid \forall\, x : \mathbb{N} \bullet x \leq F\, x\}$

proof [Infl$domainCheck]
prove;
■

theorem rule InflLe
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid F \in Infl \bullet x \leq F\, x$

proof
prove;
invoke Infl;
prove;
instantiate $x__0 == x$;
prove;
■

$SqBelow == \{F, G : \mathbb{N} \to \mathbb{N} \mid \forall\, x : \mathbb{N} \bullet G\, x * G\, x \leq F\, x\}$

proof [SqBelow$domainCheck]
prove;
■

theorem rule SqBelowLe
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid (F, G) \in SqBelow \bullet G\, x * G\, x \leq F\, x$

proof
prove;
invoke SqBelow;
prove;
instantiate $x__0 == x$;
prove;
■

3.7 Lemma 6.2: $F^- = E\,S\,(M_3FE)^+(\mathrm{Sq}\,F^+)^+$

Definition 6.1 defines $F^-(x) = F(x) - x$, meaningful when $F(x) \geq x$; Lemmas 6.2 and 6.4 are stated accordingly: the closure contains a function with those values.

exprMinusI == $(\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet$ *comp*(*comp*(*comp*(*funE*, *funS*), *opPlusI* (*comp*(*comp*(*funM*3, *F*), *funE*))), *opPlusI* (*comp*(*funSq*, *opPlusI F*))))

proof [exprMinusI$domainCheck]
prove;
■

theorem rule exprMinusIFun
$\mathit{exprMinusI} \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprMinusI;
apply FunIntro to predicate (λ *F* : $\mathbb{N} \to \mathbb{N} \bullet$ *comp*(*comp*(*comp*(*funE*, *funS*), *opPlusI* (*comp*(*comp*(*funM*3, *F*), *funE*))), *opPlusI* (*comp*(*funSq*, *opPlusI F*)))) $\in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem disabled rule exprMinusIApp
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprMinusI}\ F = \mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funE}, \mathit{funS}), \mathit{opPlusI}\ (\mathit{comp}(\mathit{comp}(\mathit{funM}3, F), \mathit{funE}))), \mathit{opPlusI}\ (\mathit{comp}(\mathit{funSq}, \mathit{opPlusI}\ F)))$

proof
use exprMinusIFun;
use pairInFunction[$\mathbb{N} \to \mathbb{N}$, $\mathbb{N} \to \mathbb{N}$][*f* := *exprMinusI*, *x* := *F*, *y* := *comp*(*comp*(*comp*(*funE*, *funS*), *opPlusI* (*comp*(*comp*(*funM*3, *F*), *funE*))), *opPlusI* (*comp*(*funSq*, *opPlusI F*)))];
invoke exprMinusI;
prove by reduce;
■

theorem rule MinusIIn
$\forall\, F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet \mathit{exprMinusI}\ F \in C6$

proof
with enabled (*exprMinusIApp*) *prove*;
■

theorem rule MinusIVal
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid x \leq F\,x \bullet \mathit{exprMinusI}\ F\ x = F\,x - x$

proof
with enabled (*exprMinusIApp*) *prove*;
use ESqAdd[*m* := *F x* + *x*, *k* := *x*];
use RobDiffE[*a* := *F x*, *b* := *x*];
prove;
■

theorem Lemma62
$\forall\, F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge F \in \mathit{Infl} \bullet \exists\, G : \mathbb{N} \to \mathbb{N} \mid G \in C6 \bullet \forall\, x : \mathbb{N} \bullet G\,x = F\,x - x$

proof
prove;
instantiate G == *exprMinusI F*;
prove;
■

3.8 Lemma 6.3: $\mathrm{Hf} = ((S\,\mathrm{Mod}_3^+)^\Box)^-$ and $\mathrm{Rt} = \mathrm{Hf}\,R^-$

HfAuxSet == $\{q : \mathbb{N} \mid opBox\,(comp(funS, opPlusI\,funMod3))\,(2 * q) = 3 * q \wedge opBox\,(comp(funS, opPlusI\,funMod3))\,(1 + 2 * q) = 1 + 3 * q\}$

proof [HfAuxSet$domainCheck]
prove;
■

theorem rule HfAuxBase
$0 \in$ *HfAuxSet*

proof
invoke HfAuxSet;
prove;
use DivMod[$x := 0,\ d := 3,\ q := 0,\ r := 0$];
prove;
■

theorem rule HfAuxStep
$\forall\, q : HfAuxSet \bullet 1 + q \in HfAuxSet$

proof
invoke HfAuxSet;
prove;
use BoxSuc[$F__0 := comp(funS,\ opPlusI\,funMod3),\ x := 1 + 2 * q$];
use BoxSuc[$F__0 := comp(funS,\ opPlusI\,funMod3),\ x := 2 + 2 * q$];
use DivMod[$x := 1 + 3 * q,\ d := 3,\ q := q,\ r := 1$];
use DivMod[$x := 3 + 3 * q,\ d := 3,\ q := q + 1,\ r := 0$];
prove;
■

theorem HfAuxSub
$\mathbb{N} \subseteq$ *HfAuxSet*

proof
apply natInduction to predicate $\mathbb{N} \subseteq HfAuxSet$;
prove;
invoke HfAuxSet;
prove;
■

theorem HfAuxAll
$\forall\, q : \mathbb{N} \bullet opBox\,(comp(funS, opPlusI\,funMod3))\,(2 * q) = 3 * q \wedge opBox\,(comp(funS, opPlusI\,funMod3))\,(1 + 2 * q) = 1 + 3 * q$

proof
prove;
use HfAuxSub;
invoke HfAuxSet;
with enabled (*inPower*) *prove*;
instantiate $e == q$;
prove;
■

theorem HfBoxVal
$\forall\, x : \mathbb{N} \bullet opBox\,(comp(funS, opPlusI\,funMod3))\,x = x + x \operatorname{div} 2$

proof
prove;
use DivModOf[$x := x,\ d := 2$];
use HfAuxAll[$q := x$ div 2];
split x mod $2 = 0$;
cases;
prove;
next;
prove;
next;
■

theorem HfExpr
$exprMinusI\ (opBox\ (comp(funS, opPlusI\ funMod3))) = funHf$

proof
apply UFExt to predicate $exprMinusI\ (opBox\ (comp(funS,\ opPlusI\ funMod3))) = funHf$;
prove;
use HfBoxVal[$x := x$];
use MinusIVal[$F := opBox\ (comp(funS,\ opPlusI\ funMod3)),\ x := x$];
prove;
■

theorem RtExpr
$comp(funHf, exprMinusI\ funR) = funRt$

proof
apply UFExt to predicate $comp(funHf,\ exprMinusI\ funR) = funRt$;
prove;
use MinusIVal[$F := funR,\ x := x$];
use DivMod[$x := 2 * funRt\ x,\ d := 2,\ q := funRt\ x,\ r := 0$];
use UFNonneg[$f := funRt,\ x := x$];
prove;
■

theorem HfInExpr
$exprMinusI\ (opBox\ (comp(funS, opPlusI\ funMod3))) \in C6$

proof
prove;
■

theorem rule HfIn
$funHf \in C6$

proof
use HfInExpr;
use HfExpr;
prove;
■

theorem RtInExpr
$comp(funHf, exprMinusI\ funR) \in C6$

proof
prove;
■

theorem rule RtIn
funRt $\in C6$

proof
use RtInExpr;
use RtExpr;
prove;
■

3.9 Lemma 6.4: $F \oplus G = (F^- E)^+ (\mathrm{Sq}\ G)^+$ and $F \ominus G = (F^+ E)^- (\mathrm{Sq}\ G)^+$

$(F \oplus G)(x) = F(x) + G(x)^2$ (Definition 6.1); $F \ominus G$, meaningful when $F(x) \geq G(x)^2$, is treated like F^-.

$opOplus : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ rule OplusVal $\rangle\!\rangle$
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opOplus(F, G)\, x = F\, x + G\, x * G\, x$

proof [opOplus$domainCheck]
prove;
■

theorem rule OplusInner
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\, x + G\, x * G\, x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F\, x + G\, x * G\, x) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

$OplusWit == (\lambda F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\, x + G\, x * G\, x))$

proof [OplusWit$domainCheck]
prove;
■

theorem OplusWitFun
$OplusWit \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke OplusWit;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x * G\, x)) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem OplusWitApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet OplusWit(F, G) = (\lambda x : \mathbb{N} \bullet F\, x + G\, x * G\, x)$

proof
use OplusWitFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := OplusWit,\ x := (F,\ G),\ y := (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x * G\, x)]$;
invoke OplusWit;
prove by reduce;
■

theorem rule OplusWitVal
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{OplusWit}(F, G)\, x = F\, x + G\, x * G\, x$

proof
prove;
use $\mathit{OplusWitApp}[F := F,\ G := G]$;
use $\mathit{pairInFunction}[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet F\, x + G\, x * G\, x),\ x := x,\ y := F\, x + G\, x * G\, x]$;
prove;
■

theorem OplusEx
$\exists\, \mathit{opOplus} : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opOplus}(F, G)\, x = F\, x + G\, x * G\, x$

proof
instantiate $\mathit{opOplus} == \mathit{OplusWit}$;
use *OplusWitFun*;
prove;
■

$\mathit{exprOplus} == (\lambda F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{comp}(\mathit{opPlusI}\,(\mathit{comp}(\mathit{exprMinusI}\, F, \mathit{funE})), \mathit{opPlusI}\,(\mathit{comp}(\mathit{funSq}, G))))$

proof [exprOplus$domainCheck]
prove;
■

theorem rule exprOplusFun
$\mathit{exprOplus} \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke *exprOplus*;
apply *FunIntro to predicate* $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet \mathit{comp}(\mathit{opPlusI}\,(\mathit{comp}(\mathit{exprMinusI}\, F,\ \mathit{funE})),\ \mathit{opPlusI}\,(\mathit{comp}(\mathit{funSq},\ G)))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(\mathit{RelPow},\ \mathit{inPower})$ *prove*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem disabled rule exprOplusApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprOplus}(F, G) = \mathit{comp}(\mathit{opPlusI}\,(\mathit{comp}(\mathit{exprMinusI}\, F, \mathit{funE})), \mathit{opPlusI}\,(\mathit{comp}(\mathit{funSq}, G)))$

proof
use *exprOplusFun*;
use $\mathit{pairInFunction}[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}),\ \mathbb{N} \to \mathbb{N}][f := \mathit{exprOplus},\ x := (F,\ G),\ y := \mathit{comp}(\mathit{opPlusI}\,(\mathit{comp}(\mathit{exprMinusI}\, F,\ \mathit{funE})),\ \mathit{opPlusI}\,(\mathit{comp}(\mathit{funSq},\ G)))]$;
invoke *exprOplus*;
prove by reduce;
■

$\mathit{exprOminus} == (\lambda F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{comp}(\mathit{exprMinusI}\,(\mathit{comp}(\mathit{opPlusI}\, F, \mathit{funE})), \mathit{opPlusI}\,(\mathit{comp}(\mathit{funSq}, G))))$

proof [exprOminus$domainCheck]
prove;
■

theorem rule exprOminusFun
$exprOminus \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprOminus;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet\ comp(exprMinusI\,(comp(opPlusI\,F,\ funE)),$
$opPlusI\,(comp(funSq,\ G)))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem disabled rule exprOminusApp
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet exprOminus(F, G) = comp(exprMinusI\,(comp(opPlusI\,F, funE)),$
$opPlusI\,(comp(funSq, G)))$

proof
use exprOminusFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}),\ \mathbb{N} \to \mathbb{N}][f := exprOminus,\ x := (F,\ G),\ y :=$
$comp(exprMinusI\,(comp(opPlusI\,F,\ funE)),\ opPlusI\,(comp(funSq,\ G)))]$;
invoke exprOminus;
prove by reduce;
■

theorem rule OplusIn
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet exprOplus(F, G) \in C6$

proof
with enabled $(exprOplusApp)$ *prove*;
■

theorem rule OminusIn
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet exprOminus(F, G) \in C6$

proof
with enabled $(exprOminusApp)$ *prove*;
■

theorem rule OplusExprVal
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid x \leq F\,x \wedge x \leq G\,x \bullet exprOplus(F, G)\,x = F\,x + G\,x * G\,x$

proof
with enabled $(exprOplusApp)$ *prove*;
use ESqAdd$[m := G\,x,\ k := x]$;
use MinusIVal$[F := F,\ x := x]$;
prove;
■

theorem rule OminusExprVal
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid x \leq G\,x \wedge G\,x * G\,x \leq F\,x \bullet exprOminus(F, G)\,x = F\,x - G\,x * G\,x$

proof
with enabled $(exprOminusApp)$ *prove*;
use ESqAdd$[m := G\,x,\ k := x]$;
use MinusIVal$[F := comp(opPlusI\,F,\ funE),\ x := G\,x * G\,x + x]$;
prove;
■

theorem OplusEq
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in \mathit{Infl} \land G \in \mathit{Infl} \bullet \mathit{exprOplus}(F, G) = \mathit{opOplus}(F, G)$

proof
apply UFExt to predicate $\mathit{exprOplus}(F,\ G) = \mathit{opOplus}(F,\ G)$;
prove;
■

theorem Lemma64Plus
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \land F \in \mathit{Infl} \land G \in \mathit{Infl} \bullet \mathit{opOplus}(F, G) \in C6$

proof
use $\mathit{OplusEq}[F := F,\ G := G]$;
prove;
equality substitute $\mathit{opOplus}(F,\ G)$;
rewrite;
■

theorem Lemma64Minus
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \land G \in \mathit{Infl} \land (F,$
$G) \in \mathit{SqBelow} \bullet \exists H : \mathbb{N} \to \mathbb{N} \mid H \in C6 \bullet \forall x : \mathbb{N} \bullet H\, x = F\, x - G\, x * G\, x$

proof
prove;
instantiate $H == \mathit{exprOminus}(F,\ G)$;
prove;
■

theorem Lemma64
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \land F \in \mathit{Infl} \land G \in \mathit{Infl} \bullet \mathit{opOplus}(F, G) \in C6 \land ((F,$
$G) \in \mathit{SqBelow} \Rightarrow (\exists H : \mathbb{N} \to \mathbb{N} \mid H \in C6 \bullet \forall x : \mathbb{N} \bullet H\, x = F\, x - G\, x * G\, x))$

proof
use $\mathit{Lemma64Plus}[F := F,\ G := G]$;
use $\mathit{Lemma64Minus}[F := F,\ G := G]$;
prove;
instantiate $H__0 == H$;
prove;
instantiate $x__0 == x$;
prove;
■

3.10 Lemma 6.5: the predecessor $\hat{P}$ and the product $F \otimes G$

theorem MulNonneg
$\forall a, b : \mathbb{N} \bullet a * b \geq 0$

proof
prove;
■

theorem SqProd
$\forall a, b : \mathbb{Z} \bullet (a * b) * (a * b) = a * (a * (b * b))$

proof
prove;
■

theorem RtSqProd
$\forall\, a, b : \mathbb{N} \bullet \mathit{funRt}\,(a * (a * (b * b))) = a * b$

proof
prove;
use $\mathit{RtI}[r := a * b,\ x := a * (a * (b * b))]$;
use $\mathit{MulNonneg}[a := a,\ b := b]$;
prove;
■

theorem LeSq
$\forall\, n : \mathbb{N} \bullet n \leq n * n$

proof
prove;
split $n = 0$;
cases;
prove;
next;
use $\mathit{timesMonotonic}1[a := n,\ b := 1,\ c := n]$;
prove;
next;
■

$\hat{P} = ((\mathrm{Sq}^+)^+)^+ \ominus S$ is a predecessor: $\hat{P}(x+1) = x$.

$\mathit{exprPhat} == \mathit{exprOminus}(\mathit{opPlusI}\,(\mathit{opPlusI}\,(\mathit{opPlusI}\,\mathit{funSq})), \mathit{funS})$

proof [exprPhat$domainCheck]
prove;
■

theorem rule PhatFun
$\mathit{exprPhat} \in \mathbb{N} \to \mathbb{N}$

proof
invoke *exprPhat*;
prove;
■

theorem rule PhatIn
$\mathit{exprPhat} \in C6$

proof
invoke *exprPhat*;
prove;
■

theorem rule PhatVal
$\forall\, x : \mathbb{N} \bullet \mathit{exprPhat}\,(1 + x) = x$

proof
prove;
invoke *exprPhat*;
use $\mathit{OminusExprVal}[F := \mathit{opPlusI}\,(\mathit{opPlusI}\,(\mathit{opPlusI}\,\mathit{funSq})),\ G := \mathit{funS},\ x := 1 + x]$;
prove;
■

$(F \otimes G)(x) = F(x)\, G(x)$ (Definition 6.1), and its expression in Lemma 6.5.

$$opOtimes : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$$

$\langle\!\langle$ rule OtimesVal $\rangle\!\rangle$
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opOtimes(F, G)\, x = F\, x * G\, x$

proof [opOtimes$domainCheck]
prove;
■

theorem rule OtimesInner
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\, x * G\, x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F\, x * G\, x) \in \mathbb{N} \to \mathbb{N}$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
■

$OtimesWit == (\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\, x * G\, x))$

proof [OtimesWit$domainCheck]
prove;
■

theorem OtimesWitFun
$OtimesWit \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke OtimesWit;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\, x * G\, x)) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(RelPow,\ inPower)$ *prove by reduce*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem OtimesWitApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet OtimesWit(F, G) = (\lambda\, x : \mathbb{N} \bullet F\, x * G\, x)$

proof
use OtimesWitFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := OtimesWit,\ x := (F,\ G),\ y := (\lambda\ x : \mathbb{N} \bullet F\, x * G\, x)]$;
invoke OtimesWit;
prove by reduce;
■

theorem rule OtimesWitVal
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet OtimesWit(F, G)\, x = F\, x * G\, x$

proof
prove;
use OtimesWitApp$[F := F,\ G := G]$;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet F\, x * G\, x),\ x := x,\ y := F\, x * G\, x]$;
prove;
■

theorem OtimesEx
$\exists\, opOtimes : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opOtimes(F, G)\, x = F\, x * G\, x$

proof
instantiate opOtimes == OtimesWit;
use OtimesWitFun;
prove;
■

$exprOtimes == (\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet comp(funRt, exprOminus(comp(comp(funHf, funHf), exprOminus(comp(comp(funSq, exprPhat), exprOminus(exprOplus(comp(comp(comp(funSq, funS), funSq), F), G), comp(funSq, F))), comp(funSq, G))), comp(funSq, F))))$

proof [exprOtimes$domainCheck]
prove;
■

theorem rule exprOtimesFun
$exprOtimes \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprOtimes;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet comp(funRt,\ exprOminus(comp(comp(funHf,\ funHf),\ exprOminus(comp(comp(funSq,\ exprPhat),\ exprOminus(exprOplus(comp(comp(comp(funSq,\ funS),\ funSq),\ F),\ G),\ comp(funSq,\ G))),\ comp(funSq,\ F)))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem disabled rule exprOtimesApp
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet exprOtimes(F, G) = comp(funRt, exprOminus(comp(comp(funHf, funHf), exprOminus(comp(comp(funSq, exprPhat), exprOminus(exprOplus(comp(comp(comp(funSq, funS), funSq), F), G), comp(funSq, F))), comp(funSq, G))), comp(funSq, F)))$

proof
use exprOtimesFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := exprOtimes,\ x := (F,\ G),\ y := comp(funRt,\ exprOminus(comp(comp(funHf,\ funHf),\ exprOminus(comp(comp(funSq,\ exprPhat),\ exprOminus(exprOplus(comp(comp(comp(funSq,\ funS),\ funSq),\ F),\ G),\ comp(funSq,\ F))),\ comp(funSq,\ G))),\ comp(funSq,\ F)))]$;
invoke exprOtimes;
prove by reduce;
■

theorem rule OtimesIn
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet exprOtimes(F, G) \in C6$

proof
with enabled (*exprOtimesApp*) *prove*;
■

theorem rule OtimesExprVal
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid x \leq F\, x \wedge x \leq G\, x \bullet exprOtimes(F, G)\, x = F\, x * G\, x$

proof
use $LeSq[n := F\,x]$;
use $LeSq[n := G\,x]$;
use $LeSq[n := 1 + F\,x * F\,x]$;
use $MulNonneg[a := F\,x,\ b := F\,x]$;
use $MulNonneg[a := G\,x,\ b := G\,x]$;
use $MulNonneg[a := F\,x * F\,x,\ b := G\,x * G\,x]$;
use $MulNonneg[a := F\,x * F\,x,\ b := F\,x * F\,x]$;
with enabled (*exprOtimesApp*) *prove*;
use $DivMod[x := 4 * (F\,x * F\,x * F\,x * F\,x + F\,x * F\,x * G\,x * G\,x),\ d := 2,$
$q := 2 * (F\,x * F\,x * F\,x * F\,x + F\,x * F\,x * G\,x * G\,x),\ r := 0]$;
prove;
use $DivMod[x := 2 * (F\,x * F\,x * F\,x * F\,x + F\,x * F\,x * G\,x * G\,x),\ d := 2,$
$q := F\,x * F\,x * F\,x * F\,x + F\,x * F\,x * G\,x * G\,x,\ r := 0]$;
prove;
use $RtSqProd[a := F\,x,\ b := G\,x]$;
prove;
■

theorem OtimesEq
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in \mathit{Infl} \wedge G \in \mathit{Infl} \bullet \mathit{exprOtimes}(F, G) = \mathit{opOtimes}(F, G)$

proof
apply UFExt to predicate $\mathit{exprOtimes}(F,\ G) = \mathit{opOtimes}(F,\ G)$;
prove;
■

theorem Lemma65
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \wedge F \in \mathit{Infl} \wedge G \in \mathit{Infl} \bullet \mathit{opOtimes}(F, G) \in C6$

proof
use $OtimesEq[F := F,\ G := G]$;
prove;
equality substitute $\mathit{opOtimes}(F,\ G)$;
rewrite;
■

3.11 Lemma 6.6: $F + G$ and the difference $F - G$

theorem LeProd
$\forall\, x, a, b : \mathbb{N} \mid x \le a \wedge x \le b \bullet x \le a * b$

proof
prove;
split $x = 0$;
cases;
use $MulNonneg[a := a,\ b := b]$;
prove;
next;
use $timesMonotonic1[a := a,\ b := 1,\ c := b]$;
prove;
next;
■

theorem RtSqSum
$\forall\, a, b : \mathbb{N} \bullet \mathit{funRt}\,(2 * (a * b) + a * a + b * b) = a + b$

proof
prove;
use RtI[$r := a + b,\ x := 2 * (a * b) + a * a + b * b$];
prove;
■

$F + G = ((\mathrm{Rt}((D(F^+ \otimes G^+) \oplus F^+) \oplus G^+))^-)^-.$

exprPlus == $(\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet$ *exprMinusI* (*exprMinusI* (*comp*(*funRt*,
exprOplus(*exprOplus*(*comp*(*funD*, *exprOtimes*(*opPlusI F*, *opPlusI G*)), *opPlusI F*), *opPlusI G*)))))

proof [exprPlus$domainCheck]
prove;
■

theorem rule exprPlusFun
exprPlus $\in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprPlus;
apply FunIntro to predicate ($\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet$ *exprMinusI* (*exprMinusI* (*comp*(*funRt*,
exprOplus(*exprOplus*(*comp*(*funD*, *exprOtimes*(*opPlusI F*, *opPlusI G*)), *opPlusI F*),
opPlusI G))))) $\in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove*;
instantiate F == *x*.1, *G* == *x*.2;
prove;
■

theorem disabled rule exprPlusApp
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet$ *exprPlus*(F, G) = *exprMinusI* (*exprMinusI* (*comp*(*funRt*,
exprOplus(*exprOplus*(*comp*(*funD*, *exprOtimes*(*opPlusI F*, *opPlusI G*)), *opPlusI F*), *opPlusI G*))))

proof
use exprPlusFun;
use pairInFunction[$(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}$][$f :=$ *exprPlus*, $x := (F,\ G),\ y :=$
exprMinusI (*exprMinusI* (*comp*(*funRt*, *exprOplus*(*exprOplus*(*comp*(*funD*,
exprOtimes(*opPlusI F*, *opPlusI G*)), *opPlusI F*), *opPlusI G*))))];
invoke exprPlus;
prove by reduce;
■

theorem rule PlusExprIn
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet$ *exprPlus*$(F, G) \in C6$

proof
with enabled (*exprPlusApp*) *prove*;
■

theorem rule PlusExprVal
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet$ *exprPlus*$(F, G)\, x = F\, x + G\, x$

proof
use UFNonneg[$f := F,\ x := x$];
use UFNonneg[$f := G,\ x := x$];
use LeProd[$x := x,\ a := F\, x + x,\ b := G\, x + x$];
use MulNonneg[$a := F\, x + x,\ b := G\, x + x$];
use MulNonneg[$a := F\, x + x,\ b := F\, x + x$];
use RtSqSum[$a := F\, x + x,\ b := G\, x + x$];
with enabled (*exprPlusApp*) *prove*;
■

theorem PlusExprEq
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprPlus}(F, G) = \mathit{opPlus}(F, G)$

proof
apply UFExt to predicate $\mathit{exprPlus}(F,\ G) = \mathit{opPlus}(F,\ G)$;
prove;
■

The difference function of footnote 8: it equals $F - G$ where $F \geq G$.

$\mathit{opDiffE} : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

$\langle\!\langle$ rule DiffEVal $\rangle\!\rangle$
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opDiffE}(F, G)\, x = \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)$

proof [opDiffE$domainCheck]
prove;
■

theorem rule DiffEInner
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)) \in \mathbb{N} \to \mathbb{N}$;
with enabled $(\mathit{RelPow},\ \mathit{inPower})$ *prove by reduce*;
■

$\mathit{DiffEWit} == (\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)))$

proof [DiffEWit$domainCheck]
prove;
■

theorem DiffEWitFun
$\mathit{DiffEWit} \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke DiffEWit;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled $(\mathit{RelPow},\ \mathit{inPower})$ *prove by reduce*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem DiffEWitApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{DiffEWit}(F, G) = (\lambda\, x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1))$

proof
use DiffEWitFun;
use pairInFunction$[(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}), \mathbb{N} \to \mathbb{N}][f := \mathit{DiffEWit},\ x := (F,\ G),\ y := (\lambda\ x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1))]$;
invoke DiffEWit;
prove by reduce;
■

theorem rule DiffEWitVal
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{DiffEWit}(F, G)\, x = \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)$

proof
prove;
use DiffEWitApp[$F := F,\ G := G$];
use pairInFunction[$\mathbb{N},\ \mathbb{N}$][$f := (\lambda\ x : \mathbb{N} \bullet \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)),\ x := x,\ y := \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)$];
prove;
■

theorem DiffEEx
$\exists\, \mathit{opDiffE} : (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}) \bullet \forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet \mathit{opDiffE}(F, G)\, x = \mathit{funE}\,((F\, x + G\, x) * (F\, x + G\, x) + 3 * F\, x + G\, x + 1)$

proof
instantiate opDiffE $==$ *DiffEWit*;
use DiffEWitFun;
prove;
■

$F - G = E\, S\, (\mathrm{Sq}(F + G) + M_3 F + G).$

$\mathit{exprMinus} == (\lambda\, F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{comp}(\mathit{comp}(\mathit{funE}, \mathit{funS}), \mathit{exprPlus}(\mathit{exprPlus}(\mathit{comp}(\mathit{funSq}, \mathit{exprPlus}(F, G)), \mathit{comp}(\mathit{funM3}, F)), G)))$

proof [exprMinus$domainCheck]
prove;
■

theorem rule exprMinusFun
$\mathit{exprMinus} \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke exprMinus;
apply FunIntro to predicate $(\lambda\ F,\ G : \mathbb{N} \to \mathbb{N} \bullet \mathit{comp}(\mathit{comp}(\mathit{funE},\ \mathit{funS}),\ \mathit{exprPlus}(\mathit{exprPlus}(\mathit{comp}(\mathit{funSq},\ \mathit{exprPlus}(F,\ G)),\ \mathit{comp}(\mathit{funM3},\ F)),\ G))) \in (\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove*;
instantiate $F == x.1,\ G == x.2$;
prove;
■

theorem disabled rule exprMinusApp
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprMinus}(F, G) = \mathit{comp}(\mathit{comp}(\mathit{funE}, \mathit{funS}), \mathit{exprPlus}(\mathit{exprPlus}(\mathit{comp}(\mathit{funSq}, \mathit{exprPlus}(F, G)), \mathit{comp}(\mathit{funM3}, F)), G))$

proof
use exprMinusFun;
use pairInFunction[$(\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}),\ \mathbb{N} \to \mathbb{N}$][$f := \mathit{exprMinus},\ x := (F,\ G),\ y := \mathit{comp}(\mathit{comp}(\mathit{funE},\ \mathit{funS}),\ \mathit{exprPlus}(\mathit{exprPlus}(\mathit{comp}(\mathit{funSq},\ \mathit{exprPlus}(F,\ G)),\ \mathit{comp}(\mathit{funM3},\ F)),\ G))$];
invoke exprMinus;
prove by reduce;
■

theorem rule MinusExprIn
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet \mathit{exprMinus}(F, G) \in C6$

proof
with enabled (*exprMinusApp*) *prove*;
■

theorem MinusExprEq
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprMinus}(F, G) = \mathit{opDiffE}(F, G)$

proof
apply UFExt to predicate $\mathit{exprMinus}(F,\ G) = \mathit{opDiffE}(F,\ G)$;
with enabled (*exprMinusApp*) *prove*;
■

theorem DiffEIsDifference
$\forall F, G : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid G\, x \leq F\, x \bullet \mathit{opDiffE}(F, G)\, x = F\, x - G\, x$

proof
prove;
use RobDiffE$[a\ :=\ F\, x,\ b\ :=\ G\, x]$;
prove;
■

theorem Lemma66
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \bullet \mathit{opPlus}(F, G) \in C6 \land \mathit{opDiffE}(F, G) \in C6$

proof
use PlusExprEq$[F\ :=\ F,\ G\ :=\ G]$;
use MinusExprEq$[F\ :=\ F,\ G\ :=\ G]$;
prove;
equality substitute $\mathit{opPlus}(F,\ G)$;
equality substitute $\mathit{opDiffE}(F,\ G)$;
rewrite;
■

theorem rule C6Plus
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \bullet \mathit{opPlus}(F, G) \in C6$

proof
*use Lemma*66$[F\ :=\ F,\ G\ :=\ G]$;
prove;
■

theorem rule C6DiffE
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \land G \in C6 \bullet \mathit{opDiffE}(F, G) \in C6$

proof
*use Lemma*66$[F\ :=\ F,\ G\ :=\ G]$;
prove;
■

4 Theorem 6.7 (closure part)

Robinson [15, Sec. 7, Thm. 3] proved that $\langle S, E, F + G, FG, F^{\Box} \rangle$ is suitable; here we show that it generates exactly $\mathrm{clos}\,\mathcal{F}$, so $\mathrm{clos}\,\mathcal{F} = \mathrm{Prim}(\mathbb{N}, \mathbb{N})$ follows from Robinson's theorem.

$\mathit{RobinsonClos} == \mathit{clos}\,\{\mathit{funS}, \mathit{funE}\}\,\{\mathit{opBox}\}\,\{\mathit{opPlus}, \mathit{comp}\}$

proof [RobinsonClos$domainCheck]
prove;
■

theorem disabled rule RobDef
$RobinsonClos = clos\,\{funS, funE\}\,\{opBox\}\,\{opPlus, comp\}$

proof
invoke RobinsonClos;
prove;
■

theorem disabled rule RobUF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in RobinsonClos \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use ClosElemUF[$I := \{ funS, funE \}$, $U := \{ opBox \}$, $B := \{ opPlus, comp \}$, $F := F$];
with enabled (*RobDef*) *prove*;
■

theorem rule RobS
$funS \in RobinsonClos$

proof
with enabled (*RobDef*) *prove*;
■

theorem rule RobE
$funE \in RobinsonClos$

proof
with enabled (*RobDef*) *prove*;
■

theorem rule RobBox
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in RobinsonClos \bullet opBox\, F \in RobinsonClos$

proof
with enabled (*RobDef*) *prove*;
■

theorem rule RobPlus
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in RobinsonClos \land G \in RobinsonClos \bullet opPlus(F, G) \in RobinsonClos$

proof
with enabled (*RobDef*) *prove*;
■

theorem rule RobComp
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in RobinsonClos \land G \in RobinsonClos \bullet comp(F, G) \in RobinsonClos$

proof
with enabled (*RobDef*) *prove*;
■

$BoxESet == \{n : \mathbb{N} \mid opBox\ funE\ n = 0\}$

proof [BoxESet$domainCheck]
prove;
■

theorem rule BoxEBase
$0 \in BoxESet$

proof
invoke BoxESet;
prove;
■

theorem rule BoxEStep
$\forall n : BoxESet \bullet 1 + n \in BoxESet$

proof
invoke BoxESet;
prove;
■

theorem BoxESub
$\mathbb{N} \subseteq BoxESet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BoxESet$;
prove;
invoke BoxESet;
prove;
■

theorem BoxEAll
$\forall n : \mathbb{N} \bullet opBox\ funE\ n = 0$

proof
prove;
use BoxESub;
invoke BoxESet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
■

theorem BoxEExpr
$opBox\ funE = cst\ 0$

proof
apply UFExt to predicate $opBox\ funE = cst\ 0$;
prove;
use $BoxEAll[n := x]$;
prove;
■

$BoxSSet == \{n : \mathbb{N} \mid opBox\ funS\ n = n\}$

proof [BoxSSet$domainCheck]
prove;
■

theorem rule BoxSBase
$0 \in BoxSSet$

proof
invoke BoxSSet;
prove;
■

theorem rule BoxSStep
$\forall\, n : BoxSSet \bullet 1 + n \in BoxSSet$

proof
invoke BoxSSet;
prove;
■

theorem BoxSSub
$\mathbb{N} \subseteq BoxSSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BoxSSet$;
prove;
invoke BoxSSet;
prove;
■

theorem BoxSAll
$\forall\, n : \mathbb{N} \bullet opBox\, funS\, n = n$

proof
prove;
use BoxSSub;
invoke BoxSSet;
with enabled (*inPower*) *prove*;
instantiate e == *n*;
prove;
■

theorem BoxSExpr
$opBox\, funS = funI$

proof
apply UFExt to predicate $opBox\, funS = funI$;
prove;
use BoxSAll[$n := x$];
prove;
■

theorem OneExpr
$comp(funS, cst\, 0) = cst\, 1$

proof
apply UFExt to predicate $comp(funS,\ cst\,0) = cst\,1$;
prove;
■

theorem PlusIEq
$\forall F : \mathbb{N} \to \mathbb{N} \bullet opPlusI\ F = opPlus(F, funI)$

proof
apply UFExt to predicate $opPlusI\ F = opPlus(F,\ funI)$;
prove;
■

theorem RobIExpr
$opBox\ funS \in RobinsonClos$

proof
prove;
■

theorem rule RobI
$funI \in RobinsonClos$

proof
use RobIExpr;
use BoxSExpr;
prove;
■

theorem RobZeroExpr
$opBox\ funE \in RobinsonClos$

proof
prove;
■

theorem rule RobZero
$cst\,0 \in RobinsonClos$

proof
use BoxEExpr;
equality substitute $cst\,0$;
rewrite;
■

theorem RobOneExpr
$comp(funS, cst\,0) \in RobinsonClos$

proof
prove;
■

theorem rule RobOne
$cst\,1 \in RobinsonClos$

proof
use *RobOneExpr*;
use *OneExpr*;
prove;
■

theorem rule RobPlusI
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in RobinsonClos \bullet opPlusI\ F \in RobinsonClos$

proof
use *PlusIEq*[$F := F$];
prove;
■

theorem C6InPower
$C6 \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use *ClosInPow*[$I := \{ cst\,1,\ funE \},\ U := \{ opPlusI,\ opBox \},\ B := \{ comp \}$];
with enabled ($C6Def$) *prove*;
■

theorem RobInPower
$RobinsonClos \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use *ClosInPow*[$I := \{ funS,\ funE \},\ U := \{ opBox \},\ B := \{ opPlus,\ comp \}$];
with enabled ($RobDef$) *prove*;
■

theorem RobUClosed
$RobinsonClos \in UClosed\,\{opPlusI, opBox\}$

proof
use *RobInPower*;
with enabled ($UClosedDef,\ C6UF,\ RobUF$) *prove*;
split $u = opPlusI$;
cases;
prove;
next;
prove;
next;
■

theorem RobBClosed
$RobinsonClos \in BClosed\,\{comp\}$

proof
use *RobInPower*;
with enabled ($BClosedDef,\ C6UF,\ RobUF$) *prove*;
■

theorem C6UClosed
$C6 \in UClosed\,\{opBox\}$

proof
use $C6InPower$;
with enabled $(UClosedDef,\ C6UF,\ RobUF)$ *prove*;
■

theorem C6BClosed
$C6 \in BClosed\,\{opPlus, comp\}$

proof
use $C6InPower$;
with enabled $(BClosedDef,\ C6UF,\ RobUF)$ *prove*;
split $b = opPlus$;
cases;
prove;
next;
prove;
next;
■

theorem C6SubRob
$C6 \subseteq RobinsonClos$

proof
use $ClosLeast[I := \{\ cst\,1,\ funE\ \},\ U := \{\ opPlusI,\ opBox\ \},\ B := \{\ comp\ \},\ Z := RobinsonClos]$;
use $RobInPower$;
use $RobUClosed$;
use $RobBClosed$;
with enabled $(C6Def)$ *prove*;
■

theorem RobSubC6
$RobinsonClos \subseteq C6$

proof
use $ClosLeast[I := \{\ funS,\ funE\ \},\ U := \{\ opBox\ \},\ B := \{\ opPlus,\ comp\ \},\ Z := C6]$;
use $C6InPower$;
use $C6UClosed$;
use $C6BClosed$;
with enabled $(RobDef)$ *prove*;
■

theorem Theorem67
$C6 = RobinsonClos$

proof
use $C6SubRob$;
use $RobSubC6$;
use $C6InPower$;
use $RobInPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := C6,\ B := RobinsonClos]$;
prove;
■

5 Cantor pairing functions

$A(x) = x(x+1)/2$, V its inverse ($A(V(x)) \leq x < A(V(x)+1)$), $K(x) = x - A(V(x))$ and $L(x) = A(V(x)+1) - x - 1$.

$$
\begin{array}{|l}
funA : \mathbb{N} \to \mathbb{N} \\
\hline
\langle\!\langle \text{ disabled rule AVal } \rangle\!\rangle \\
\forall\, x : \mathbb{N} \bullet funA\, x = (x * x + x) \operatorname{div} 2
\end{array}
$$

proof [funA$domainCheck]
prove;
■

$AWit == (\lambda\, x : \mathbb{N} \bullet (x * x + x) \operatorname{div} 2)$

theorem AWitFun
$AWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke AWit;
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet (x * x + x) \operatorname{div} 2) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule AWitVal
$\forall\, x : \mathbb{N} \bullet AWit\, x = (x * x + x) \operatorname{div} 2$

proof
prove;
use AWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := AWit, x := x, y := (x * x + x) \operatorname{div} 2]$;
invoke AWit;
prove by reduce;
■

theorem AEx
$\exists\, funA : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funA\, x = (x * x + x) \operatorname{div} 2$

proof
instantiate funA $==$ *AWit*;
use AWitFun;
prove;
■

proof [AWit$domainCheck]
prove;
■

$TwoASet == \{n : \mathbb{N} \mid 2 * funA\, n = n * n + n\}$

proof [TwoASet$domainCheck]
prove;
■

theorem rule TwoABase
$0 \in TwoASet$

proof
invoke $TwoASet$;
prove;
use $DivMod[x := 0,\ d := 2,\ q := 0,\ r := 0]$;
with enabled $(AVal)$ *prove*;
■

theorem rule TwoAStep
$\forall n : TwoASet \bullet 1 + n \in TwoASet$

proof
invoke $TwoASet$;
prove;
use $DivMod[x := (1 + n) * (1 + n) + (1 + n),\ d := 2,\ q := funA\, n + n + 1,\ r := 0]$;
with enabled $(AVal)$ *prove*;
■

theorem TwoASub
$\mathbb{N} \subseteq TwoASet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq TwoASet$;
prove;
invoke $TwoASet$;
prove;
■

theorem TwoAAll
$\forall n : \mathbb{N} \bullet 2 * funA\, n = n * n + n$

proof
prove;
use $TwoASub$;
invoke $TwoASet$;
with enabled $(inPower)$ *prove*;
instantiate $e == n$;
prove;
■

theorem TwoA
$\forall x : \mathbb{N} \bullet 2 * funA\, x = x * x + x$

proof
prove;
use $TwoAAll[n := x]$;
prove;
■

theorem ASuc
$\forall x : \mathbb{N} \bullet funA\, (1 + x) = funA\, x + x + 1$

proof
prove;
use *TwoA*[$x := x$];
use *TwoA*[$x := 1 + x$];
use *UFNonneg*[$f := funA,\ x := x$];
use *UFNonneg*[$f := funA,\ x := 1 + x$];
prove;
■

theorem AZero
$funA\,0 = 0 \land funA\,1 = 1$

proof
use *TwoA*[$x := 0$];
use *TwoA*[$x := 1$];
use *UFNonneg*[$f := funA,\ x := 0$];
use *UFNonneg*[$f := funA,\ x := 1$];
prove;
■

theorem AMono
$\forall\, m, n : \mathbb{N} \mid m \leq n \bullet funA\,m \leq funA\,n$

proof
prove;
use *TwoA*[$x := m$];
use *TwoA*[$x := n$];
use *timesMonotonic2*[$a := m,\ b := m,\ c := n,\ d := n$];
use *UFNonneg*[$f := funA,\ x := m$];
use *UFNonneg*[$f := funA,\ x := n$];
prove;
■

theorem AIndexUq
$\forall\, r, s, x : \mathbb{N} \mid funA\,r \leq x \land x < funA\,(1 + r) \land funA\,s \leq x \land x < funA\,(1 + s) \bullet r = s$

proof
prove;
split $r < s$;
cases;
use *AMono*[$m := 1 + r,\ n := s$];
prove;
next;
split $s < r$;
cases;
use *AMono*[$m := 1 + s,\ n := r$];
prove;
next;
prove;
next;
■

$AIdxSet == \{x : \mathbb{N} \mid \exists\, t : \mathbb{N} \bullet funA\,t \leq x \land x < funA\,(1 + t)\}$

proof [AIdxSet$domainCheck]
prove;
■

theorem rule AIdxBase
$0 \in AIdxSet$

proof
invoke AIdxSet;
prove;
use AZero;
instantiate $t == 0$;
prove;
■

theorem rule AIdxStep
$\forall x : AIdxSet \bullet 1 + x \in AIdxSet$

proof
invoke AIdxSet;
prove;
use ASuc$[x := 1 + t]$;
use UFNonneg$[f := funA,\ x := t]$;
use UFNonneg$[f := funA,\ x := 1 + t]$;
use UFNonneg$[f := funA,\ x := 2 + t]$;
use LeOrGt$[a := funA\,(1 + t),\ b := 1 + x]$;
split $funA\,(1 + t) \leq 1 + x$;
cases;
instantiate $t_0 == 1 + t$;
prove;
next;
instantiate $t_1 == t$;
prove;
next;
■

theorem AIdxSub
$\mathbb{N} \subseteq AIdxSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq AIdxSet$;
prove;
invoke AIdxSet;
prove;
■

theorem AIndexEx
$\forall x : \mathbb{N} \bullet \exists t : \mathbb{N} \bullet funA\ t \leq x \land x < funA\,(1 + t)$

proof
prove;
use AIdxSub;
with enabled (*inPower*) *prove*;
instantiate $e == x$;
invoke AIdxSet;
prove;
■

$funV : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ VSpec $\rangle\!\rangle$
$\forall x : \mathbb{N} \bullet funA\,(funV\ x) \leq x \land x < funA\,(1 + funV\ x)$

proof [funV$domainCheck]
prove;
■

$VWit == \{x, t : \mathbb{N} \mid funA\, t \leq x \wedge x < funA\,(1 + t) \bullet (x, t)\}$

proof [VWit$domainCheck]
prove;
■

theorem VWitFun
$VWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke VWit;
apply FunIntro to predicate $\{\ x,\ t : \mathbb{N} \mid funA\, t \leq x \wedge x < funA\,(1 + t) \bullet (x,\ t)\ \}$
$\in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use AIndexEx[$x := x$];
use AIndexUq[$r := y,\ s := z,\ x := x__0$];
prove;
instantiate $y__0 == t$;
prove;
■

theorem rule VWitLe
$\forall x : \mathbb{N} \bullet funA\,(VWit\, x) \leq x$

proof
prove;
use AIndexEx[$x := x$];
use VWitFun;
prove;
use pairInFunction[$\mathbb{N},\ \mathbb{N}$][$f := VWit,\ x := x,\ y := t$];
invoke VWit;
prove by reduce;
■

theorem rule VWitLt
$\forall x : \mathbb{N} \bullet x < funA\,(1 + VWit\, x)$

proof
prove;
use AIndexEx[$x := x$];
use VWitFun;
prove;
use pairInFunction[$\mathbb{N},\ \mathbb{N}$][$f := VWit,\ x := x,\ y := t$];
invoke VWit;
prove by reduce;
■

theorem VEx
$\exists funV : \mathbb{N} \to \mathbb{N} \bullet \forall x : \mathbb{N} \bullet funA\,(funV\, x) \leq x \wedge x < funA\,(1 + funV\, x)$

proof
instantiate $funV == VWit$;
use VWitFun;
prove;
■

theorem VChar
$\forall\, t, x : \mathbb{N} \mid funA\, t \leq x \wedge x < funA\,(1 + t) \bullet funV\, x = t$

proof
prove;
use $VSpec[x := x]$;
use $AIndexUq[r := funV\, x,\ s := t,\ x := x]$;
use $UFNonneg[f := funV,\ x := x]$;
prove;
■

$funK : \mathbb{N} \to \mathbb{N}$

⟪ disabled rule KVal ⟫
$\forall\, x : \mathbb{N} \bullet funK\, x = x - funA\,(funV\, x)$

proof [funK$domainCheck]
prove;
■

$KWit == (\lambda\, x : \mathbb{N} \bullet x - funA\,(funV\, x))$

proof [KWit$domainCheck]
prove;
■

theorem KWitRel
$(\lambda\, x : \mathbb{N} \bullet x - funA\,(funV\, x)) \in \mathbb{P}(\mathbb{N} \times \mathbb{N})$

proof
with enabled (*inPower*) *prove by reduce*;
use $VSpec[x := x]$;
prove;
■

theorem KWitFun
$KWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke KWit;
use KWitRel;
apply FunIntro to predicate $(\lambda\, x : \mathbb{N} \bullet x - funA\,(funV\, x)) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use $VSpec[x := x]$;
prove;
■

theorem rule KWitVal
$\forall\, x : \mathbb{N} \bullet KWit\, x = x - funA\,(funV\, x)$

proof
prove;
use KWitFun;
use $pairInFunction[\mathbb{N},\ \mathbb{N}][f := KWit,\ x := x,\ y := x - funA\,(funV\, x)]$;
invoke KWit;
prove by reduce;
■

theorem KEx
$\exists\, funK : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funK\, x = x - funA\,(funV\, x)$

proof
instantiate $funK == KWit$;
use *KWitFun*;
prove;
■

$funL : \mathbb{N} \to \mathbb{N}$

$\langle\!\langle$ disabled rule LVal $\rangle\!\rangle$
$\forall\, x : \mathbb{N} \bullet funL\, x = funA\,(1 + funV\, x) - x - 1$

proof [funL$domainCheck]
prove;
■

$LWit == (\lambda\, x : \mathbb{N} \bullet funA\,(1 + funV\, x) - x - 1)$

proof [LWit$domainCheck]
prove;
■

theorem LWitRel
$(\lambda\, x : \mathbb{N} \bullet funA\,(1 + funV\, x) - x - 1) \in \mathbb{P}(\mathbb{N} \times \mathbb{N})$

proof
with enabled (*inPower*) *prove by reduce*;
use $VSpec[x := x]$;
prove;
■

theorem LWitFun
$LWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke *LWit*;
use *LWitRel*;
apply FunIntro to predicate $(\lambda\, x : \mathbb{N} \bullet funA\,(1 + funV\, x) - x - 1) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use $VSpec[x := x]$;
prove;
■

theorem rule LWitVal
$\forall\, x : \mathbb{N} \bullet LWit\, x = funA\,(1 + funV\, x) - x - 1$

proof
prove;
use *LWitFun*;
use *pairInFunction*$[\mathbb{N}, \mathbb{N}][f := LWit,\ x := x,\ y := funA\,(1 + funV\, x) - x - 1]$;
invoke *LWit*;
prove by reduce;
■

theorem LEx
$\exists\, funL : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funL\, x = funA\,(1 + funV\, x) - x - 1$

proof
instantiate funL == LWit;
use LWitFun;
prove;
■

Values on a diagonal of the Cantor table: for $k \leq t$, $V(A(t)+k) = t$, $K(A(t)+k) = k$ and $L(A(t)+k) = t-k$.

theorem Diag
$\forall\, k, t : \mathbb{N} \mid k \leq t \bullet funV\,(funA\, t + k) = t \wedge funK\,(funA\, t + k) = k \wedge funL\,(funA\, t + k) = t - k$

proof
prove;
use ASuc[$x := t$];
use UFNonneg[$f := funA,\ x := t$];
use VChar[$t := t,\ x := funA\, t + k$];
with enabled (*KVal*, *LVal*) *prove*;
■

theorem Decomp
$\forall\, x : \mathbb{N} \bullet x = funA\,(funV\, x) + funK\, x \wedge funK\, x \leq funV\, x$

proof
prove;
use VSpec[$x := x$];
use ASuc[$x := funV\, x$];
use UFNonneg[$f := funA,\ x := funV\, x$];
use UFNonneg[$f := funA,\ x := 1 + funV\, x$];
use UFNonneg[$f := funV,\ x := x$];
with enabled (*KVal*) *prove*;
■

5.1 A, V, K and L are in $\operatorname{clos}\mathcal{F}$

theorem AExpr
$comp(funHf, opPlusI\, funSq) = funA$

proof
apply UFExt to predicate $comp(funHf,\ opPlusI\, funSq) = funA$;
prove;
use TwoA[$x := x$];
use UFNonneg[$f := funA,\ x := x$];
use DivMod[$x := x * x + x,\ d := 2,\ q := funA\, x,\ r := 0$];
prove;
■

theorem RtPos
$\forall\, y : \mathbb{N} \mid 1 \leq y \bullet 1 \leq funRt\, y$

proof
prove;
use RtSpec[$x := y$];
use UFNonneg[$f := funRt,\ x := y$];
split $funRt\, y = 0$;
cases;
prove;
next;
prove;
next;
■

theorem PhatPos
$\forall\, y : \mathbb{N} \mid 1 \leq y \bullet \mathit{exprPhat}\ y = y - 1$

proof
prove;
use $\mathit{PhatVal}[x := y - 1]$;
prove;
■

theorem VFormula
$\forall\, x : \mathbb{N} \bullet \mathit{funV}\ x = (\mathit{funRt}\ (1 + 8 * x) - 1)\ \mathsf{div}\ 2$

proof
prove;
use $\mathit{VSpec}[x := x]$;
use $\mathit{TwoA}[x := \mathit{funV}\ x]$;
use $\mathit{TwoA}[x := 1 + \mathit{funV}\ x]$;
use $\mathit{UFNonneg}[f := \mathit{funA},\ x := \mathit{funV}\ x]$;
use $\mathit{UFNonneg}[f := \mathit{funA},\ x := 1 + \mathit{funV}\ x]$;
use $\mathit{UFNonneg}[f := \mathit{funV},\ x := x]$;
use $\mathit{LeOrGt}[a := (2 * \mathit{funV}\ x + 2) * (2 * \mathit{funV}\ x + 2),\ b := 1 + 8 * x]$;
split $(2 * \mathit{funV}\ x + 2) * (2 * \mathit{funV}\ x + 2) \leq 1 + 8 * x$;
cases;
use $\mathit{RtI}[r := 2 * \mathit{funV}\ x + 2,\ x := 1 + 8 * x]$;
prove;
use $\mathit{DivMod}[x := 2 * \mathit{funV}\ x + 1,\ d := 2,\ q := \mathit{funV}\ x,\ r := 1]$;
prove;
next;
use $\mathit{RtI}[r := 2 * \mathit{funV}\ x + 1,\ x := 1 + 8 * x]$;
prove;
use $\mathit{DivMod}[x := 2 * \mathit{funV}\ x,\ d := 2,\ q := \mathit{funV}\ x,\ r := 0]$;
prove;
next;
■

theorem VExpr
$\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funHf}, \mathit{exprPhat}), \mathit{funRt}), \mathit{funS}), \mathit{funD}), \mathit{funD}), \mathit{funD}) = \mathit{funV}$

proof
apply UFExt to predicate $\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funHf},\ \mathit{exprPhat}),\ \mathit{funRt}),\ \mathit{funS}),\ \mathit{funD}),\ \mathit{funD}),\ \mathit{funD}) = \mathit{funV}$;
prove;
use $\mathit{VFormula}[x := x]$;
use $\mathit{RtPos}[y := 1 + 8 * x]$;
use $\mathit{PhatPos}[y := \mathit{funRt}\ (1 + 8 * x)]$;
prove;
■

theorem IInExpr
$\mathit{opBox}\ \mathit{funS} \in C6$

proof
prove;
■

theorem rule IIn
$\mathit{funI} \in C6$

proof
use IInExpr;
use BoxSExpr;
prove;
■

theorem AInExpr
$comp(funHf, opPlusI\ funSq) \in C6$

proof
prove;
■

theorem rule AIn
$funA \in C6$

proof
use AInExpr;
use AExpr;
prove;
■

theorem VInExpr
$comp(comp(comp(comp(comp(comp(funHf, exprPhat), funRt), funS), funD), funD), funD) \in C6$

proof
prove;
■

theorem rule VIn
$funV \in C6$

proof
use VInExpr;
use VExpr;
prove;
■

theorem KExpr
$opDiffE(funI, comp(funA, funV)) = funK$

proof
apply UFExt to predicate $opDiffE(funI,\ comp(funA,\ funV)) = funK$;
prove;
use VSpec$[x := x]$;
use DiffEIsDifference$[F := funI,\ G := comp(funA,\ funV),\ x := x]$;
with enabled $(KVal)$ *prove*;
■

theorem LExpr
$opDiffE(comp(funA, comp(funS, funV)), funS) = funL$

proof
apply UFExt to predicate $opDiffE(comp(funA,\ comp(funS,\ funV)),\ funS) = funL$;
prove;
use VSpec$[x := x]$;
use DiffEIsDifference$[F := comp(funA,\ comp(funS,\ funV)),\ G := funS,\ x := x]$;
with enabled $(LVal)$ *prove*;
■

theorem KInExpr
$opDiffE(funI, comp(funA, funV)) \in C6$

proof
prove;
■

theorem rule KIn
$funK \in C6$

proof
use KInExpr;
use KExpr;
prove;
■

theorem LInExpr
$opDiffE(comp(funA, comp(funS, funV)), funS) \in C6$

proof
prove;
■

theorem rule LIn
$funL \in C6$

proof
use LInExpr;
use LExpr;
prove;
■

6 Theorem 6.8: the bases with K and L

6.1 Auxiliary functions (Robinson [16, p. 664])

theorem KAt
$\forall m, t, k : \mathbb{N} \mid m = funA\, t + k \wedge k \leq t \bullet funK\, m = k$

proof
prove;
use Diag$[k := k,\ t := t]$;
prove;
■

theorem RtLe
$\forall x : \mathbb{N} \bullet funRt\, x \leq x$

proof
prove;
use RtSpec$[x := x]$;
use LeSq$[n := funRt\, x]$;
prove;
■

$Z(x) = x(x+3)/2 = A(x) + x$ and $Y(x) = 2x - \mathrm{Rt}(x)$.

$$
\begin{array}{|l}
funZ : \mathbb{N} \to \mathbb{N} \\
\hline
\langle\!\langle \text{rule ZVal} \rangle\!\rangle \\
\forall\, x : \mathbb{N} \bullet funZ\, x = funA\, x + x
\end{array}
$$

proof [funZ$domainCheck]
prove;
■

$ZWit == (\lambda\, x : \mathbb{N} \bullet funA\, x + x)$

proof [ZWit$domainCheck]
prove;
■

theorem ZWitFun
$ZWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke ZWit;
apply FunIntro to predicate $(\lambda\, x : \mathbb{N} \bullet funA\, x + x) \in \mathbb{N} \to \mathbb{N}$;
prove by reduce;
with enabled (*RelPow*) *prove by reduce*;
■

theorem rule ZWitVal
$\forall\, x : \mathbb{N} \bullet ZWit\, x = funA\, x + x$

proof
prove;
use ZWitFun;
use pairInFunction$[\mathbb{N}, \mathbb{N}][f := ZWit, x := x, y := funA\, x + x]$;
invoke ZWit;
prove by reduce;
■

theorem ZEx
$\exists\, funZ : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funZ\, x = funA\, x + x$

proof
instantiate funZ $==$ *ZWit*;
use ZWitFun;
prove;
■

$$
\begin{array}{|l}
funY : \mathbb{N} \to \mathbb{N} \\
\hline
\langle\!\langle \text{disabled rule YVal} \rangle\!\rangle \\
\forall\, x : \mathbb{N} \bullet funY\, x = 2 * x - funRt\, x
\end{array}
$$

proof [funY$domainCheck]
prove;
■

$YWit == (\lambda\, x : \mathbb{N} \bullet 2 * x - funRt\, x)$

proof [YWit$domainCheck]
prove;
■

theorem YWitRel
$(\lambda\, x : \mathbb{N} \bullet 2 * x - funRt\, x) \in \mathbb{P}(\mathbb{N} \times \mathbb{N})$

proof
with enabled (*inPower*) *prove by reduce*;
use $RtLe[x := x]$;
prove;
■

theorem YWitFun
$YWit \in \mathbb{N} \to \mathbb{N}$

proof
invoke YWit;
use YWitRel;
apply FunIntro to predicate $(\lambda\, x : \mathbb{N} \bullet 2 * x - funRt\, x) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*) *prove by reduce*;
use $RtLe[x := x]$;
prove;
■

theorem rule YWitVal
$\forall\, x : \mathbb{N} \bullet YWit\, x = 2 * x - funRt\, x$

proof
prove;
use YWitFun;
use pairInFunction $[\mathbb{N}, \mathbb{N}][f := YWit,\ x := x,\ y := 2 * x - funRt\, x]$;
invoke YWit;
prove by reduce;
■

theorem YEx
$\exists\, funY : \mathbb{N} \to \mathbb{N} \bullet \forall\, x : \mathbb{N} \bullet funY\, x = 2 * x - funRt\, x$

proof
instantiate $funY == YWit$;
use YWitFun;
prove;
■

$ZBoxSet == \{n : \mathbb{N} \mid opBox\, (comp(comp(funS, funS), opPlusI\, funK))\, n = funZ\, n\}$

proof [ZBoxSet$domainCheck]
prove;
■

theorem rule ZBoxBase
$0 \in ZBoxSet$

proof
invoke ZBoxSet;
prove;
use AZero;
prove;
■

theorem rule ZBoxStep
$\forall\, n : ZBoxSet \bullet 1 + n \in ZBoxSet$

proof
invoke ZBoxSet;
prove;
use Diag$[k := n,\ t := n]$;
use ASuc$[x := n]$;
prove;
■

theorem ZBoxSub
$\mathbb{N} \subseteq ZBoxSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq ZBoxSet$;
prove;
invoke ZBoxSet;
prove;
■

theorem ZBoxAll
$\forall\, n : \mathbb{N} \bullet opBox\,(comp(comp(funS, funS), opPlusI\, funK))\, n = funZ\, n$

proof
prove;
use ZBoxSub;
invoke ZBoxSet;
with enabled (*inPower*) *prove*;
instantiate e $==$ n;
prove;
■

theorem ZExpr
$opBox\,(comp(comp(funS, funS), opPlusI\, funK)) = funZ$

proof
apply UFExt to predicate $opBox\,(comp(comp(funS,\ funS),\ opPlusI\, funK)) = funZ$;
prove;
use ZBoxAll$[n := x]$;
prove;
■

Robinson's formula: $K(Z(a + b) + 2a + 3) = a - b$ if $a \geq b$.

theorem RobDiffK
$\forall\, a, b : \mathbb{N} \mid b \leq a \bullet funK\,(funZ\,(a + b) + 2 * a + 3) = a - b$

proof
prove;
use $ASuc[x := a + b]$;
use $ASuc[x := 1 + a + b]$;
use $KAt[m := funA\,(a + b) + a + b + 2 * a + 3,\ t := a + b + 2,\ k := a - b]$;
prove;
■

$I - F = K\ S\ S\ S\ ((ZF^+)^+)^+$ where $F(x) \leq x$, and $D - F = K\ S\ S\ S\ ((((Z(F^+)^+)^+)^+)^+)^+$ where $F(x) \leq 2x$.

$exprIminusF == (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(comp(comp(funK, funS), funS), funS), funS),$
$opPlusI\ (opPlusI\ (comp(funZ, opPlusI\ F)))))$

proof [exprIminusF$domainCheck]
prove;
■

theorem rule exprIminusFFun
$exprIminusF \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke *exprIminusF*;
apply *FunIntro* *to* *predicate* $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(comp(comp(funK, funS), funS), funS), funS),$
$opPlusI\ (opPlusI\ (comp(funZ, opPlusI\ F)))))\ \in\ (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with *enabled* $(RelPow, inPower)$ *prove* *by* *reduce*;
■

theorem disabled rule exprIminusFApp
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet exprIminusF\ F = comp(comp(comp(comp(funK, funS), funS), funS), funS),$
$opPlusI\ (opPlusI\ (comp(funZ, opPlusI\ F))))$

proof
use *exprIminusFFun*;
use *pairInFunction*$[\mathbb{N} \to \mathbb{N}, \mathbb{N} \to \mathbb{N}][f := exprIminusF,\ x := F,\ y :=$
$comp(comp(comp(comp(funK, funS), funS), funS), opPlusI\ (opPlusI\ (comp(funZ,$
$opPlusI\ F))))]$;
invoke *exprIminusF*;
prove *by* *reduce*;
■

$exprDminusF == (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(comp(comp(funK, funS), funS), funS), funS),$
$opPlusI\ (opPlusI\ (opPlusI\ (opPlusI\ (comp(funZ, opPlusI\ (opPlusI\ F)))))))$

proof [exprDminusF$domainCheck]
prove;
■

theorem rule exprDminusFFun
$exprDminusF \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
invoke *exprDminusF*;
apply *FunIntro* *to* *predicate* $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(comp(comp(funK, funS), funS), funS), funS),$
$opPlusI\ (opPlusI\ (opPlusI\ (opPlusI\ (comp(funZ, opPlusI\ (opPlusI\ F)))))))\ \in\ (\mathbb{N} \to \mathbb{N}) \to$
$(\mathbb{N} \to \mathbb{N})$;
with *enabled* $(RelPow, inPower)$ *prove* *by* *reduce*;
■

theorem disabled rule exprDminusFApp
$\forall\, F : \mathbb{N} \to \mathbb{N} \bullet \mathit{exprDminusF}\ F = \mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funK}, \mathit{funS}), \mathit{funS}), \mathit{funS}),$
$\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{comp}(\mathit{funZ}, \mathit{opPlusI}\ (\mathit{opPlusI}\ F)))))))$

proof
use exprDminusFFun;
use pairInFunction$[\mathbb{N} \to \mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := \mathit{exprDminusF},\ x := F,\ y :=$
$\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{comp}(\mathit{funK},\ \mathit{funS}),\ \mathit{funS}),\ \mathit{funS}),$
$\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{opPlusI}\ (\mathit{comp}(\mathit{funZ},\ \mathit{opPlusI}\ (\mathit{opPlusI}\ F)))))))]$;
invoke exprDminusF;
prove by reduce;
■

theorem IminusFVal
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid F\ x \leq x \bullet \mathit{exprIminusF}\ F\ x = x - F\ x$

proof
with enabled (*exprIminusFApp*) *prove*;
use RobDiffK$[a := x,\ b := F\ x]$;
prove;
■

theorem DminusFVal
$\forall\, F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid F\ x \leq 2 * x \bullet \mathit{exprDminusF}\ F\ x = 2 * x - F\ x$

proof
with enabled (*exprDminusFApp*) *prove*;
use RobDiffK$[a := 2 * x,\ b := F\ x]$;
prove;
■

$Y = ((\mathrm{Sgn}\, K)^{+} S)^{\square}$: $Y(n+1) = Y(n) + 1 + \mathrm{Sgn}\, K(Y(n)+1)$, and $K(Y(n)+1) = 0$ exactly when $n+1$ is a square ($Y(n)+1 = 2n - r + 1$ with $r = \mathrm{Rt}(n)$ lies on the diagonal of $A(2r+1)$, $A(2r)$ or $A(2r-1)$).

theorem YStep
$\forall\, n : \mathbb{N} \bullet \mathit{opPlusI}\ (\mathit{comp}(\mathit{funSgn}, \mathit{funK}))\ (\mathit{funS}\ (\mathit{funY}\ n)) = \mathit{funY}\ (1 + n)$

proof
with enabled (*YVal*) *prove*;
use *RtSpec*[$x := n$];
use *RtLe*[$x := n$];
use *UFNonneg*[$f := funRt,\ x := n$];
use *LeOrGt*[$a := (1 + funRt\ n) * (1 + funRt\ n),\ b := 1 + n$];
split $(1 + funRt\ n) * (1 + funRt\ n) \leq 1 + n$;
cases;
use *RtI*[$r := 1 + funRt\ n,\ x := 1 + n$];
use *TwoA*[$x := 1 + 2 * funRt\ n$];
use *UFNonneg*[$f := funA,\ x := 1 + 2 * funRt\ n$];
use *KAt*[$m := 1 + 2 * n - funRt\ n,\ t := 1 + 2 * funRt\ n,\ k := 0$];
prove;
next;
use *RtI*[$r := funRt\ n,\ x := 1 + n$];
use *LeOrGt*[$a := funRt\ n * funRt\ n + funRt\ n,\ b := n$];
split $funRt\ n * funRt\ n + funRt\ n \leq n$;
cases;
use *TwoA*[$x := 2 * funRt\ n$];
use *UFNonneg*[$f := funA,\ x := 2 * funRt\ n$];
use *KAt*[$m := 1 + 2 * n - funRt\ n,\ t := 2 * funRt\ n,\ k := 1 + 2 * n - 2 * funRt\ n * funRt\ n - 2 * funRt\ n$];
prove;
next;
use *TwoA*[$x := 2 * funRt\ n - 1$];
use *UFNonneg*[$f := funA,\ x := 2 * funRt\ n - 1$];
use *KAt*[$m := 1 + 2 * n - funRt\ n,\ t := 2 * funRt\ n - 1,\ k := 1 + 2 * n - 2 * funRt\ n * funRt\ n$];
prove;
next;
■

$YBoxSet == \{n : \mathbb{N} \mid opBox\ (comp(opPlusI\ (comp(funSgn, funK)), funS))\ n = funY\ n\}$

proof [YBoxSet$domainCheck]
prove;
■

theorem rule YBoxBase
$0 \in YBoxSet$

proof
invoke *YBoxSet*;
prove;
use *RtSq*[$m := 0$];
with enabled (*YVal*) *prove*;
■

theorem rule YBoxStep
$\forall\ n : YBoxSet \bullet 1 + n \in YBoxSet$

proof
invoke *YBoxSet*;
prove;
use *YStep*[$n := n$];
prove;
■

theorem YBoxSub
$\mathbb{N} \subseteq YBoxSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq YBoxSet$;
prove;
invoke YBoxSet;
prove;
■

theorem YBoxAll
$\forall n : \mathbb{N} \bullet opBox\,(comp(opPlusI\,(comp(funSgn, funK)), funS))\,n = funY\,n$

proof
prove;
use YBoxSub;
invoke YBoxSet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
■

theorem YExpr
$opBox\,(comp(opPlusI\,(comp(funSgn, funK)), funS)) = funY$

proof
apply UFExt to predicate $opBox\,(comp(opPlusI\,(comp(funSgn,\ funK)),\ funS)) = funY$;
prove;
use $YBoxAll[n := x]$;
prove;
■

$\mathrm{Rt} = D - Y$, $\mathrm{Sq} = (((\bar{1}^{+})^{+}\mathrm{Rt})^{+})^{\square}$ and $E = I - \mathrm{Sq}\,\mathrm{Rt}$.

theorem RtExprK
$exprDminusF\,funY = funRt$

proof
apply UFExt to predicate $exprDminusF\,funY = funRt$;
prove;
use $RtLe[x := x]$;
use $UFNonneg[f := funRt,\ x := x]$;
use $DminusFVal[F := funY,\ x := x]$;
with enabled (*YVal*) *prove*;
■

$SqKSet == \{n : \mathbb{N} \mid opBox\,(opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)), funRt)))\,n = n * n\}$

proof [SqKSet$domainCheck]
prove;
■

theorem rule SqKBase
$0 \in SqKSet$

proof
invoke SqKSet;
prove;
■

theorem rule SqKStep
$\forall\, n : SqKSet \bullet 1 + n \in SqKSet$

proof
invoke SqKSet;
prove;
equality substitute opBox (*opPlusI* (*comp*(*opPlusI* (*opPlusI* (*cst* 1)), *funRt*))) *n*;
rewrite;
■

theorem SqKSub
$\mathbb{N} \subseteq SqKSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq SqKSet$;
prove;
invoke SqKSet;
prove;
■

theorem SqKAll
$\forall\, n : \mathbb{N} \bullet opBox\,(opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)), funRt)))\,n = n * n$

proof
prove;
use SqKSub;
invoke SqKSet;
with enabled (*inPower*) *prove*;
instantiate e == *n*;
prove;
■

theorem SqExprK
$opBox\,(opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)), funRt))) = funSq$

proof
apply UFExt to predicate opBox (*opPlusI* (*comp*(*opPlusI* (*opPlusI* (*cst* 1)), *funRt*))) = *funSq*;
prove;
use SqKAll[*n* := *x*];
prove;
■

theorem EExprK
$exprIminusF\,(comp(funSq, funRt)) = funE$

proof
apply UFExt to predicate exprIminusF (*comp*(*funSq*, *funRt*)) = *funE*;
prove;
use RtSpec[*x* := *x*];
use IminusFVal[*F* := *comp*(*funSq*, *funRt*), *x* := *x*];
with enabled (*EVal*) *prove*;
■

6.2 The basis $\mathcal{F}' = \langle \bar{1}, K, F^+, FG, F^\square \rangle$

$C6K == clos\,\{cst\,1, funK\}\,\{opPlusI, opBox\}\,\{comp\}$

proof [C6K$domainCheck]
prove;
■

theorem disabled rule C6KDef
$C6K = clos\,\{cst\,1, funK\}\,\{opPlusI, opBox\}\,\{comp\}$

proof
invoke $C6K$;
prove;
■

theorem C6KInPower
$C6K \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use $ClosInPow[I := \{ cst\,1,\ funK \},\ U := \{ opPlusI,\ opBox \},\ B := \{ comp \}]$;
with enabled $(C6KDef)$ *prove*;
■

theorem disabled rule C6KUF
$\forall F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in C6K \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $ClosElemUF[I := \{ cst\,1,\ funK \},\ U := \{ opPlusI,\ opBox \},\ B := \{ comp \},\ F := F]$;
with enabled $(C6KDef)$ *prove*;
■

theorem rule C6KInit0
$cst\,1 \in C6K$

proof
with enabled $(C6KDef)$ *prove*;
■

theorem rule C6KInit1
$funK \in C6K$

proof
with enabled $(C6KDef)$ *prove*;
■

theorem rule C6KUnop0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6K \bullet opPlusI\ F \in C6K$

proof
with enabled $(C6KDef)$ *prove*;
■

theorem rule C6KUnop1
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6K \bullet opBox\ F \in C6K$

proof
with enabled $(C6KDef)$ *prove*;
■

theorem rule C6KBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6K \land G \in C6K \bullet comp(F, G) \in C6K$

proof
with enabled $(C6KDef)$ *prove*;
■

theorem SInKExpr
$opPlusI\ (cst\ 1) \in C6K$

proof
prove;
■

theorem rule SInK
$funS \in C6K$

proof
use SInKExpr;
use SExpr;
prove;
■

theorem SgnInKExpr
$opBox\ (cst\ 1) \in C6K$

proof
prove;
■

theorem rule SgnInK
$funSgn \in C6K$

proof
use SgnInKExpr;
use SgnExpr;
prove;
■

theorem ZInKExpr
$opBox\ (comp(comp(funS, funS), opPlusI\ funK)) \in C6K$

proof
prove;
■

theorem rule ZInK
funZ ∈ *C*6*K*

proof
use ZInKExpr;
use ZExpr;
prove;
■

theorem YInKExpr
opBox (*comp*(*opPlusI* (*comp*(*funSgn*, *funK*)), *funS*)) ∈ *C*6*K*

proof
prove;
■

theorem rule YInK
funY ∈ *C*6*K*

proof
use YInKExpr;
use YExpr;
prove;
■

theorem RtInKExpr
exprDminusF funY ∈ *C*6*K*

proof
with enabled (*exprDminusFApp*) *prove*;
■

theorem rule RtInK
funRt ∈ *C*6*K*

proof
use RtInKExpr;
use RtExprK;
prove;
■

theorem SqInKExpr
opBox (*opPlusI* (*comp*(*opPlusI* (*opPlusI* (*cst* 1)), *funRt*))) ∈ *C*6*K*

proof
prove;
■

theorem rule SqInK
funSq ∈ *C*6*K*

proof
use SqInKExpr;
use SqExprK;
prove;
■

theorem EInKExpr
$exprIminusF\,(comp(funSq, funRt)) \in C6K$

proof
with enabled (*exprIminusFApp*) *prove*;
■

theorem rule EInK
$funE \in C6K$

proof
use EInKExpr;
use EExprK;
prove;
■

theorem Theorem68KSub1U
$C6 \in UClosed\,\{opPlusI, opBox\}$

proof
use C6InPower;
with enabled (*UClosedDef*, *C6UF*) *prove*;
split u = opPlusI;
cases;
prove;
next;
prove;
next;
■

theorem Theorem68KSub1B
$C6 \in BClosed\,\{comp\}$

proof
use C6InPower;
with enabled (*BClosedDef*, *C6UF*) *prove*;
prove;
■

theorem Theorem68KSub1
$C6K \subseteq C6$

proof
use ClosLeast[$I := \{ cst\,1, funK \}$, $U := \{ opPlusI, opBox \}$, $B := \{ comp \}$, $Z := C6$];
use C6InPower;
use Theorem68KSub1U;
use Theorem68KSub1B;
with enabled (*C6KDef*) *prove*;
■

theorem Theorem68KSub2U
$C6K \in UClosed\,\{opPlusI, opBox\}$

proof
use $C6KInPower$;
with enabled $(UClosedDef,\ C6KUF)$ *prove*;
split $u\ =\ opPlusI$;
cases;
prove;
next;
prove;
next;
■

theorem Theorem68KSub2B
$C6K \in BClosed\,\{comp\}$

proof
use $C6KInPower$;
with enabled $(BClosedDef,\ C6KUF)$ *prove*;
prove;
■

theorem Theorem68KSub2
$C6 \subseteq C6K$

proof
use $ClosLeast[I\ :=\ \{\ cst\,1,\ funE\ \},\ U\ :=\ \{\ opPlusI,\ opBox\ \},\ B\ :=\ \{\ comp\ \},\ Z\ :=\ C6K]$;
use $C6KInPower$;
use $Theorem68KSub2U$;
use $Theorem68KSub2B$;
with enabled $(C6Def)$ *prove*;
■

theorem Theorem68K
$C6K = C6$

proof
use $Theorem68KSub1$;
use $Theorem68KSub2$;
use $C6KInPower$;
use $C6InPower$;
use $SubAntisym[\mathbb{N}\ \to\ \mathbb{N}][A\ :=\ C6K,\ B\ :=\ C6]$;
prove;
■

6.3 The basis $\mathcal{F}'' = \langle \bar{1}, L, F^+, FG, F^\Box \rangle$

theorem LAt
$\forall\, m, t, k : \mathbb{N} \mid m = funA\,t + k \wedge k \leq t \bullet funL\,m = t - k$

proof
prove;
use $Diag[k\ :=\ k,\ t\ :=\ t]$;
prove;
■

$K = L\,\bar{1}^+((\bar{1}^+)^+L)^+$, i.e. $K(x) = L(x + 2L(x) + 2)$.

theorem KExprL
$comp(comp(funL, opPlusI\,(cst\,1)), opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)), funL))) = funK$

proof
apply UFExt to predicate $comp(comp(funL,\ opPlusI\,(cst\,1)),$
$opPlusI\,(comp(opPlusI\,(opPlusI\,(cst\,1)),\ funL)))\ =\ funK$;
prove;
use $Decomp[x\ :=\ x]$;
use $UFNonneg[f\ :=\ funV,\ x\ :=\ x]$;
use $UFNonneg[f\ :=\ funA,\ x\ :=\ funV\,x]$;
use $ASuc[x\ :=\ funV\,x]$;
use $LAt[m\ :=\ x,\ t\ :=\ funV\,x,\ k\ :=\ funK\,x]$;
prove;
use $UFNonneg[f\ :=\ funK,\ x\ :=\ x]$;
use $LAt[m\ :=\ 2\ +\ x\ -\ 2\ *\ funK\,x\ +\ 2\ *\ funV\,x,\ t\ :=\ 1\ +\ funV\,x,\ k\ :=\ funV\,x\ -\ funK\,x$
$+\ 1]$;
prove;
■

$C6L == clos\,\{cst\,1, funL\}\,\{opPlusI, opBox\}\,\{comp\}$

proof [C6L$domainCheck]
prove;
■

theorem disabled rule C6LDef
$C6L = clos\,\{cst\,1, funL\}\,\{opPlusI, opBox\}\,\{comp\}$

proof
invoke $C6L$;
prove;
■

theorem C6LInPower
$C6L \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
use $ClosInPow[I\ :=\ \{\ cst\,1,\ funL\ \},\ U\ :=\ \{\ opPlusI,\ opBox\ \},\ B\ :=\ \{\ comp\ \}]$;
with enabled $(C6LDef)$ *prove*;
■

theorem disabled rule C6LUF
$\forall\, F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in C6L \bullet F \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $ClosElemUF[I\ :=\ \{\ cst\,1,\ funL\ \},\ U\ :=\ \{\ opPlusI,\ opBox\ \},\ B\ :=\ \{\ comp\ \},\ F\ :=\ F]$;
with enabled $(C6LDef)$ *prove*;
■

theorem rule C6LInit0
$cst\,1 \in C6L$

proof
with enabled $(C6LDef)$ *prove*;
■

theorem rule C6LInit1
$funL \in C6L$

proof
with enabled (*C6LDef*) *prove*;
■

theorem rule C6LUnop0
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6L \bullet opPlusI\ F \in C6L$

proof
with enabled (*C6LDef*) *prove*;
■

theorem rule C6LUnop1
$\forall F : \mathbb{N} \to \mathbb{N} \mid F \in C6L \bullet opBox\ F \in C6L$

proof
with enabled (*C6LDef*) *prove*;
■

theorem rule C6LBinop0
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6L \land G \in C6L \bullet comp(F, G) \in C6L$

proof
with enabled (*C6LDef*) *prove*;
■

theorem KInLExpr
$comp(comp(funL, opPlusI\ (cst\ 1)), opPlusI\ (comp(opPlusI\ (opPlusI\ (cst\ 1)), funL))) \in C6L$

proof
prove;
■

theorem rule KInL
$funK \in C6L$

proof
use KInLExpr;
use KExprL;
prove;
■

theorem Theorem68LSub1U
$C6 \in UClosed\ \{opPlusI, opBox\}$

proof
use C6InPower;
with enabled (*UClosedDef*, *C6UF*) *prove*;
split u = opPlusI;
cases;
prove;
next;
prove;
next;
■

theorem Theorem68LSub1B
$C6 \in BClosed\,\{comp\}$

proof
use $C6InPower$;
with enabled $(BClosedDef,\ C6UF)$ *prove*;
prove;
■

theorem Theorem68LSub1
$C6L \subseteq C6$

proof
use $ClosLeast[I := \{\ cst\,1,\ funL\ \},\ U := \{\ opPlusI,\ opBox\ \},\ B := \{\ comp\ \},\ Z := C6]$;
use $C6InPower$;
use $Theorem68LSub1U$;
use $Theorem68LSub1B$;
with enabled $(C6LDef)$ *prove*;
■

theorem Theorem68LSubKU
$C6L \in UClosed\,\{opPlusI, opBox\}$

proof
use $C6LInPower$;
with enabled $(UClosedDef,\ C6LUF)$ *prove*;
split $u = opPlusI$;
cases;
prove;
next;
prove;
next;
■

theorem Theorem68LSubKB
$C6L \in BClosed\,\{comp\}$

proof
use $C6LInPower$;
with enabled $(BClosedDef,\ C6LUF)$ *prove*;
prove;
■

theorem Theorem68LSubK
$C6K \subseteq C6L$

proof
use $ClosLeast[I := \{\ cst\,1,\ funK\ \},\ U := \{\ opPlusI,\ opBox\ \},\ B := \{\ comp\ \},\ Z := C6L]$;
use $C6LInPower$;
use $Theorem68LSubKU$;
use $Theorem68LSubKB$;
with enabled $(C6KDef)$ *prove*;
■

theorem Theorem68LSub2
$C6 \subseteq C6L$

proof
use $Theorem68LSubK$;
use $Theorem68K$;
prove;
■

theorem Theorem68L
$C6L = C6$

proof
use $Theorem68LSub1$;
use $Theorem68LSub2$;
use $C6LInPower$;
use $C6InPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := C6L,\ B := C6]$;
prove;
■

theorem Theorem68
$C6K = C6 \wedge C6L = C6$

proof
use $Theorem68K$;
use $Theorem68L$;
prove;
■

7 Remark 6.9 and iteration from any $a \geq 1$

7.1 The product, multiples and iteration from a in $\operatorname{clos}\mathcal{F}$

$FG = \mathrm{Hf}((F+G)^2 - (F^2+G^2))$ for arbitrary F, G.

theorem TimesEq
$\forall F, G : \mathbb{N} \to \mathbb{N} \bullet comp(funHf, opDiffE(comp(funSq, opPlus(F, G)), opPlus(comp(funSq, F), comp(funSq, G)))) = opOtimes(F, G)$

proof
apply UFExt to predicate $comp(funHf,\ opDiffE(comp(funSq,\ opPlus(F,\ G)),\ opPlus(comp(funSq,\ F),\ comp(funSq,\ G)))) = opOtimes(F,\ G)$;
prove;
use $MulNonneg[a := F\,x,\ b := G\,x]$;
use $DiffEIsDifference[F := comp(funSq,\ opPlus(F,\ G)),\ G := opPlus(comp(funSq,\ F), comp(funSq,\ G)),\ x := x]$;
prove;
use $DivMod[x := 2 * (F\,x * G\,x),\ d := 2,\ q := F\,x * G\,x,\ r := 0]$;
prove;
■

theorem rule C6Times
$\forall F, G : \mathbb{N} \to \mathbb{N} \mid F \in C6 \wedge G \in C6 \bullet opOtimes(F, G) \in C6$

proof
use $TimesEq[F := F,\ G := G]$;
prove;
■

Multiples $M_r(x) = rx$, and $M_r = (\ldots(\bar{0}^+)^+\ldots)^+$ (r times) in every closure containing $\bar{0}$ and closed under F^+.

$$
\begin{array}{|l}
funM : \mathbb{N} \to (\mathbb{N} \to \mathbb{N}) \\
\hline
\langle\!\langle \text{rule MVal} \rangle\!\rangle \\
\forall\, r, x : \mathbb{N} \bullet funM\ r\ x = r * x
\end{array}
$$

proof [funM$domainCheck]
prove;
■

theorem rule MInner
$\forall\, r : \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet r * x) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet r * x) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

$MWit == (\lambda\, r : \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet r * x))$

theorem MWitFun
$MWit \in \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$

proof
invoke MWit;
apply FunIntro to predicate $(\lambda\ r : \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet r * x)) \in \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem MWitApp
$\forall\, r : \mathbb{N} \bullet MWit\ r = (\lambda\, x : \mathbb{N} \bullet r * x)$

proof
use MWitFun;
use pairInFunction$[\mathbb{N},\ \mathbb{N} \to \mathbb{N}][f := MWit,\ x := r,\ y := (\lambda\ x : \mathbb{N} \bullet r * x)]$;
invoke MWit;
prove by reduce;
■

theorem rule MWitVal
$\forall\, r, x : \mathbb{N} \bullet MWit\ r\ x = r * x$

proof
prove;
use MWitApp$[r := r]$;
use pairInFunction$[\mathbb{N},\ \mathbb{N}][f := (\lambda\ x : \mathbb{N} \bullet r * x),\ x := x,\ y := r * x]$;
prove;
■

theorem MEx
$\exists\, funM : \mathbb{N} \to (\mathbb{N} \to \mathbb{N}) \bullet \forall\, r, x : \mathbb{N} \bullet funM\ r\ x = r * x$

proof
instantiate funM == MWit;
use MWitFun;
prove;
■

theorem rule NatUFApp
$\forall\, g : \mathbb{N} \to (\mathbb{N} \to \mathbb{N});\ r : \mathbb{N} \bullet g\ r \in \mathbb{N} \to \mathbb{N}$

proof
prove;
■

theorem MZero
$funM\ 0 = cst\ 0$

proof
apply UFExt to predicate funM 0 = cst 0;
prove;
■

theorem MSuc
$\forall\, r : \mathbb{N} \bullet opPlusI\ (funM\ r) = funM\ (1 + r)$

proof
apply UFExt to predicate opPlusI (funM r) = funM (1 + r);
prove;
■

$MSet == \{r : \mathbb{N} \mid \forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \mid$
$cst\ 0 \in clos\ I\ U\ B \wedge opPlusI \in U \bullet funM\ r \in clos\ I\ U\ B\}$

proof [MSet$domainCheck]
prove;
■

theorem rule MBase
$0 \in MSet$

proof
invoke MSet;
use MZero;
prove;
■

theorem rule MStep
$\forall\, r : MSet \bullet 1 + r \in MSet$

proof
invoke MSet;
prove;
instantiate I__0 == I, U__0 == U, B__0 == B;
use MSuc[$r := r$];
prove;
equality substitute funM (1 + r);
rewrite;
■

theorem MSub
$\mathbb{N} \subseteq MSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq MSet$;
prove;
invoke MSet;
prove;
■

theorem MInClos
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ r : \mathbb{N} \mid cst\, 0 \in clos\ I\ U\ B \wedge opPlusI \in U \bullet funM\ r \in clos\ I\ U\ B$

proof
prove;
use MSub;
apply subsetDef to predicate $\mathbb{N} \subseteq MSet$;
apply inPower to predicate $\mathbb{N} \in \mathbb{P}\ MSet$;
prove;
instantiate $e == r$;
invoke MSet;
prove;
instantiate $I__1 == I,\ U__1 == U,\ B__1 == B$;
prove;
■

theorem rule MInC6
$\forall r : \mathbb{N} \bullet funM\ r \in C6$

proof
prove;
use MInClos$[I := \{ cst\, 1,\ funE \},\ U := \{ opPlusI,\ opBox \},\ B := \{ comp \},\ r := r]$;
use ZeroIn;
with enabled $(C6Def)$ *prove*;
■

Soundness for iteration from a: $F^{\Box(a)} = \hat{P}\, T^{\Box} S$ with $T(0) = a + 1$ and $T(y+1) = F(y) + 1$, namely $T = \mathrm{Sgn} \cdot (SF\hat{P}) + M_{a+1} O$.

theorem BoxATVal
$\forall a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ y : \mathbb{N} \bullet opPlus(opOtimes(funSgn, comp(comp(funS, F), exprPhat)), comp(funM\ (1 + a), funO))\ 0 = 1 + a \wedge opPlus(opOtimes(funSgn, comp(comp(funS, F), exprPhat)), comp(funM\ (1 + a), funO))\ (1 + y) = 1 + F\ y$

proof
prove;
■

$BoxASet == \{n : \mathbb{N} \mid \forall a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet opBox\ (opPlus(opOtimes(funSgn, comp(comp(funS, F), exprPhat)), comp(funM\ (1 + a), funO)))\ (1 + n) = 1 + opBoxA\ a\ F\ n\}$

proof [BoxASet$domainCheck]
prove;
■

theorem rule BoxABase
$0 \in BoxASet$

proof
invoke $BoxASet$;
prove;
■

theorem rule BoxAStep
$\forall\, n : BoxASet \bullet 1 + n \in BoxASet$

proof
invoke $BoxASet$;
with disabled ($BoxSuc$) *prove*;
instantiate $a__0 == a,\ F__0 == F$;
use $BoxSuc[F__0 := opPlus(opOtimes(funSgn,\ comp(comp(funS,\ F),\ exprPhat)),\ comp(funM\ (1 + a),\ funO)),\ x := 1 + n]$;
use $BoxATVal[a := a,\ F := F,\ y := opBoxA\ a\ F\ n]$;
with disabled ($BoxSuc,\ PlusVal$) *prove*;
■

theorem BoxASub
$\mathbb{N} \subseteq BoxASet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BoxASet$;
prove;
invoke $BoxASet$;
prove;
■

theorem BoxAExprEq
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet comp(comp(exprPhat, opBox\ (opPlus(opOtimes(funSgn, comp(comp(funS, F), exprPhat)), comp(funM\ (1 + a), funO)))), funS) = opBoxA\ a\ F$

proof
apply UFExt to predicate $comp(comp(exprPhat,\ opBox\ (opPlus(opOtimes(funSgn,\ comp(comp(funS,\ F),\ exprPhat)),\ comp(funM\ (1 + a),\ funO)))),\ funS) = opBoxA\ a\ F$;
with disabled ($BoxSuc$) *prove*;
use $BoxASub$;
apply subsetDef to predicate $\mathbb{N} \subseteq BoxASet$;
apply inPower to predicate $\mathbb{N} \in \mathbb{P}\ BoxASet$;
with disabled ($BoxSuc$) *prove*;
instantiate $e == x$;
invoke $BoxASet$;
with disabled ($BoxSuc$) *prove*;
instantiate $a__1 == a,\ F__1 == F$;
with disabled ($BoxSuc$) *prove*;
equality substitute $opBox\ (opPlus(opOtimes(funSgn,\ comp(comp(funS,\ F),\ exprPhat)),\ comp(funM\ (1 + a),\ funO)))\ (1 + x)$;
with disabled ($BoxSuc$) *rewrite*;
■

theorem rule BoxAIn
$\forall\, a : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet opBoxA\ a\ F \in C6$

proof
use $BoxAExprEq[a := a,\ F := F]$;
prove;
■

7.2 Powers under composition

F^k is the Toolkit's iterate $F^k =$ `iter k F`.

$$IterSSet == \{k : \mathbb{N} \mid \forall x : \mathbb{N} \bullet iter[\mathbb{N}]\, k\, funS\, x = x + k\}$$

proof [IterSSet$domainCheck]
prove;
■

theorem rule IterSBase
$0 \in IterSSet$

proof
invoke IterSSet;
prove;
■

theorem rule IterSStep
$\forall k : IterSSet \bullet 1 + k \in IterSSet$

proof
invoke IterSSet;
prove;
instantiate $x__0 == x$;
prove;
■

theorem IterSSub
$\mathbb{N} \subseteq IterSSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq IterSSet$;
prove;
invoke IterSSet;
prove;
■

theorem IterSAll
$\forall k : \mathbb{N} \bullet \forall x : \mathbb{N} \bullet iter[\mathbb{N}]\, k\, funS\, x = x + k$

proof
prove;
use IterSSub;
invoke IterSSet;
with enabled (*inPower*) *prove*;
instantiate $e == k$;
prove;
instantiate $x__1 == x$;
prove;
■

Predecessors ($P(x+1) = x$) undo successors: $P^k(x+k) = x$.

$$Preds == \{P : \mathbb{N} \to \mathbb{N} \mid \forall x : \mathbb{N} \bullet P\,(1 + x) = x\}$$

proof [Preds$domainCheck]
prove;
■

theorem rule PredsVal
$\forall P : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid P \in \mathit{Preds} \bullet P\,(1 + x) = x$

proof
prove;
invoke Preds;
prove;
instantiate $x__0 == x$;
prove;
■

$\mathit{IterPSet} == \{k : \mathbb{N} \mid \forall P : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid P \in \mathit{Preds} \bullet \mathit{iter}[\mathbb{N}]\,k\,P\,(x + k) = x\}$

proof [IterPSet$domainCheck]
prove;
■

theorem rule IterPBase
$0 \in \mathit{IterPSet}$

proof
invoke IterPSet;
prove;
■

theorem rule IterPStep
$\forall k : \mathit{IterPSet} \bullet 1 + k \in \mathit{IterPSet}$

proof
invoke IterPSet;
prove;
instantiate $P__0 == P,\ x__0 == 1 + x$;
use IterSuc$[F := P,\ n := k,\ a := 1 + x + k]$;
prove;
equality substitute $\mathit{iter}[\mathbb{N}]\,k\,P\,(1 + (k + x))$;
rewrite;
■

theorem IterPSub
$\mathbb{N} \subseteq \mathit{IterPSet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{IterPSet}$;
prove;
invoke IterPSet;
prove;
■

theorem IterPAll
$\forall k : \mathbb{N} \bullet \forall P : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \mid P \in \mathit{Preds} \bullet \mathit{iter}[\mathbb{N}]\,k\,P\,(x + k) = x$

proof
prove;
use IterPSub;
invoke IterPSet;
with enabled (*inPower*) *prove*;
instantiate $e == k$;
prove;
instantiate $P__1 == P,\ x__1 == x$;
prove;
■

$F^k\, G$ stays in every closure containing composition.

$PowSet == \{k : \mathbb{N} \mid \forall\, I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F, G : \mathbb{N} \to \mathbb{N} \mid comp \in B \wedge F \in clos\ I\ U\ B \wedge G \in clos\ I\ U\ B \bullet comp(iter[\mathbb{N}]\, k\, F, G) \in clos\ I\ U\ B\}$

proof [PowSet$domainCheck]
prove;
■

theorem IterCompZero
$\forall\, F, G : \mathbb{N} \to \mathbb{N} \bullet comp(iter[\mathbb{N}]\, 0\, F, G) = G$

proof
apply UFExt to predicate $comp(iter[\mathbb{N}]\, 0\, F,\ G) = G$;
prove;
■

theorem IterCompSuc
$\forall\, F, G : \mathbb{N} \to \mathbb{N};\ k : \mathbb{N} \bullet comp(iter[\mathbb{N}]\, (1 + k)\, F, G) = comp(F, comp(iter[\mathbb{N}]\, k\, F, G))$

proof
apply UFExt to predicate $comp(iter[\mathbb{N}]\, (1 + k)\, F,\ G) = comp(F,\ comp(iter[\mathbb{N}]\, k\, F,\ G))$;
prove;
■

theorem rule PowBase
$0 \in PowSet$

proof
invoke PowSet;
prove;
use IterCompZero$[F := F,\ G := G]$;
prove;
■

theorem rule PowStep
$\forall\, k : PowSet \bullet 1 + k \in PowSet$

proof
invoke PowSet;
prove;
instantiate $I__0 == I,\ U__0 == U,\ B__0 == B,\ F__0 == F,\ G__0 == G$;
use IterCompSuc$[F := F,\ G := G,\ k := k]$;
prove;
■

theorem PowSub
$\mathbb{N} \subseteq PowSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq PowSet$;
prove;
invoke PowSet;
prove;
■

theorem PowIn
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F,$
$G : \mathbb{N} \to \mathbb{N};\ k : \mathbb{N} \mid comp \in B \wedge F \in clos\ I\ U\ B \wedge G \in clos\ I\ U\ B \bullet comp(iter[\mathbb{N}]\ k\ F,$
$G) \in clos\ I\ U\ B$

proof
prove;
use PowSub;
apply subsetDef to predicate $\mathbb{N} \subseteq PowSet$;
apply inPower to predicate $\mathbb{N} \in \mathbb{P}\ PowSet$;
prove;
instantiate $e == k$;
invoke PowSet;
prove;
instantiate $I__1 == I,\ U__1 == U,\ B__1 == B,\ F__1 == F,\ G__1 == G$;
prove;
■

Recovering iteration from 0 out of iteration from a (as in Remark 5.10 of the paper): $F^{\square} = P^a(S^a F P^a)^{\square(a)}$ for a predecessor P.

$BoxFromASet == \{n : \mathbb{N} \mid \forall a : \mathbb{N};\ F, P : \mathbb{N} \to \mathbb{N} \mid P \in Preds \bullet opBoxA\ a\ (comp(comp(iter[\mathbb{N}]\ a\ funS, F),$
$iter[\mathbb{N}]\ a\ P))\ n = opBox\ F\ n + a\}$

proof [BoxFromASet$domainCheck]
prove;
■

theorem rule BoxFromABase
$0 \in BoxFromASet$

proof
invoke BoxFromASet;
prove;
■

theorem rule BoxFromAStep
$\forall n : BoxFromASet \bullet 1 + n \in BoxFromASet$

proof
invoke BoxFromASet;
prove;
instantiate $a__0 == a,\ F__0 == F,\ P__0 == P$;
use IterPAll$[k := a,\ P := P,\ x := opBox\ F\ n]$;
use IterSAll$[k := a,\ x := F\ (opBox\ F\ n)]$;
prove;
■

theorem BoxFromASub
$\mathbb{N} \subseteq \mathit{BoxFromASet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{BoxFromASet}$;
prove;
invoke BoxFromASet;
prove;
■

theorem BoxFromAAll
$\forall n : \mathbb{N} \bullet \forall a : \mathbb{N};\ F, P : \mathbb{N} \to \mathbb{N} \mid P \in \mathit{Preds} \bullet \mathit{opBoxA}\ a\ (\mathit{comp}(\mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ \mathit{funS}, F),$
$\mathit{iter}[\mathbb{N}]\ a\ P))\ n = \mathit{opBox}\ F\ n + a$

proof
prove;
use BoxFromASub;
invoke BoxFromASet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $a__1 == a,\ F__1 == F,\ P__1 == P$;
prove;
■

theorem BoxFromBoxA
$\forall a : \mathbb{N};\ F, P : \mathbb{N} \to \mathbb{N} \mid P \in \mathit{Preds} \bullet \mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ P, \mathit{opBoxA}\ a\ (\mathit{comp}(\mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ \mathit{funS}, F),$
$\mathit{iter}[\mathbb{N}]\ a\ P))) = \mathit{opBox}\ F$

proof
apply UFExt to predicate $\mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ P,\ \mathit{opBoxA}\ a\ (\mathit{comp}(\mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ \mathit{funS},\ F),\ \mathit{iter}[\mathbb{N}]\ a\ P)))$
$= \mathit{opBox}\ F$;
prove;
use $\mathit{BoxFromAAll}[n := x,\ a := a,\ F := F,\ P := P]$;
use $\mathit{IterPAll}[k := a,\ P := P,\ x := \mathit{opBox}\ F\ x]$;
prove;
■

7.3 The bases $\langle X, F^+, FG, F^{\square(a)} \rangle$: the common part

X admits multipliers if $X(0) = 0$ and for every $i \geq 1$ some r has $X(r(i+1)) = i$; then $T_j = XM_{r(j)}$ maps j to $j-1$ and 0 to 0.

$\mathit{Mult} == \{X : \mathbb{N} \to \mathbb{N} \mid X\ 0 = 0 \land (\forall i : \mathbb{N} \mid 1 \leq i \bullet \exists r : \mathbb{N} \bullet X\ (r * (1+i)) = i)\}$

proof [Mult$domainCheck]
prove;
■

theorem MultZero
$\forall X : \mathbb{N} \to \mathbb{N} \mid X \in \mathit{Mult} \bullet X\ 0 = 0$

proof
invoke Mult;
prove;
■

theorem MultEx
$\forall X : \mathbb{N} \to \mathbb{N};\ i : \mathbb{N} \mid X \in \mathit{Mult} \wedge 1 \leq i \bullet \exists r : \mathbb{N} \bullet X\,(r * (1 + i)) = i$

proof
prove;
invoke Mult;
prove;
instantiate $i__0 == i$;
prove;
instantiate $r__1 == r$;
prove;
■

The chain $N = T_2 \cdots T_{k+1}$ maps $k + 1$ to 1 and 0 to 0.

$\mathit{ChainSet} == \{k : \mathbb{N} \mid \forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid X \in \mathit{Mult} \wedge \mathit{cst}\,0 \in \mathit{clos}\,\{X\}\,\{\mathit{opPlusI},\ \mathit{opBoxA}\,a\}\,\{\mathit{comp}\} \bullet \exists W : \mathbb{N} \to \mathbb{N} \mid W \in \mathit{clos}\,\{X\}\,\{\mathit{opPlusI},\ \mathit{opBoxA}\,a\}\,\{\mathit{comp}\} \bullet W\,(1 + k) = 1 \wedge W\,0 = 0\}$

proof [ChainSet$domainCheck]
prove;
■

theorem rule ChainBase
$0 \in \mathit{ChainSet}$

proof
invoke ChainSet;
prove;
use MInClos[$I := \{ X \},\ U := \{ \mathit{opPlusI},\ \mathit{opBoxA}\,a \},\ B := \{ \mathit{comp} \},\ r := 1$];
prove;
instantiate $W == \mathit{funM}\,1$;
prove;
■

theorem rule ChainStep
$\forall k : \mathit{ChainSet} \bullet 1 + k \in \mathit{ChainSet}$

proof
invoke ChainSet;
prove;
instantiate $X__0 == X,\ a__0 == a$;
prove;
use MultEx[$X := X,\ i := 1 + k$];
use MultZero[$X := X$];
prove;
use MInClos[$I := \{ X \},\ U := \{ \mathit{opPlusI},\ \mathit{opBoxA}\,a \},\ B := \{ \mathit{comp} \},\ r := r$];
prove;
instantiate $W__1 == \mathit{comp}(W,\ \mathit{comp}(X,\ \mathit{funM}\,r))$;
prove;
■

theorem ChainSub
$\mathbb{N} \subseteq \mathit{ChainSet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{ChainSet}$;
prove;
invoke ChainSet;
prove;
■

theorem ChainAll
$\forall X : \mathbb{N} \to \mathbb{N};\ a, k : \mathbb{N} \mid X \in Mult \wedge cst\, 0 \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\, a\}\, \{comp\} \bullet \exists W : \mathbb{N} \to \mathbb{N} \mid W \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\, a\}\, \{comp\} \bullet W\, (1 + k) = 1 \wedge W\, 0 = 0$

proof
prove;
use ChainSub;
apply subsetDef to predicate $\mathbb{N} \subseteq ChainSet$;
apply inPower to predicate $\mathbb{N} \in \mathbb{P}\ ChainSet$;
prove;
instantiate $e == k$;
invoke ChainSet;
prove;
instantiate $X__1 == X,\ a__1 == a$;
prove;
instantiate $W__2 == W$;
prove;
■

theorem MOne
$funM\, 1 = funI \wedge funM\, 2 = funD$

proof
use $UFExt[F := funM\, 1,\ G := funI]$;
use $UFExt[F := funM\, 2,\ G := funD]$;
prove;
■

$BoxAISet == \{n : \mathbb{N} \mid \forall a : \mathbb{N} \bullet opBoxA\, a\, funI\, n = a\}$

proof [BoxAISet$domainCheck]
prove;
■

theorem rule BoxAIBase
$0 \in BoxAISet$

proof
invoke BoxAISet;
prove;
■

theorem rule BoxAIStep
$\forall n : BoxAISet \bullet 1 + n \in BoxAISet$

proof
invoke BoxAISet;
prove;
■

theorem BoxAISub
$\mathbb{N} \subseteq BoxAISet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BoxAISet$;
prove;
invoke BoxAISet;
prove;
■

theorem BoxAIAll
$\forall n : \mathbb{N} \bullet \forall a : \mathbb{N} \bullet opBoxA\ a\ funI\ n = a$

proof
prove;
use BoxAISub;
invoke BoxAISet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $a__1 == a$;
prove;
■

theorem OneFromW
$\forall W : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid W\ a = 1 \bullet comp(W, opBoxA\ a\ funI) = cst\ 1$

proof
apply UFExt to predicate $comp(W,\ opBoxA\ a\ funI) = cst\ 1$;
prove;
use $BoxAIAll[n := x]$;
prove;
■

theorem OFromW
$\forall W : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid W\ a = 1 \land W\ 0 = 0 \bullet comp(W, opBoxA\ a\ (cst\ 0)) = funO$

proof
apply UFExt to predicate $comp(W,\ opBoxA\ a\ (cst\ 0)) = funO$;
prove;
split $x = 0$;
cases;
prove;
next;
use $BoxASuc[a__0 := a,\ F__0 := cst\ 0,\ x := x - 1]$;
prove;
next;
■

theorem ExtAI
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid X \in Mult \land cst\ 0 \in clos\ \{X\}\ \{opPlusI,$
$opBoxA\ a\}\ \{comp\} \bullet funI \in clos\ \{X\}\ \{opPlusI, opBoxA\ a\}\ \{comp\} \land funD \in clos\ \{X\}\ \{opPlusI,$
$opBoxA\ a\}\ \{comp\}$

proof
prove;
use $MInClos[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ r := 1]$;
use $MInClos[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ r := 2]$;
use MOne;
prove;
■

theorem OneInCB
$\forall X, W : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid W \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge funI \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge W\ a = 1 \bullet cst\ 1 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
use $OneFromW[W := W,\ a := a]$;
prove;
■

theorem OExprInCB
$\forall X, W : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid W \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge cst\ 0 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \bullet comp(W, opBoxA\ a\ (cst\ 0)) \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
■

theorem OInCB
$\forall X, W : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid W \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge cst\ 0 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge W\ a = 1 \wedge W\ 0 = 0 \bullet funO \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
use $OExprInCB[X := X,\ W := W,\ a := a]$;
use $OFromW[W := W,\ a := a]$;
prove;
■

theorem ExtAOne
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid 1 \leq a \wedge X \in Mult \wedge cst\ 0 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \bullet cst\ 1 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge funO \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $ExtAI[X := X,\ a := a]$;
use $ChainAll[X := X,\ a := a,\ k := a - 1]$;
prove;
use $OneInCB[X := X,\ W := W,\ a := a]$;
use $OInCB[X := X,\ W := W,\ a := a]$;
prove;
■

theorem SExprInCB
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid cst\ 1 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \bullet opPlusI\ (cst\ 1) \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
■

theorem ExtAS
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid 1 \leq a \wedge X \in Mult \wedge cst\ 0 \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \bullet funS \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use ExtAOne[$X := X,\ a := a$];
prove;
use SExprInCB[$X := X,\ a := a$];
use SExpr;
prove;
■

theorem IterCompI
$\forall F : \mathbb{N} \to \mathbb{N};\ k : \mathbb{N} \bullet comp(iter[\mathbb{N}]\, k\, F, funI) = iter[\mathbb{N}]\, k\, F$

proof
apply UFExt to predicate $comp(iter[\mathbb{N}]\, k\, F,\ funI) = iter[\mathbb{N}]\, k\, F$;
prove;
■

theorem IterIn
$\forall I : \mathbb{P}(\mathbb{N} \to \mathbb{N});\ U : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ B : \mathbb{P}((\mathbb{N} \to \mathbb{N}) \times (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}));\ F : \mathbb{N} \to \mathbb{N};\ k : \mathbb{N} \mid comp \in B \wedge F \in clos\, I\ U\ B \wedge funI \in clos\, I\ U\ B \bullet iter[\mathbb{N}]\, k\, F \in clos\, I\ U\ B$

proof
prove;
use PowIn[$I := I,\ U := U,\ B := B,\ F := F,\ G := funI,\ k := k$];
use IterCompI[$F := F,\ k := k$];
prove;
■

With a predecessor in the closure, iteration from 0 is available.

theorem ExtABox
$\forall X, P, F : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid 1 \leq a \wedge X \in Mult \wedge cst\, 0 \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\ a\}\, \{comp\} \wedge P \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\ a\}\, \{comp\} \wedge P \in Preds \wedge F \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\ a\}\, \{comp\} \bullet opBox\, F \in clos\, \{X\}\, \{opPlusI, opBoxA\ a\}\, \{comp\}$

proof
prove;
use ExtAI[$X := X,\ a := a$];
use ExtAS[$X := X,\ a := a$];
use IterIn[$I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := P,\ k := a$];
use IterIn[$I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := funS,\ k := a$];
use BoxFromBoxA[$a := a,\ F := F,\ P := P$];
prove;
■

7.4 Simulation

theorem CBUClosed
$\forall X, P : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid 1 \leq a \wedge X \in Mult \wedge cst\, 0 \in clos\, \{X\}\, \{opPlusI,$
$opBoxA\ a\}\, \{comp\} \wedge P \in clos\, \{X\}\, \{opPlusI, opBoxA\ a\}\, \{comp\} \wedge P \in Preds \bullet clos\, \{X\}\, \{opPlusI,$
$opBoxA\ a\}\, \{comp\} \in UClosed\, \{opPlusI, opBox\}$

proof
prove;
use ClosInPow[$I := \{ X \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$];
with enabled (*UClosedDef*) *prove*;
use ClosElemUF[$I := \{ X \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$, $F := F$];
split $u = opPlusI$;
cases;
prove;
next;
use ExtABox[$X := X$, $P := P$, $F := F$, $a := a$];
prove;
next;
■

theorem CBBClosed
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \bullet clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \in BClosed\,\{comp\}$

proof
prove;
use ClosInPow[$I := \{ X \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$];
with enabled (*BClosedDef*) *prove*;
■

theorem ExtASup
$\forall X, P : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid 1 \leq a \wedge X \in Mult \wedge cst\,0 \in clos\,\{X\}\,\{opPlusI,$
$opBoxA\ a\}\,\{comp\} \wedge P \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \wedge P \in Preds \bullet clos\,\{cst\,1,$
$X\}\,\{opPlusI, opBox\}\,\{comp\} \subseteq clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use ClosLeast[$I := \{ cst\,1,\ X \}$, $U := \{ opPlusI,\ opBox \}$, $B := \{ comp \}$, $Z := clos\,\{ X \}\,\{$
$opPlusI,\ opBoxA\ a \}\,\{ comp \}$];
use ClosInPow[$I := \{ X \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$];
use CBUClosed[$X := X$, $P := P$, $a := a$];
use CBBClosed[$X := X$, $a := a$];
use ExtAOne[$X := X$, $a := a$];
prove;
■

theorem C6UClosedA
$\forall a : \mathbb{N} \bullet C6 \in UClosed\,\{opPlusI, opBoxA\ a\}$

proof
prove;
use C6InPower;
with enabled (*UClosedDef*, *C6UF*) *prove*;
use InPair[$\mathbb{P}\,(\mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}) \times \mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}))$][$u := u$, $p := opPlusI$, $q := opBoxA\ a$];
split $u = opBoxA\ a$;
cases;
prove;
next;
prove;
next;
■

theorem C6BClosedComp
$C6 \in BClosed\,\{comp\}$

proof
use $C6InPower$;
with enabled $(BClosedDef,\ C6UF)$ *prove*;
■

theorem ExtASub6
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid X \in C6 \bullet clos\,\{X\}\,\{opPlusI, opBoxA\,a\}\,\{comp\} \subseteq C6$

proof
prove;
use $ClosLeast[I := \{ X \},\ U := \{ opPlusI,\ opBoxA\,a \},\ B := \{ comp \},\ Z := C6]$;
use $C6InPower$;
use $C6UClosedA[a := a]$;
use $C6BClosedComp$;
prove;
■

7.5 Zero for a decreasing X: $\bar{0} = X^{a} X^{\square(a)}$

$Decr == \{X : \mathbb{N} \to \mathbb{N} \mid X\,0 = 0 \land (\forall y : \mathbb{N} \mid 0 < y \bullet X\,y < y)\}$

proof [Decr$domainCheck]
prove;
■

theorem DecrZero
$\forall X : \mathbb{N} \to \mathbb{N} \mid X \in Decr \bullet X\,0 = 0$

proof
invoke $Decr$;
prove;
■

theorem DecrLt
$\forall X : \mathbb{N} \to \mathbb{N};\ y : \mathbb{N} \mid X \in Decr \land 0 < y \bullet X\,y < y \land X\,0 = 0$

proof
prove;
invoke $Decr$;
prove;
instantiate $y_0 == y$;
prove;
■

$DecBoundSet == \{n : \mathbb{N} \mid \forall a : \mathbb{N};\ X : \mathbb{N} \to \mathbb{N} \mid X \in Decr \bullet opBoxA\,a\,X\,n \leq a\}$

proof [DecBoundSet$domainCheck]
prove;
■

theorem rule DecBoundBase
$0 \in DecBoundSet$

proof
invoke $DecBoundSet$;
prove;
■

theorem rule DecBoundStep
$\forall\, n : \mathit{DecBoundSet} \bullet 1 + n \in \mathit{DecBoundSet}$

proof
invoke *DecBoundSet*;
prove;
instantiate $a__0 == a,\ X__0 == X$;
use $\mathit{DecrZero}[X := X]$;
use $\mathit{DecrLt}[X := X,\ y := \mathit{opBoxA}\ a\ X\ n]$;
use $\mathit{UFNonneg}[f := X,\ x := \mathit{opBoxA}\ a\ X\ n]$;
use $\mathit{UFNonneg}[f := \mathit{opBoxA}\ a\ X,\ x := n]$;
split $\mathit{opBoxA}\ a\ X\ n = 0$;
cases;
prove;
next;
prove;
next;
■

theorem DecBoundSub
$\mathbb{N} \subseteq \mathit{DecBoundSet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{DecBoundSet}$;
prove;
invoke *DecBoundSet*;
prove;
■

theorem DecBoundAll
$\forall\, n : \mathbb{N} \bullet \forall\, a : \mathbb{N};\ X : \mathbb{N} \to \mathbb{N} \mid X \in \mathit{Decr} \bullet \mathit{opBoxA}\ a\ X\ n \leq a$

proof
prove;
use *DecBoundSub*;
invoke *DecBoundSet*;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $a__1 == a,\ X__1 == X$;
prove;
■

$\mathit{DecIterSet} == \{k : \mathbb{N} \mid \forall\, X : \mathbb{N} \to \mathbb{N};\ y : \mathbb{N} \mid X \in \mathit{Decr} \land y \leq k \bullet \mathit{iter}[\mathbb{N}]\ k\ X\ y = 0\}$

proof [DecIterSet$domainCheck]
prove;
■

theorem rule DecIterBase
$0 \in \mathit{DecIterSet}$

proof
invoke *DecIterSet*;
prove;
■

theorem rule DecIterStep
$\forall\, k : \mathit{DecIterSet} \bullet 1 + k \in \mathit{DecIterSet}$

proof
invoke DecIterSet;
prove;
instantiate $X__0 == X,\ y__0 == X\ y$;
use IterSucR$[F := X,\ k := k,\ a := y]$;
use DecrZero$[X := X]$;
use DecrLt$[X := X,\ y := y]$;
use UFNonneg$[f := X,\ x := y]$;
split $y = 0$;
cases;
prove;
next;
prove;
next;
■

theorem DecIterSub
$\mathbb{N} \subseteq \mathit{DecIterSet}$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{DecIterSet}$;
prove;
invoke DecIterSet;
prove;
■

theorem DecIterAll
$\forall\, k : \mathbb{N} \bullet \forall\, X : \mathbb{N} \to \mathbb{N};\ y : \mathbb{N} \mid X \in \mathit{Decr} \land y \leq k \bullet \mathit{iter}[\mathbb{N}]\ k\ X\ y = 0$

proof
prove;
use DecIterSub;
invoke DecIterSet;
with enabled (*inPower*) *prove*;
instantiate $e == k$;
prove;
instantiate $X__1 == X,\ y__1 == y$;
prove;
■

theorem DecZeroEq
$\forall\, X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid X \in \mathit{Decr} \bullet \mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ X, \mathit{opBoxA}\ a\ X) = \mathit{cst}\ 0$

proof
apply UFExt to predicate $\mathit{comp}(\mathit{iter}[\mathbb{N}]\ a\ X,\ \mathit{opBoxA}\ a\ X) = \mathit{cst}\ 0$;
prove;
use DecBoundAll$[n := x,\ a := a,\ X := X]$;
use DecIterAll$[k := a,\ X := X,\ y := \mathit{opBoxA}\ a\ X\ x]$;
prove;
■

theorem ZeroInCB
$\forall\, X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \mid X \in \mathit{Decr} \bullet \mathit{cst}\ 0 \in \mathit{clos}\ \{X\}\ \{\mathit{opPlusI}, \mathit{opBoxA}\ a\}\ \{\mathit{comp}\}$

proof
prove;
use PowIn[$I := \{ X \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$, $F := X$, $G := opBoxA\ a\ X$, $k := a$];
use DecZeroEq[$X := X$, $a := a$];
prove;
■

7.6 The case $X = E$

theorem ELt
$\forall\, y : \mathbb{N} \mid 0 < y \bullet funE\ y < y$

proof
prove;
use RtSpec[$x := y$];
use RtPos[$y := y$];
use timesMonotonic2[$a := 1$, $b := 1$, $c := funRt\ y$, $d := funRt\ y$];
with enabled (*EVal*) *prove*;
■

theorem EDecr
$funE \in Decr$

proof
invoke Decr;
prove;
use ELt[$y := y$];
prove;
■

theorem EMult
$funE \in Mult$

proof
invoke Mult;
prove;
instantiate $r == i$;
use ESqAdd[$m := i$, $k := i$];
prove;
■

theorem rule IterSVal
$\forall\, k, x : \mathbb{N} \bullet iter[\mathbb{N}]\ k\ funS\ x = x + k$

proof
prove;
use IterSAll[$k := k$, $x := x$];
prove;
■

theorem RRec
$\forall\, t : \mathbb{N} \bullet funR\ (1 + t) = 2 * funQ\ (4 + funR\ t) + funR\ t + 1$

proof
prove;
use RStep[$n := t$];
prove;
■

With $c = a^2 + a + 4$: $B = (S(DQS^c)^+)^{\Box(a)}$ satisfies $B(n) = R(n + a^2) - a^2 - a$, $G = (S^{a+1}B)^{\Box(a)}$ satisfies $G(n) = n^2 + 2an + a$, and $\hat{P} = E\, S^{a^2-a-1} G^+$ is a predecessor.

$BEValSet == \{n : \mathbb{N} \mid \forall\, a : \mathbb{N} \bullet opBoxA\ a\ (comp(funS, opPlusI\ (comp(comp(funD, funQ),$
$iter[\mathbb{N}]\ (a * a + a + 4)\ funS)))\ n = funR\ (n + a * a) - a * a - a\}$

proof [BEValSet$domainCheck]
prove;
■

theorem rule BEValBase
$0 \in BEValSet$

proof
invoke BEValSet;
prove;
■

theorem rule BEValStep
$\forall\, n : BEValSet \bullet 1 + n \in BEValSet$

proof
invoke BEValSet;
with disabled (*QVal*, *RVal*) *prove*;
instantiate a__0 == a;
use RRec[$t := n + a * a$];
with disabled (*QVal*, *RVal*) *prove*;
■

theorem BEValSub
$\mathbb{N} \subseteq BEValSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BEValSet$;
prove;
invoke BEValSet;
prove;
■

theorem BEValAll
$\forall\, n : \mathbb{N} \bullet \forall\, a : \mathbb{N} \bullet opBoxA\ a\ (comp(funS, opPlusI\ (comp(comp(funD, funQ),$
$iter[\mathbb{N}]\ (a * a + a + 4)\ funS)))\ n = funR\ (n + a * a) - a * a - a$

proof
prove;
use BEValSub;
invoke BEValSet;
with enabled (*inPower*) *prove*;
instantiate e == n;
prove;
instantiate a__1 == a;
prove;
■

$GEValSet == \{n : \mathbb{N} \mid \forall\, a : \mathbb{N} \bullet opBoxA\ a\ (comp(iter[\mathbb{N}]\ (1 + a)\ funS, opBoxA\ a\ (comp(funS,$
$opPlusI\ (comp(comp(funD, funQ), iter[\mathbb{N}]\ (a * a + a + 4)\ funS))))))\ n = n * n + 2 * a * n + a\}$

proof [GEValSet$domainCheck]
prove;
■

theorem rule GEValBase
$0 \in GEValSet$

proof
invoke GEValSet;
prove;
■

theorem rule GEValStep
$\forall\, n : GEValSet \bullet 1 + n \in GEValSet$

proof
invoke GEValSet;
prove;
instantiate a__0 == a;
use BEValAll[$n := n * n + 2 * a * n + a,\ a := a$];
use RtSqAdd[$m := n + a,\ k := a$];
prove;
■

theorem GEValSub
$\mathbb{N} \subseteq GEValSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq GEValSet$;
prove;
invoke GEValSet;
prove;
■

theorem GEValAll
$\forall\, n : \mathbb{N} \bullet \forall\, a : \mathbb{N} \bullet opBoxA\ a\ (comp(iter[\mathbb{N}]\ (1 + a)\ funS, opBoxA\ a\ (comp(funS,$
$opPlusI\ (comp(comp(funD, funQ), iter[\mathbb{N}]\ (a * a + a + 4)\ funS))))))\ n = n * n + 2 * a * n + a$

proof
prove;
use GEValSub;
invoke GEValSet;
with enabled (*inPower*) *prove*;
instantiate e == n;
prove;
instantiate a__1 == a;
prove;
■

theorem PEVal
$\forall\, a, k : \mathbb{N} \mid 2 \leq a \bullet comp(comp(funE, iter[\mathbb{N}]\ (a * a - a - 1)\ funS),$
$opPlusI\ (opBoxA\ a\ (comp(iter[\mathbb{N}]\ (1 + a)\ funS, opBoxA\ a\ (comp(funS, opPlusI\ (comp(comp(funD,$
$funQ), iter[\mathbb{N}]\ (a * a + a + 4)\ funS)))))))))\ (1 + k) = k$

proof
prove;
use $timesMonotonic2[a := 2,\ b := a,\ c := a,\ d := a]$;
use $GEValAll[n := 1 + k,\ a := a]$;
use $ESqAdd[m := 1 + k + a,\ k := k]$;
prove;
■

theorem CBComp
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N};\ F, G : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{X\}\,\{opPlusI,$
$opBoxA\ a\}\,\{comp\} \wedge G \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \bullet comp(F,$
$G) \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $ClosElemUF[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := F]$;
use $ClosElemUF[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := G]$;
prove;
■

theorem CBPlusI
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N};\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{X\}\,\{opPlusI,$
$opBoxA\ a\}\,\{comp\} \bullet opPlusI\ F \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $ClosElemUF[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := F]$;
prove;
■

theorem CBBoxA
$\forall X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N};\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{X\}\,\{opPlusI,$
$opBoxA\ a\}\,\{comp\} \bullet opBoxA\ a\ F \in clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $ClosElemUF[I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ F := F]$;
prove;
■

theorem QExprInCE
$\forall a : \mathbb{N} \mid 1 \leq a \bullet comp(funO, funE) \in clos\,\{funE\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $ZeroInCB[X := funE,\ a := a]$;
use $EDecr$;
use $EMult$;
use $ExtAOne[X := funE,\ a := a]$;
prove;
■

theorem QInCE
$\forall a : \mathbb{N} \mid 1 \leq a \bullet funQ \in clos\,\{funE\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use $QExprInCE[a := a]$;
use $QExpr$;
prove;
■

theorem PEInCE
$\forall\, a : \mathbb{N} \mid 2 \leq a \bullet comp(comp(funE, iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\,a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS, opBoxA\,a\,(comp(funS, opPlusI\,(comp(comp(funD,$
$funQ), iter[\mathbb{N}]\,(a * a + a + 4)\,funS)))))))) \in clos\,\{funE\}\,\{opPlusI, opBoxA\,a\}\,\{comp\}$

proof
prove;
use $timesMonotonic2[a := 2,\ b := a,\ c := a,\ d := a]$;
use $ZeroInCB[X := funE,\ a := a]$;
use $EDecr$;
use $EMult$;
use $ExtAI[X := funE,\ a := a]$;
use $ExtAS[X := funE,\ a := a]$;
use $QInCE[a := a]$;
prove;
use $IterIn[I := \{\ funE\ \},\ U := \{\ opPlusI,\ opBoxA\,a\ \},\ B := \{\ comp\ \},\ F := funS,\ k := a * a + a + 4]$;
use $IterIn[I := \{\ funE\ \},\ U := \{\ opPlusI,\ opBoxA\,a\ \},\ B := \{\ comp\ \},\ F := funS,\ k := 1 + a]$;
use $IterIn[I := \{\ funE\ \},\ U := \{\ opPlusI,\ opBoxA\,a\ \},\ B := \{\ comp\ \},\ F := funS,\ k := a * a - a - 1]$;
use $CBComp[X := funE,\ a := a,\ F := funD,\ G := funQ]$;
prove;
use $CBComp[X := funE,\ a := a,\ F := comp(funD,\ funQ),\ G := iter[\mathbb{N}]\,(a * a + a + 4)\,funS]$;
prove;
use $CBPlusI[X := funE,\ a := a,\ F := comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,(a * a + a + 4)\,funS)]$;
prove;
■

theorem PEUF
$\forall\, a : \mathbb{N} \mid 2 \leq a \bullet comp(comp(funE, iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\,a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS, opBoxA\,a\,(comp(funS, opPlusI\,(comp(comp(funD,$
$funQ), iter[\mathbb{N}]\,(a * a + a + 4)\,funS)))))))) \in \mathbb{N} \to \mathbb{N}$

proof
prove;
use $PEInCE[a := a]$;
use $ClosInPow[I := \{\ funE\ \},\ U := \{\ opPlusI,\ opBoxA\,a\ \},\ B := \{\ comp\ \}]$;
apply inPower to predicate $clos\,\{\ funE\ \}\,\{\ opPlusI,\ opBoxA\,a\ \}\,\{\ comp\ \} \in \mathbb{P}\,(\mathbb{N} \to \mathbb{N})$;
instantiate e $== comp(comp(funE,\ iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\,a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS,\ opBoxA\,a\,(comp(funS,$
$opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,(a * a + a + 4)\,funS)))))))$;
prove;
■

theorem PEPred
$\forall\, a : \mathbb{N} \mid 2 \leq a \bullet comp(comp(funE, iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\,a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS, opBoxA\,a\,(comp(funS, opPlusI\,(comp(comp(funD,$
$funQ), iter[\mathbb{N}]\,(a * a + a + 4)\,funS)))))))) \in Preds$

proof
prove;
use PEInCE[$a := a$];
use ClosInPow[$I := \{ funE \}$, $U := \{ opPlusI,\ opBoxA\ a \}$, $B := \{ comp \}$];
apply inPower to predicate clos $\{ funE \} \{ opPlusI,\ opBoxA\ a \} \{ comp \} \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$;
instantiate e $==$ $comp(comp(funE,\ iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\ a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS,\ opBoxA\ a\,(comp(funS,$
$opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,(a * a + a + 4)\,funS))))))))$;
prove;
invoke Preds;
prove;
use PEVal[$a := a,\ k := x$];
prove;
■

theorem ExtAE
$\forall\, a : \mathbb{N} \mid 2 \le a \bullet clos\,\{funE\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} = C6$

proof
prove;
use EDecr;
use EMult;
use ZeroInCB[$X := funE,\ a := a$];
use PEInCE[$a := a$];
use PEUF[$a := a$];
use PEPred[$a := a$];
use ExtASup[$X := funE,\ P := comp(comp(funE,\ iter[\mathbb{N}]\,(a * a - a - 1)\,funS),$
$opPlusI\,(opBoxA\ a\,(comp(iter[\mathbb{N}]\,(1 + a)\,funS,\ opBoxA\ a\,(comp(funS,$
$opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,(a * a + a + 4)\,funS)))))))),\ a := a$];
*use ExtASub*6[$X := funE,\ a := a$];
use C6Def;
use ClosInPow[$I := \{ funE \},\ U := \{ opPlusI,\ opBoxA\ a \},\ B := \{ comp \}$];
use C6InPower;
use SubAntisym[$\mathbb{N} \to \mathbb{N}$][$A := clos\,\{ funE \}\,\{ opPlusI,\ opBoxA\ a \}\,\{ comp \},\ B := C6$];
prove;
■

7.7 The case $X = K$

theorem KLt
$\forall\, y : \mathbb{N} \mid 0 < y \bullet funK\ y < y$

proof
prove;
use Decomp[$x := y$];
use UFNonneg[$f := funV,\ x := y$];
use UFNonneg[$f := funK,\ x := y$];
prove;
split $funV\ y = 0$;
cases;
use AZero;
prove;
next;
use AMono[$m := 1,\ n := funV\ y$];
use AZero;
prove;
next;
■

theorem KZero
$funK\ 0 = 0$

proof
use $KAt[m := 0,\ t := 0,\ k := 0]$;
use *AZero*;
prove;
■

theorem KDecr
$funK \in Decr$

proof
invoke *Decr*;
use *KZero*;
prove;
use $KLt[y := y]$;
prove;
■

$K(2i(i+1)) = i$, since $2i(i+1) = A(2i) + i$.

theorem KMult
$funK \in Mult$

proof
invoke *Mult*;
use *KZero*;
prove;
instantiate $r == 2 * i$;
use $TwoA[x := 2 * i]$;
use $KAt[m := 2 * i * (1 + i),\ t := 2 * i,\ k := i]$;
prove;
■

With $c = A(a)$: $B = (S^2(KS^c)^+)^{\Box(a)}$ satisfies $B(n) + A(a) + 1 = A(n + a + 1)$, and $\hat{P} = KS^cB^+$ is a predecessor.

$BKValSet == \{n : \mathbb{N} \mid \forall\, a : \mathbb{N} \bullet (opBoxA\ a\ (comp(iter[\mathbb{N}]\ 2\ funS, opPlusI\ (comp(funK,$
$iter[\mathbb{N}]\ (funA\ a)\ funS))))\ n + funA\ a + 1 = funA\ (n + a + 1)\}$

proof [BKValSet$domainCheck]
prove;
■

theorem rule BKValBase
$0 \in BKValSet$

proof
invoke *BKValSet*;
prove;
use $ASuc[x := a]$;
prove;
■

theorem rule BKValStep
$\forall\, n : BKValSet \bullet 1 + n \in BKValSet$

proof
invoke BKValSet;
prove;
instantiate $a_0 == a$;
use $ASuc[x := n + a]$;
use $ASuc[x := 1 + n + a]$;
use $KAt[m := (opBoxA\, a\, (comp(iter[\mathbb{N}]\, 2\, funS,\ opPlusI\, (comp(funK,\ iter[\mathbb{N}]\, (funA\, a)\, funS))))\, n +$
$funA\, a,\ t := n + a,\ k := n + a]$;
prove;
■

theorem BKValSub
$\mathbb{N} \subseteq BKValSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq BKValSet$;
prove;
invoke BKValSet;
prove;
■

theorem BKValAll
$\forall\, n, a : \mathbb{N} \bullet (opBoxA\, a\, (comp(iter[\mathbb{N}]\, 2\, funS, opPlusI\, (comp(funK,$
$iter[\mathbb{N}]\, (funA\, a)\, funS))))\, n + funA\, a + 1 = funA\, (n + a + 1)$

proof
prove;
use BKValSub;
invoke BKValSet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $a_1 == a$;
prove;
■

theorem PKVal
$\forall\, a, k : \mathbb{N} \bullet comp(comp(funK, iter[\mathbb{N}]\, (funA\, a)\, funS), opPlusI\, (opBoxA\, a\, (comp(iter[\mathbb{N}]\, 2\, funS,$
$opPlusI\, (comp(funK, iter[\mathbb{N}]\, (funA\, a)\, funS))))))\, (1 + k) = k$

proof
prove;
use $BKValAll[n := 1 + k,\ a := a]$;
use $KAt[m := (opBoxA\, a\, (comp(iter[\mathbb{N}]\, 2\, funS,\ opPlusI\, (comp(funK,\ iter[\mathbb{N}]\, (funA\, a)\, funS))))\, (1$
$+ k) + (1 + k) + funA\, a,\ t := k + a + 2,\ k := k]$;
prove;
■

theorem PKInCK
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet comp(comp(funK, iter[\mathbb{N}]\, (funA\, a)\, funS), opPlusI\, (opBoxA\, a\, (comp(iter[\mathbb{N}]\, 2\, funS,$
$opPlusI\, (comp(funK, iter[\mathbb{N}]\, (funA\, a)\, funS))))) \in clos\, \{funK\}\, \{opPlusI, opBoxA\, a\}\, \{comp\}$

proof
prove;
use *ZeroInCB*[$X := funK,\ a := a$];
use *KDecr*;
use *KMult*;
use *ExtAI*[$X := funK,\ a := a$];
use *ExtAS*[$X := funK,\ a := a$];
prove;
use *IterIn*[$I := \{ funK \},\ U := \{ opPlusI,\ opBoxA\ a \},\ B := \{ comp \},\ F := funS,\ k :=$ $funA\ a$];
use *IterIn*[$I := \{ funK \},\ U := \{ opPlusI,\ opBoxA\ a \},\ B := \{ comp \},\ F := funS,\ k := 2$];
prove;
■

theorem PKUF
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet comp(comp(funK, iter[\mathbb{N}]\,(funA\ a)\,funS), opPlusI\ (opBoxA\ a\ (comp(iter[\mathbb{N}]\,2\,funS,$ $opPlusI\ (comp(funK, iter[\mathbb{N}]\,(funA\ a)\,funS))))) \in \mathbb{N} \to \mathbb{N}$

proof
prove;
■

theorem PKPred
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet comp(comp(funK, iter[\mathbb{N}]\,(funA\ a)\,funS), opPlusI\ (opBoxA\ a\ (comp(iter[\mathbb{N}]\,2\,funS,$ $opPlusI\ (comp(funK, iter[\mathbb{N}]\,(funA\ a)\,funS))))) \in Preds$

proof
prove;
use *PKUF*[$a := a$];
invoke *Preds*;
prove;
use *PKVal*[$a := a,\ k := x$];
prove;
■

theorem ExtAK
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet clos\,\{funK\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} = C6$

proof
prove;
use *KDecr*;
use *KMult*;
use *ZeroInCB*[$X := funK,\ a := a$];
use *PKInCK*[$a := a$];
use *PKUF*[$a := a$];
use *PKPred*[$a := a$];
use *ExtASup*[$X := funK,\ P := comp(comp(funK,\ iter[\mathbb{N}]\,(funA\ a)\,funS),$ $opPlusI\ (opBoxA\ a\ (comp(iter[\mathbb{N}]\,2\,funS,\ opPlusI\ (comp(funK,\ iter[\mathbb{N}]\,(funA\ a)\,funS))))),\ a :=$ a];
use *ExtASub*6[$X := funK,\ a := a$];
use *Theorem*68K;
use *C*6*KDef*;
use *ClosInPow*[$I := \{ funK \},\ U := \{ opPlusI,\ opBoxA\ a \},\ B := \{ comp \}$];
use *C*6*InPower*;
use *SubAntisym*[$\mathbb{N} \to \mathbb{N}$][$A := clos\,\{ funK \}\,\{ opPlusI,\ opBoxA\ a \}\,\{ comp \},\ B := C6$];
prove;
■

7.8 The case $X = L$

$\bar{0} = LL^+$, because $L(x + L(x)) = 0$.

theorem LZeroExpr
$comp(funL, opPlusI\ funL) = cst\ 0$

proof
apply UFExt to predicate $comp(funL,\ opPlusI\ funL)\ =\ cst\ 0$;
prove;
use Decomp$[x\ :=\ x]$;
prove;
use LAt$[m\ :=\ x,\ t\ :=\ funV\ x,\ k\ :=\ funK\ x]$;
prove;
use LAt$[m\ :=\ funA\ (funV\ x)\ +\ funV\ x,\ t\ :=\ funV\ x,\ k\ :=\ funV\ x]$;
prove;
■

theorem LZeroExprInCL
$\forall\ a : \mathbb{N} \bullet comp(funL, opPlusI\ funL) \in clos\ \{funL\}\ \{opPlusI, opBoxA\ a\}\ \{comp\}$

proof
prove;
■

theorem ZeroInCL
$\forall\ a : \mathbb{N} \bullet cst\ 0 \in clos\ \{funL\}\ \{opPlusI, opBoxA\ a\}\ \{comp\}$

proof
prove;
use LZeroExprInCL$[a\ :=\ a]$;
use LZeroExpr;
prove;
■

$L(2(i + 1)(i + 1)) = i$, since $2(i + 1)(i + 1) = A(2i + 1) + i + 1$.

theorem LMult
$funL \in Mult$

proof
invoke Mult;
use LAt$[m\ :=\ 0,\ t\ :=\ 0,\ k\ :=\ 0]$;
use AZero;
prove;
instantiate $r\ ==\ 2\ +\ 2\ *\ i$;
use TwoA$[x\ :=\ 1\ +\ 2\ *\ i]$;
use LAt$[m\ :=\ (2\ +\ 2\ *\ i)\ *\ (1\ +\ i),\ t\ :=\ 1\ +\ 2\ *\ i,\ k\ :=\ 1\ +\ i]$;
prove;
■

$K = L\bar{1}^+((\bar{1}^+)^+L)^+$ (Theorem 6.8), so the case of L reduces to that of K.

theorem KExprInCL
$\forall\ a : \mathbb{N} \mid 1 \leq a \bullet comp(comp(funL, opPlusI\ (cst\ 1)), opPlusI\ (comp(opPlusI\ (opPlusI\ (cst\ 1)),$
$funL))) \in clos\ \{funL\}\ \{opPlusI, opBoxA\ a\}\ \{comp\}$

proof
prove;
use *ZeroInCL*[$a := a$];
use *LMult*;
use *ExtAOne*[$X := funL,\ a := a$];
prove;
■

theorem KInCL
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet funK \in clos\,\{funL\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use *KExprInCL*[$a := a$];
use *KExprL*;
prove;
■

theorem CBUClosedA
$\forall\, X : \mathbb{N} \to \mathbb{N};\ a : \mathbb{N} \bullet clos\,\{X\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \in UClosed\,\{opPlusI, opBoxA\ a\}$

proof
prove;
use *ClosInPow*[$I := \{\ X\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \}$];
with enabled (*UClosedDef*) *prove*;
■

theorem CKSubCL
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet clos\,\{funK\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} \subseteq clos\,\{funL\}\,\{opPlusI, opBoxA\ a\}\,\{comp\}$

proof
prove;
use *KInCL*[$a := a$];
use *ClosLeast*[$I := \{\ funK\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \},\ Z := clos\,\{\ funL\ \}\,\{\ opPlusI,\ opBoxA\ a\ \}\,\{\ comp\ \}$];
use *ClosInPow*[$I := \{\ funL\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \}$];
use *CBUClosedA*[$X := funL,\ a := a$];
use *CBBClosed*[$X := funL,\ a := a$];
prove;
■

theorem ExtAL
$\forall\, a : \mathbb{N} \mid 1 \leq a \bullet clos\,\{funL\}\,\{opPlusI, opBoxA\ a\}\,\{comp\} = C6$

proof
prove;
use *ExtAK*[$a := a$];
use *CKSubCL*[$a := a$];
use *ExtASub*6[$X := funL,\ a := a$];
use *ClosInPow*[$I := \{\ funL\ \},\ U := \{\ opPlusI,\ opBoxA\ a\ \},\ B := \{\ comp\ \}$];
use *C*6*InPower*;
use *SubAntisym*[$\mathbb{N} \to \mathbb{N}$][$A := clos\,\{\ funL\ \}\,\{\ opPlusI,\ opBoxA\ a\ \}\,\{\ comp\ \},\ B := C6$];
prove;
■

7.9 Every starting value $a \geq 1$ (Remark 6.9)

For $a \geq 1$ the constructions for K and L above already work (ExtAK, ExtAL). For E and $a = 1$: $G = (S^2B)^{\square(1)}$ (the G above with $a = 1$) satisfies $G(n) = (n+1)^2$, and $\hat{P} = ES^2((G^+)^+)^+$ is a predecessor, since $(x+2)^2 + 3(x+1) + 2 = (x+3)^2 + x$.

theorem PEOneVal
$\forall\, k : \mathbb{N} \bullet comp(comp(funE, iter[\mathbb{N}]\, 2\, funS), opPlusI\ (opPlusI\ (opPlusI\ (opBoxA\, 1\ (comp(iter[\mathbb{N}]\, 2\, funS,$
$opBoxA\, 1\ (comp(funS, opPlusI\ (comp(comp(funD, funQ), iter[\mathbb{N}]\, 6\, funS)))))))))))\ (1 + k) = k$

proof
prove;
use $GEValAll[n := 1 + k,\ a := 1]$;
use $ESqAdd[m := 3 + k,\ k := k]$;
prove;
■

theorem PEOneInCE
$comp(comp(funE, iter[\mathbb{N}]\, 2\, funS), opPlusI\ (opPlusI\ (opPlusI\ (opBoxA\, 1\ (comp(iter[\mathbb{N}]\, 2\, funS,$
$opBoxA\, 1\ (comp(funS, opPlusI\ (comp(comp(funD, funQ),$
$iter[\mathbb{N}]\, 6\, funS)))))))))) \in clos\ \{funE\}\ \{opPlusI, opBoxA\, 1\}\ \{comp\}$

proof
use $ZeroInCB[X := funE,\ a := 1]$;
use $EDecr$;
use $EMult$;
use $ExtAI[X := funE,\ a := 1]$;
use $ExtAS[X := funE,\ a := 1]$;
use $QInCE[a := 1]$;
prove;
use $IterIn[I := \{ funE \},\ U := \{ opPlusI,\ opBoxA\,1 \},\ B := \{ comp \},\ F := funS,\ k := 6]$;
use $IterIn[I := \{ funE \},\ U := \{ opPlusI,\ opBoxA\,1 \},\ B := \{ comp \},\ F := funS,\ k := 2]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := funD,\ G := funQ]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := comp(funD,\ funQ),\ G := iter[\mathbb{N}]\,6\,funS]$;
prove;
use $CBPlusI[X := funE,\ a := 1,\ F := comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS)]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := funS,\ G := opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))]$;
prove;
use $CBBoxA[X := funE,\ a := 1,\ F := comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS)))]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := iter[\mathbb{N}]\,2\,funS,\ G := opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))]$;
prove;
use $CBBoxA[X := funE,\ a := 1,\ F := comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS)))))]$;
prove;
use $CBPlusI[X := funE,\ a := 1,\ F := opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))]$;
prove;
use $CBPlusI[X := funE,\ a := 1,\ F := opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))),]$;
prove;
use $CBPlusI[X := funE,\ a := 1,\ F := opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))))]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := funE,\ G := iter[\mathbb{N}]\,2\,funS]$;
prove;
use $CBComp[X := funE,\ a := 1,\ F := comp(funE,\ iter[\mathbb{N}]\,2\,funS),\ G := opPlusI\,(opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))))))]$;
prove;
■

theorem PEOneUF
$comp(comp(funE, iter[\mathbb{N}]\,2\,funS), opPlusI\,(opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS, opPlusI\,(comp(comp(funD, funQ), iter[\mathbb{N}]\,6\,funS))))))))))) \in \mathbb{N} \to \mathbb{N}$

proof
use $PEOneInCE$;
use $ClosInPow[I := \{ funE \},\ U := \{ opPlusI,\ opBoxA\,1 \},\ B := \{ comp \}]$;
apply inPower to predicate $clos\,\{ funE \}\,\{ opPlusI,\ opBoxA\,1 \}\,\{ comp \} \in \mathbb{P}\,(\mathbb{N} \to \mathbb{N})$;
instantiate $e == comp(comp(funE,\ iter[\mathbb{N}]\,2\,funS),\ opPlusI\,(opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,\ opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))))))$;
prove;
■

theorem PEOnePred
$comp(comp(funE, iter[\mathbb{N}]\,2\,funS), opPlusI\,(opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,$
$opBoxA\,1\,(comp(funS, opPlusI\,(comp(comp(funD, funQ), iter[\mathbb{N}]\,6\,funS))))))))))) \in Preds$

proof
use $PEOneUF$;
invoke $Preds$;
prove;
use $PEOneVal[k := x]$;
prove;
■

theorem ExtAEOne
$clos\,\{funE\}\,\{opPlusI, opBoxA\,1\}\,\{comp\} = C6$

proof
use $EDecr$;
use $EMult$;
use $ZeroInCB[X := funE,\ a := 1]$;
use $PEOneInCE$;
use $PEOneUF$;
use $PEOnePred$;
use $ExtASup[X := funE,\ P := comp(comp(funE,\ iter[\mathbb{N}]\,2\,funS),$
$opPlusI\,(opPlusI\,(opPlusI\,(opBoxA\,1\,(comp(iter[\mathbb{N}]\,2\,funS,\ opBoxA\,1\,(comp(funS,$
$opPlusI\,(comp(comp(funD,\ funQ),\ iter[\mathbb{N}]\,6\,funS))))))))),\ a := 1]$;
use $ExtASub6[X := funE,\ a := 1]$;
use $C6Def$;
use $ClosInPow[I := \{\ funE\ \},\ U := \{\ opPlusI,\ opBoxA\,1\ \},\ B := \{\ comp\ \}]$;
use $C6InPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := clos\,\{\ funE\ \}\,\{\ opPlusI,\ opBoxA\,1\ \}\,\{\ comp\ \},\ B := C6]$;
prove;
■

theorem Remark69
$clos\,\{funE\}\,\{opPlusI, opBoxA\,1\}\,\{comp\} = C6 \wedge clos\,\{funK\}\,\{opPlusI,$
$opBoxA\,1\}\,\{comp\} = C6 \wedge clos\,\{funL\}\,\{opPlusI, opBoxA\,1\}\,\{comp\} = C6$

proof
use $ExtAEOne$;
use $ExtAK[a := 1]$;
use $ExtAL[a := 1]$;
prove;
■

theorem ExtAAll
$\forall\, a : \mathbb{N};\ X : \mathbb{N} \to \mathbb{N} \mid 1 \leq a \wedge X \in \{funE, funK, funL\} \bullet clos\,\{X\}\,\{opPlusI, opBoxA\,a\}\,\{comp\} = C6$

proof
prove;
use ExtAK[$a := a$];
use ExtAL[$a := a$];
split $a = 1$;
cases;
use ExtAEOne;
prove;
split $X = funE$;
cases;
prove;
next;
split $X = funK$;
cases;
prove;
next;
prove;
next;
next;
use ExtAE[$a := a$];
prove;
split $X = funE$;
cases;
prove;
next;
split $X = funK$;
cases;
prove;
next;
prove;
next;
next;
■

theorem rule UFInPow
$\forall F : \mathbb{N} \to \mathbb{N} \bullet F \in \mathbb{P}(\mathbb{Z} \times \mathbb{Z})$

proof
prove;
■

8 The shifted operator F^{+b} (open question 2.2)

For fixed b, $F^{+b}(x) = F(x) + x + b$. The basis $\langle E, F^{+b}, FG, F^{\square} \rangle$ generates the same functions as Robinson's basis iff $b \geq 1$.

$opPlusIb : \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$

$\langle\!\langle$ rule PlusIbVal $\rangle\!\rangle$
$\forall b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlusIb\ b\ F\ x = F\ x + x + b$

proof [opPlusIb$domainCheck]
prove;
■

theorem rule PlusIbInner
$\forall b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda x : \mathbb{N} \bullet F\ x + x + b) \in \mathbb{N} \to \mathbb{N}$

proof
apply FunIntro to predicate $(\lambda\ x : \mathbb{N} \bullet F\,x + x + b) \in \mathbb{N} \to \mathbb{N}$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem rule PlusIbMid
$\forall\, b : \mathbb{N} \bullet (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\,x + x + b)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$

proof
apply FunIntro to predicate $(\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\,x + x + b)) \in (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

$PlusIbWit == (\lambda\, b : \mathbb{N} \bullet (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\,x + x + b)))$

proof [PlusIbWit$domainCheck]
prove;
■

theorem PlusIbWitFun
$PlusIbWit \in \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$

proof
invoke PlusIbWit;
apply FunIntro to predicate $(\lambda\ b : \mathbb{N} \bullet (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\,x + x + b))) \in \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N}))$;
with enabled (*RelPow*, *inPower*) *prove by reduce*;
■

theorem PlusIbWitApp
$\forall\, b : \mathbb{N} \bullet PlusIbWit\ b = (\lambda\, F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\, x : \mathbb{N} \bullet F\,x + x + b))$

proof
use PlusIbWitFun;
use pairInFunction$[\mathbb{N}, (\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})][f := PlusIbWit, x := b, y := (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\,x + x + b))]$;
invoke PlusIbWit;
prove by reduce;
■

theorem PlusIbWitApp2
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet PlusIbWit\ b\ F = (\lambda\, x : \mathbb{N} \bullet F\,x + x + b)$

proof
prove;
use PlusIbWitApp$[b := b]$;
use pairInFunction$[\mathbb{N} \to \mathbb{N}, \mathbb{N} \to \mathbb{N}][f := (\lambda\ F : \mathbb{N} \to \mathbb{N} \bullet (\lambda\ x : \mathbb{N} \bullet F\,x + x + b)), x := F, y := (\lambda\ x : \mathbb{N} \bullet F\,x + x + b)]$;
prove;
■

theorem rule PlusIbWitVal
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet PlusIbWit\ b\ F\ x = F\,x + x + b$

proof
prove;
use *PlusIbWitApp2*[$b := b$, $F := F$];
use *pairInFunction*[$\mathbb{N}$, $\mathbb{N}$][$f := (\lambda\ x : \mathbb{N} \bullet F\ x + x + b)$, $x := x$, $y := F\ x + x + b$];
prove;
■

theorem PlusIbEx
$\exists\ opPlusIb : \mathbb{N} \to ((\mathbb{N} \to \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})) \bullet \forall\ b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N};\ x : \mathbb{N} \bullet opPlusIb\ b\ F\ x = F\ x + x + b$

proof
instantiate *opPlusIb* $==$ *PlusIbWit*;
use *PlusIbWitFun*;
prove;
■

8.1 Shifts

$Shift == \{F : \mathbb{N} \to \mathbb{N};\ c : \mathbb{N} \mid \forall\ y : \mathbb{N} \bullet F\ y = y + c\}$

proof [Shift$domainCheck]
prove;
■

$ShiftPowSet == \{k : \mathbb{N} \mid \forall\ F : \mathbb{N} \to \mathbb{N};\ c, x : \mathbb{N} \mid (F, c) \in Shift \bullet iter[\mathbb{N}]\ k\ F\ x = x + k * c\}$

proof [ShiftPowSet$domainCheck]
prove;
■

theorem rule ShiftPowBase
$0 \in ShiftPowSet$

proof
invoke *ShiftPowSet*;
prove;
■

theorem rule ShiftPowStep
$\forall\ k : ShiftPowSet \bullet 1 + k \in ShiftPowSet$

proof
invoke *ShiftPowSet*;
prove;
instantiate $F__0 == F$, $c__0 == c$, $x__0 == x$;
invoke *Shift*;
prove;
instantiate $y__0 == iter[\mathbb{N}]\ k\ F\ x$;
prove;
■

theorem ShiftPowSub
$\mathbb{N} \subseteq ShiftPowSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq \mathit{ShiftPowSet}$;
prove;
invoke ShiftPowSet;
prove;
■

theorem ShiftPow
$\forall\, k, c, x : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid (F, c) \in \mathit{Shift} \bullet \mathit{iter}[\mathbb{N}]\, k\, F\, x = x + k * c$

proof
prove;
use ShiftPowSub;
invoke ShiftPowSet;
with enabled (*inPower*) *prove*;
instantiate $e == k$;
prove;
instantiate $F__1 == F,\ c__1 == c,\ x__1 == x$;
prove;
■

8.2 The construction, with $b = c + 1$

$\bar{0} = E^{\Box}$, $A = \bar{0}^{+b}$ ($A(x) = x+b$), $C = EA^{b-1}\bar{0}$ (the constant $b-1$), $B = C^{+b}$ ($B(x) = x+2b-1$), $T = B^{2b-2}AA$ ($T(x) = x + 4b^2 - 4b + 2$), $H = A^3(ET)^{+b}$, $G = H^{\Box}$ ($G(n) = n^2 + 4bn$), $\hat{P} = EA^{4b+1}((G^{+b})^{+b})^{+b}$ (a predecessor), $\bar{1} = \hat{P}^{b-1}A\bar{0}$, and $F^{+} = \hat{P}^{b}F^{+b}$.

theorem AShift
$\forall\, c : \mathbb{N} \bullet (\mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0), 1 + c) \in \mathit{Shift}$

proof
invoke Shift;
prove;
■

theorem rule CbVal
$\forall\, c, x : \mathbb{N} \bullet \mathit{comp}(\mathit{funE}, \mathit{comp}(\mathit{iter}[\mathbb{N}]\, c\,(\mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0)), \mathit{cst}\,0))\, x = c$

proof
prove;
use AShift$[c := c]$;
use ShiftPow$[k := c,\ c := 1 + c,\ x := 0,\ F := \mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0)]$;
use ESqAdd$[m := c,\ k := c]$;
prove;
■

theorem BShift
$\forall\, c : \mathbb{N} \bullet (\mathit{opPlusIb}\,(1+c)\,(\mathit{comp}(\mathit{funE}, \mathit{comp}(\mathit{iter}[\mathbb{N}]\, c\,(\mathit{opPlusIb}\,(1+c)\,(\mathit{cst}\,0)), \mathit{cst}\,0))), 1+2*c) \in \mathit{Shift}$

proof
invoke Shift;
prove;
■

theorem rule TbVal
$\forall\, c, x : \mathbb{N} \bullet \mathit{comp}(\mathit{iter}[\mathbb{N}]\,(2 * c)\,(\mathit{opPlusIb}\,(1 + c)\,(\mathit{comp}(\mathit{funE}, \mathit{comp}(\mathit{iter}[\mathbb{N}]\, c\,(\mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0)),$
$\mathit{cst}\,0)))), \mathit{comp}(\mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0), \mathit{opPlusIb}\,(1 + c)\,(\mathit{cst}\,0)))\, x = x + 4 * c * c + 4 * c + 2$

proof
prove;
use BShift[$c := c$];
use ShiftPow[$k := 2 * c$, $c := 1 + 2 * c$, $x := 2 + 2 * c + x$, $F := opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))$];
prove;
■

theorem rule HbVal
$\forall\, c, y : \mathbb{N} \bullet comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)), opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE, comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),$
$cst\,0)))), comp(opPlusIb\,(1 + c)\,(cst\,0),$
$opPlusIb\,(1 + c)\,(cst\,0))))))\ y = funE\,(y + 4 * c * c + 4 * c + 2) + y + 4 * c + 4$

proof
prove;
use AShift[$c := c$];
use ShiftPow[$k := 3$, $c := 1 + c$, $x := funE\,(y + 4 * c * c + 4 * c + 2) + y + 1 + c$, $F := opPlusIb\,(1 + c)\,(cst\,0)$];
prove;
■

$GbSet == \{n : \mathbb{N} \mid \forall\, c : \mathbb{N} \bullet opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)), opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE, comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)), cst\,0)))),$
$comp(opPlusIb\,(1 + c)\,(cst\,0), opPlusIb\,(1 + c)\,(cst\,0)))))))\ n = n * n + 4 * (1 + c) * n\}$

proof [GbSet$domainCheck]
prove;
■

theorem rule GbBase
$0 \in GbSet$

proof
invoke GbSet;
prove;
■

theorem rule GbStep
$\forall\, n : GbSet \bullet 1 + n \in GbSet$

proof
invoke GbSet;
prove;
instantiate c__0 == c;
use ESqAdd[$m := n + 2 * c + 1$, $k := 2 * n + 1$];
prove;
■

theorem GbSub
$\mathbb{N} \subseteq GbSet$

proof
apply natInduction to predicate $\mathbb{N} \subseteq GbSet$;
prove;
invoke GbSet;
prove;
■

theorem rule GbVal

$\forall\, n, c : \mathbb{N} \bullet opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)), opPlusIb\,(1+c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)),$
$cst\,0)))), comp(opPlusIb\,(1+c)\,(cst\,0), opPlusIb\,(1+c)\,(cst\,0)))))))\,n = n*n+4*(1+c)*n$

proof

prove;
use GbSub;
invoke GbSet;
with enabled (*inPower*) *prove*;
instantiate $e == n$;
prove;
instantiate $c__1 == c$;
prove;
■

theorem PbVal

$\forall\, c, k : \mathbb{N} \bullet comp(funE, comp(iter[\mathbb{N}]\,(4*c+5)\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)), cst\,0)))), comp(opPlusIb\,(1+c)\,(cst\,0),$
$opPlusIb\,(1+c)\,(cst\,0))))))))))))\,(1+k) = k$

proof

with disabled (*BoxSuc*, *HbVal*) *prove*;
use AShift$[c := c]$;
use ShiftPow$[k := 4*c+5,\ c := 1+c,\ x := 6+3*c+3*k+(1+k)*(1+k)+$
$4*(1+c)*(1+k),\ F := opPlusIb\,(1+c)\,(cst\,0)]$;
use ESqAdd$[m := k+2*c+4,\ k := k]$;
with disabled (*BoxSuc*, *HbVal*) *prove*;
■

theorem PbPred

$\forall\, c : \mathbb{N} \bullet comp(funE, comp(iter[\mathbb{N}]\,(4*c+5)\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)), cst\,0)))), comp(opPlusIb\,(1+c)\,(cst\,0),$
$opPlusIb\,(1+c)\,(cst\,0))))))))))))) \in Preds$

proof

prove;
invoke Preds;
prove;
use PbVal$[c := c,\ k := x]$;
prove;
■

theorem OneEqVb

$\forall\, c : \mathbb{N} \bullet comp(iter[\mathbb{N}]\,c\,(comp(funE, comp(iter[\mathbb{N}]\,(4*c+5)\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)),$
$opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)), cst\,0)))), comp(opPlusIb\,(1+c)\,(cst\,0),$
$opPlusIb\,(1+c)\,(cst\,0))))))))))))), comp(opPlusIb\,(1+c)\,(cst\,0), cst\,0)) = cst\,1$

proof

apply UFExt to predicate $comp(iter[\mathbb{N}]\,c\,(comp(funE,\ comp(iter[\mathbb{N}]\,(4\ *\ c\ +\ 5)\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2\ *\ c)\,(opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1\ +\ c)\,(cst\,0),\ opPlusIb\,(1\ +\ c)\,(cst\,0))))))))))))),\ comp(opPlusIb\,(1\ +\ c)\,(cst\,0),\ cst\,0))\ =\ cst\,1$;

prove;

use $PbPred[c\ :=\ c]$;

use $IterPAll[k\ :=\ c,\ P\ :=\ comp(funE,\ comp(iter[\mathbb{N}]\,(4\ *\ c\ +\ 5)\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2\ *\ c)\,(opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1\ +\ c)\,(cst\,0),\ opPlusIb\,(1\ +\ c)\,(cst\,0))))))))))))),\ x\ :=\ 1]$;

prove;

■

theorem PlusIEqVb

$\forall\, c : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet comp(iter[\mathbb{N}]\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(4*c+5)\,(opPlusIb\,(1+c)\,(cst\,0)), opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)), opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)), cst\,0)))), comp(opPlusIb\,(1+c)\,(cst\,0), opPlusIb\,(1+c)\,(cst\,0))))))))))))), opPlusIb\,(1+c)\,F) = opPlusI\ F$

proof

apply UFExt to predicate $comp(iter[\mathbb{N}]\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(4\ *\ c\ +\ 5)\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2\ *\ c)\,(opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1\ +\ c)\,(cst\,0),\ opPlusIb\,(1\ +\ c)\,(cst\,0))))))))))))),\ opPlusIb\,(1\ +\ c)\,F)\ =\ opPlusI\ F$;

prove;

use $PbPred[c\ :=\ c]$;

use $IterPAll[k\ :=\ 1\ +\ c,\ P\ :=\ comp(funE,\ comp(iter[\mathbb{N}]\,(4\ *\ c\ +\ 5)\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opPlusIb\,(1\ +\ c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2\ *\ c)\,(opPlusIb\,(1\ +\ c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1\ +\ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1\ +\ c)\,(cst\,0),\ opPlusIb\,(1\ +\ c)\,(cst\,0))))))))))))),\ x\ :=\ x\ +\ F\,x]$;

prove;

■

8.3 Closure steps in $\langle E, F^{+b}, FG, F^{\Box} \rangle$

theorem CVComp

$\forall\, b : \mathbb{N};\ F, G : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\} \wedge G \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\} \bullet comp(F, G) \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof

prove;

use $ClosElemUF[I\ :=\ \{\ funE\ \},\ U\ :=\ \{\ opPlusIb\ b,\ opBox\ \},\ B\ :=\ \{\ comp\ \},\ F\ :=\ F]$;

use $ClosElemUF[I\ :=\ \{\ funE\ \},\ U\ :=\ \{\ opPlusIb\ b,\ opBox\ \},\ B\ :=\ \{\ comp\ \},\ F\ :=\ G]$;

prove;

■

theorem CVPlusIb

$\forall\, b : \mathbb{N};\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\} \bullet opPlusIb\ b\ F \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof

prove;

use $ClosElemUF[I\ :=\ \{\ funE\ \},\ U\ :=\ \{\ opPlusIb\ b,\ opBox\ \},\ B\ :=\ \{\ comp\ \},\ F\ :=\ F]$;

prove;

■

theorem CVBox
$\forall\, b : \mathbb{N};\ F : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{funE\}\,\{opPlusIb\ b,$
$opBox\}\,\{comp\} \bullet opBox\ F \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof
prove;
use $ClosElemUF[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opBox\ \},\ B := \{\ comp\ \},\ F := F]$;
prove;
■

theorem CVIter
$\forall\, b, k : \mathbb{N};\ F, G : \mathbb{P}(\mathbb{Z} \times \mathbb{Z}) \mid F \in clos\,\{funE\}\,\{opPlusIb\ b,$
$opBox\}\,\{comp\} \wedge G \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\} \bullet comp(iter[\mathbb{N}]\ k\ F,$
$G) \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof
prove;
use $ClosElemUF[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opBox\ \},\ B := \{\ comp\ \},\ F := F]$;
use $ClosElemUF[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opBox\ \},\ B := \{\ comp\ \},\ F := G]$;
use $PowIn[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opBox\ \},\ B := \{\ comp\ \},\ F := F,\ G := G,\ k$
$:= k]$;
prove;
■

theorem ZeroExprInCV
$\forall\, b : \mathbb{N} \bullet opBox\ funE \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof
prove;
■

theorem ZeroInCV
$\forall\, b : \mathbb{N} \bullet cst\ 0 \in clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\}$

proof
prove;
use $ZeroExprInCV[b := b]$;
use *BoxEExpr*;
prove;
equality substitute $cst\ 0$;
prove;
■

theorem PbInCV
$\forall\, c : \mathbb{N} \bullet comp(funE, comp(iter[\mathbb{N}]\ (4 * c + 5)\ (opPlusIb\ (1 + c)\ (cst\ 0)),$
$opPlusIb\ (1 + c)\ (opPlusIb\ (1 + c)\ (opPlusIb\ (1 + c)\ (opBox\ (comp(iter[\mathbb{N}]\ 3\ (opPlusIb\ (1 + c)\ (cst\ 0)),$
$opPlusIb\ (1 + c)\ (comp(funE, comp(iter[\mathbb{N}]\ (2 * c)\ (opPlusIb\ (1 + c)\ (comp(funE,$
$comp(iter[\mathbb{N}]\ c\ (opPlusIb\ (1 + c)\ (cst\ 0)), cst\ 0)))), comp(opPlusIb\ (1 + c)\ (cst\ 0),$
$opPlusIb\ (1 + c)\ (cst\ 0)))))))))))) \in clos\,\{funE\}\,\{opPlusIb\ (1 + c), opBox\}\,\{comp\}$

proof

prove;
use *ZeroInCV*[$b := 1 + c$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := cst\,0$];
prove;
use *CVIter*[$b := 1 + c,\ k := c,\ F := opPlusIb\,(1 + c)\,(cst\,0),\ G := cst\,0$];
prove;
use *CVComp*[$b := 1 + c,\ F := funE,\ G := comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0))$];
prove;
use *CVComp*[$b := 1 + c,\ F := opPlusIb\,(1 + c)\,(cst\,0),\ G := opPlusIb\,(1 + c)\,(cst\,0)$];
prove;
use *CVIter*[$b := 1 + c,\ k := 2 * c,\ F := opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0))),\ G := comp(opPlusIb\,(1 + c)\,(cst\,0),$
$opPlusIb\,(1 + c)\,(cst\,0))$];
prove;
use *CVComp*[$b := 1 + c,\ F := funE,\ G := comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1$
$+ c)\,(cst\,0)))$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1$
$+ c)\,(cst\,0))))$];
prove;
use *CVIter*[$b := 1 + c,\ k := 3,\ F := opPlusIb\,(1 + c)\,(cst\,0),\ G := opPlusIb\,(1 +$
$c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1$
$+ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0)))))$];
prove;
use *CVBox*[$b := 1 + c,\ F := comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 +$
$c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1$
$+ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0))))))$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 +$
$c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1$
$+ c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0)))))))$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := opPlusIb\,(1 + c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 +$
$c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1$
$+ c)\,(cst\,0)))))))))$];
prove;
use *CVPlusIb*[$b := 1 + c,\ F := opPlusIb\,(1 + c)\,(opPlusIb\,(1 +$
$c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),$
$cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0))))))))))$];
prove;
use *CVIter*[$b := 1 + c,\ k := 4 * c + 5,\ F := opPlusIb\,(1 + c)\,(cst\,0),\ G := opPlusIb\,(1 +$
$c)\,(opPlusIb\,(1 + c)\,(opPlusIb\,(1 + c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),$
$opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,$
$comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1$
$+ c)\,(cst\,0))))))))))$];
prove;
■

theorem OneInCV

$\forall\, c : \mathbb{N} \bullet cst\,1 \in clos\,\{funE\}\,\{opPlusIb\,(1 + c), opBox\}\,\{comp\}$

proof
prove;
use $ZeroInCV[b := 1 + c]$;
use $PbInCV[c := c]$;
prove;
use $CVPlusIb[b := 1 + c,\ F := cst\,0]$;
prove;
use $CVComp[b := 1 + c,\ F := opPlusIb\,(1 + c)\,(cst\,0),\ G := cst\,0]$;
prove;
use $CVIter[b := 1 + c,\ k := c,\ F := comp(funE,\ comp(iter[\mathbb{N}]\,(4 * c + 5)\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(opPlusIb\,(1 + c)\,(opPlusIb\,(1 + c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0))))))))))),\ G := comp(opPlusIb\,(1 + c)\,(cst\,0),\ cst\,0)]$;
prove;
use $OneEqVb[c := c]$;
prove;
■

theorem PlusIExprInCV
$\forall\, c : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in clos\,\{funE\}\,\{opPlusIb\,(1 + c),\ opBox\}\,\{comp\} \bullet comp(iter[\mathbb{N}]\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(4*c+5)\,(opPlusIb\,(1+c)\,(cst\,0)),\ opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opPlusIb\,(1+c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1+c)\,(cst\,0)),\ opPlusIb\,(1+c)\,(comp(funE, comp(iter[\mathbb{N}]\,(2*c)\,(opPlusIb\,(1+c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1+c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1+c)\,(cst\,0),\ opPlusIb\,(1+c)\,(cst\,0))))))))))),\ opPlusIb\,(1+c)\,F) \in clos\,\{funE\}\,\{opPlusIb\,(1+c),\ opBox\}\,\{comp\}$

proof
prove;
use $PbInCV[c := c]$;
use $CVPlusIb[b := 1 + c,\ F := F]$;
prove;
use $CVIter[b := 1 + c,\ k := 1 + c,\ F := comp(funE,\ comp(iter[\mathbb{N}]\,(4 * c + 5)\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(opPlusIb\,(1 + c)\,(opPlusIb\,(1 + c)\,(opBox\,(comp(iter[\mathbb{N}]\,3\,(opPlusIb\,(1 + c)\,(cst\,0)),\ opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,(2 * c)\,(opPlusIb\,(1 + c)\,(comp(funE,\ comp(iter[\mathbb{N}]\,c\,(opPlusIb\,(1 + c)\,(cst\,0)),\ cst\,0)))),\ comp(opPlusIb\,(1 + c)\,(cst\,0),\ opPlusIb\,(1 + c)\,(cst\,0))))))))))),\ G := opPlusIb\,(1 + c)\,F]$;
prove;
■

theorem PlusIInCV
$\forall\, c : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in clos\,\{funE\}\,\{opPlusIb\,(1 + c),\ opBox\}\,\{comp\} \bullet opPlusI\ F \in clos\,\{funE\}\,\{opPlusIb\,(1 + c), opBox\}\,\{comp\}$

proof
prove;
use $PlusIExprInCV[c := c,\ F := F]$;
use $PlusIEqVb[c := c,\ F := F]$;
prove;
■

8.4 $\langle E, F^{+b}, FG, F^{\Box} \rangle$ for $b \geq 1$

theorem CVUClosed
$\forall\, c : \mathbb{N} \bullet clos\,\{funE\}\,\{opPlusIb\,(1 + c), opBox\}\,\{comp\} \in UClosed\,\{opPlusI, opBox\}$

proof
prove;
use ClosInPow[$I := \{ funE \}$, $U := \{ opPlusIb\,(1 + c),\ opBox \}$, $B := \{ comp \}$];
with enabled (*UClosedDef*) *prove*;
use ClosElemUF[$I := \{ funE \}$, $U := \{ opPlusIb\,(1 + c),\ opBox \}$, $B := \{ comp \}$, $F := F$];
split $u = opPlusI$;
cases;
use PlusIInCV[$c := c$, $F := F$];
prove;
next;
prove;
next;
■

theorem CVBClosed
$\forall\, b : \mathbb{N} \bullet clos\,\{funE\}\,\{opPlusIb\ b, opBox\}\,\{comp\} \in BClosed\,\{comp\}$

proof
prove;
use ClosInPow[$I := \{ funE \}$, $U := \{ opPlusIb\ b,\ opBox \}$, $B := \{ comp \}$];
with enabled (*BClosedDef*) *prove*;
■

theorem C6SubCV
$\forall\, c : \mathbb{N} \bullet C6 \subseteq clos\,\{funE\}\,\{opPlusIb\,(1 + c), opBox\}\,\{comp\}$

proof
prove;
use C6Def;
use ClosLeast[$I := \{ cst\,1,\ funE \}$, $U := \{ opPlusI,\ opBox \}$, $B := \{ comp \}$, $Z := clos\,\{$ *funE* $\}\,\{$ $opPlusIb\,(1 + c),\ opBox \}\,\{$ *comp* $\}$];
use ClosInPow[$I := \{ funE \}$, $U := \{ opPlusIb\,(1 + c),\ opBox \}$, $B := \{ comp \}$];
use CVUClosed[$c := c$];
use CVBClosed[$b := 1 + c$];
use OneInCV[$c := c$];
prove;
■

$F^{+b} = (S^b F)^+$, so C_6 is closed under F^{+b}.

theorem PlusIbExpr
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \bullet opPlusIb\ b\ F = opPlusI\,(comp(iter[\mathbb{N}]\ b\ funS, F))$

proof
prove;
apply UFExt to predicate $opPlusIb\ b\ F = opPlusI\,(comp(iter[\mathbb{N}]\ b\ funS,\ F))$;
prove;
■

theorem C6PlusIbExpr
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet opPlusI\,(comp(iter[\mathbb{N}]\ b\ funS, F)) \in C6$

proof
prove;
use C6Def;
use PowIn[$I := \{ cst\,1,\ funE \}$, $U := \{ opPlusI,\ opBox \}$, $B := \{ comp \}$, $F := funS$, $G := F$, $k := b$];
prove;
■

theorem C6PlusIb
$\forall\, b : \mathbb{N};\ F : \mathbb{N} \to \mathbb{N} \mid F \in C6 \bullet opPlusIb\ b\ F \in C6$

proof
prove;
use $C6PlusIbExpr[b := b,\ F := F]$;
use $PlusIbExpr[b := b,\ F := F]$;
prove;
■

theorem C6UClosedB
$\forall\, b : \mathbb{N} \bullet C6 \in UClosed\ \{opPlusIb\ b, opBox\}$

proof
prove;
use $C6InPower$;
with enabled $(UClosedDef,\ C6\,UF)$ *prove*;
use $InPair[\mathbb{P}\ (\mathbb{P}\ (\mathbb{Z} \times \mathbb{Z}) \times \mathbb{P}\ (\mathbb{Z} \times \mathbb{Z}))][u := u,\ p := opPlusIb\ b,\ q := opBox]$;
split $u = opBox$;
cases;
prove;
next;
use $C6PlusIb[b := b,\ F := F]$;
prove;
next;
■

theorem CVSubC6
$\forall\, b : \mathbb{N} \bullet clos\ \{funE\}\ \{opPlusIb\ b, opBox\}\ \{comp\} \subseteq C6$

proof
prove;
use $ClosLeast[I := \{\ funE\ \},\ U := \{\ opPlusIb\ b,\ opBox\ \},\ B := \{\ comp\ \},\ Z := C6]$;
use $C6InPower$;
use $C6\,UClosedB[b := b]$;
use $C6BClosedComp$;
prove;
■

theorem ExtVbSucc
$\forall\, c : \mathbb{N} \bullet clos\ \{funE\}\ \{opPlusIb\ (1 + c), opBox\}\ \{comp\} = C6$

proof
prove;
use $C6SubCV[c := c]$;
use $CVSubC6[b := 1 + c]$;
use $ClosInPow[I := \{\ funE\ \},\ U := \{\ opPlusIb\ (1 + c),\ opBox\ \},\ B := \{\ comp\ \}]$;
use $C6InPower$;
use $SubAntisym[\mathbb{N} \to \mathbb{N}][A := clos\ \{\ funE\ \}\ \{\ opPlusIb\ (1 + c),\ opBox\ \}\ \{\ comp\ \},\ B := C6]$;
prove;
■

theorem ExtVb
$\forall\, b : \mathbb{N} \mid 1 \leq b \bullet clos\ \{funE\}\ \{opPlusIb\ b, opBox\}\ \{comp\} = C6$

proof
prove;
use $ExtVbSucc[c := b - 1]$;
prove;
■

8.5 $b = 0$: every generated function fixes 0, so $\bar{1}$ is not generated

$FixZero == \{F : \mathbb{N} \to \mathbb{N} \mid F\,0 = 0\}$

proof [FixZero$domainCheck]
prove;
■

theorem FixZeroInPower
$FixZero \in \mathbb{P}(\mathbb{N} \to \mathbb{N})$

proof
invoke FixZero;
with enabled (*inPower*) *prove*;
■

theorem FixZeroUClosed
$FixZero \in UClosed\,\{opPlusIb\,0, opBox\}$

proof
use FixZeroInPower;
with enabled (*UClosedDef*) *prove*;
invoke FixZero;
prove;
use InPair$[\mathbb{P}\,(\mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}) \times \mathbb{P}\,(\mathbb{Z} \times \mathbb{Z}))][u := u,\ p := opPlusIb\,0,\ q := opBox]$;
split $u = opBox$;
cases;
prove;
next;
prove;
next;
■

theorem FixZeroBClosed
$FixZero \in BClosed\,\{comp\}$

proof
use FixZeroInPower;
with enabled (*BClosedDef*) *prove*;
invoke FixZero;
prove;
■

theorem CVZeroFix
$clos\,\{funE\}\,\{opPlusIb\,0, opBox\}\,\{comp\} \subseteq FixZero$

proof
use ClosLeast$[I := \{\ funE\ \},\ U := \{\ opPlusIb\,0,\ opBox\ \},\ B := \{\ comp\ \},\ Z := FixZero]$;
use FixZeroInPower;
use FixZeroUClosed;
use FixZeroBClosed;
invoke FixZero;
prove;
■

theorem OneNotCVZero
$\neg\ cst\,1 \in clos\,\{funE\}\,\{opPlusIb\,0, opBox\}\,\{comp\}$

proof
use CVZeroFix;
apply subsetDef to predicate clos { *funE* } { *opPlusIb* 0, *opBox* } { *comp* } ⊆ *FixZero*;
apply inPower to predicate clos { *funE* } { *opPlusIb* 0, *opBox* } { *comp* } ∈ ℙ *FixZero*;
instantiate e == *cst* 1;
invoke FixZero;
prove;
■

theorem OneInEqC6
$\forall\, Y : \mathbb{P}(\mathbb{N} \to \mathbb{N}) \mid Y = C6 \bullet \mathit{cst}\, 1 \in Y$

proof
prove;
■

theorem ExtVbZero
$\neg\ \mathit{clos}\,\{\mathit{funE}\}\,\{\mathit{opPlusIb}\,0, \mathit{opBox}\}\,\{\mathit{comp}\} = C6$

proof
use OneNotCVZero;
use ClosInPow[*I* := { *funE* }, *U* := { *opPlusIb* 0, *opBox* }, *B* := { *comp* }];
use OneInEqC6[*Y* := *clos* { *funE* } { *opPlusIb* 0, *opBox* } { *comp* }];
prove;
■

theorem ExtVbIff
$\forall\, b : \mathbb{N} \bullet \mathit{clos}\,\{\mathit{funE}\}\,\{\mathit{opPlusIb}\, b, \mathit{opBox}\}\,\{\mathit{comp}\} = C6 \Leftrightarrow 1 \leq b$

proof
prove;
split b = 0;
cases;
use ExtVbZero;
prove;
next;
use ExtVb[*b* := *b*];
prove;
next;
■